# Bulk versus Sequential Product Returns: Algorithms and Insights for Assortment Planning

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**Abstract**

In this paper, we present, study and compare two distinct customer choice frameworks crafted to model the step-by-step process through which consumers order products, and then subsequently decide which to keep and which to return based on their realized preferences. The first of the two models that we consider is the so-called Sequential Returns model conceived by Wagner and Martínez-de Albéniz (2020), wherein customers sequentially order products one-by-one, deciding when to stop their search process by trading off their realized utility for the product at-hand with the inconveniences of returning this item and ordering a substitute. The second model is our own creation, which we derive by tweaking the random-utility-based framework of the sequential model so as to capture what we refer to as "bulk" ordering behavior. That is, we develop the Bulk Returns model, wherein customers are allowed to order all of the products that interest them upfront, ultimately keeping their most preferred, and returning the rest.

First and foremost, we set out to study the now-classic assortment optimization problem, where the goal is to select the profit-maximizing subset of products to make available for purchase, when demand is governed by either of the two returns models, and the retailer incurs a refurbishing cost for each item that is returned. For the Sequential Returns model, we provide an optimal polynomial-time algorithm for the setting where the return costs are homogeneous across products, noting that this result improves upon the previous-best PTAS of Wagner and Martínez-de Albéniz (2020). Additionally, we present a novel hardness result for a constrained variant of the assortment problem. For the Bulk Returns model, we show that its corresponding assortment problem is NP-Hard, and present a PTAS based on a carefully-crafted knapsack-based approximation. We then augment the bulk framework to allow for multiple customer types, each distinguished by the extent to which returning items is deemed to be an inconvenience. For this multi-type extension, we present two nuanced approximate-dynamic-programming-based approaches. We conclude our analysis by offering both theoretical and empirical comparisons of the two returns model so as to better understand what sorts of returns behavior should be enforced or encouraged by retailers. Our cornerstone result along this line is that we are able to show that the sequential model surprisingly leads to universally higher profits. In the numerical experiments that follow, we set out to quantify these gains and understand if the high-level managerial insights derived from our theoretical analysis persist when we augmented our returns disutility/cost framework to add more of an air of realism.

**Keywords**: consumer choice, return, assortment optimization, approximation schemes.

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## 1 Introduction

In the realm of retail, recent trends have revealed two fundamental truths regarding customer purchasing patterns. First, the convenience of online shopping has made it pervasive and irresistible, with total U.S. online sales eclipsing $1 trillion in 2021 (Schneider, 2022a). Second, born out of the ease of online shopping, lenient return policies have made returns equally as ubiquitous; in 2021, approximately 16.5% of online sales were returned, which amounts to an excess of $200 billion. It should come as no surprise that these staggering rates of return affect retailers' bottom lines in a multitude of ways. The 2023 Forbes piece "Total Cost Of Online Returns Breaks $200 Billion—Can Retail Afford It?" (Guha, 2023) explicitly documents the many sources of costs that result from returns:

> *"First, returns represent a de-conversion where the sale is lost. On top of that come the reverse logistic costs—getting the item back, inspecting it, refurbishing and restocking it, potentially even liquidating or scrapping the item. "*

Ader et al. (2021) note that the main reason for these returns is overwhelmingly poor fit or a mismatch of style. This same article points to data-driven assortment optimization tools as a means to mitigate returns by recommending "products most suited to consumers' size and style preferences, based on their purchase history".

Given the discussion above, it should come as no surprise that a 2022 survey of 100 mid- to large-sized U.S.-based retailers ranked "reducing returns/reducing lost sales from returns" as the second highest priority behind only growth and expansion (Optoro, 2022). In fact, this is likely the first time that returns have been at the forefront of retail operations. As such, important questions linger regarding (i) how to make optimal product recommendation decisions in the face of returns and (ii) which types of returns behaviors should be encouraged/discouraged to improve revenues and ensure customers ultimately find the products they seek. In the remainder of this paper, we set our sights on addressing these two questions from both theoretical and empirical vantage points. At the core of our work are two choice models, whose dynamics are introduced next. The distinguishing feature of the two models is the fundamental nature through which they exhibit returns behavior.

**An overview of our two returns frameworks.** The first of our two choice models is the Sequential Returns model conceived by Wagner and Martínez-de Albéniz (2020). Under this model, customers are assumed to buy/order products one-by-one. After each purchase, the consumer familiarizes herself with the product during a short trial period, and then either keeps the product, exchanges it (and thus continues her search process), or returns it and abandons the search process. The second model is new, and takes the perspective of consumers who are convenience-driven, and hence do not have the time or patience to adopt the one-at-a-time ordering framework of the Sequential Returns model. Aptly named for its all-at-once ordering framework, we introduce the Bulk Returns model, where customers simultaneously order multiple products, try them each on, and then keep the one they most prefer, while returning the rest. Indeed, a recent survey by Brightpearl (Smith, 2018) found that 25% of shoppers bought multiple items with the intention of returning some. In both models, we capture the hassle of returns through a returns disutility parameter that is incurred for each returned item. In what follows, we summarize our main contributions, diving deeper into the nuances of our two focal returns models and the assortment problems they admit.

### 1.1 Contributions

In this section, we summarize our main contributions, which run the gamut from modeling innovations to algorithmic findings to managerial insights.

**A novel model for consumer returns.** Our first contribution is the development of the Bulk Returns model, which replicates the random utility framework of Wagner and Martínez-de Albéniz (2020)'s Sequential Returns model within a setting where customers order *all* products that interest them upfront, rather than one-by-one. More specifically, under the Bulk Returns model, the choice process of each customer unfolds over only two sequential stages; namely, an initial ordering stage, which is followed by a try-on and purchase stage. Informally speaking, during the initial ordering stage, the customer browses the set of offered or displayed products, and chooses a subset to take home and try-on. We model this initial ordering decision as one made with (i) only a partial sense of the extent to which the customer will value or enjoy each product, and (ii) a premonition of future return hassles, i.e. the more products ordered, the more that will eventually need to be returned. If the customer orders at least one product, she enters the try-on and purchase stage, where she familiarizes herself with each ordered product enough to finalize her preferences. We formalize the dynamics described above in Section 2.2, where we analytically characterize the ordering, and subsequent purchasing decision of each customer under the Bulk Returns model.

**The assortment planning problem.** Our main focus is the assortment optimization problem under the two returns models. In this problem, the goal is to select a subset of products to make available for purchase with the goal to maximize expected profits. A key distinction between our formulation and that of the classical assortment problem is that our formulation (usually) includes the aforementioned return costs, incurred for each product that is returned. In what follows, we summarize our algorithmic findings apropos the two resulting assortment problems.

- <u>Results for the Bulk Returns model:</u> In Section 3.1, we formulate our assortment problem of interest, and show that even in the absence of return costs, the assortment problem under the Bulk Returns model is NP-Hard. In Section 4, we present approximation schemes for two variants of our assortment problem. First, we present a polynomial time approximation scheme (PTAS), which identifies an $\epsilon$-optimal assortment by exploiting properties of the optimal solution to a linear programming relaxation of a particular knapsack problem.

- <u>Results for the Bulk Returns model - the multi-type extension:</u> From here, we propose a multi-type extension of our model, where we assume the customer population is partitioned into an arbitrary number of types, each distinguished by a distinct aversion to returning items, i.e. some customer types do not mind returning items, while others might view the process as a great inconvenience. In this case, for tractability purposes, we consider the assortment problem in the absence of return costs, and develop a carefully crafted $O(\log n)$-approximation scheme, where $n$ is the number of products available to the retailer. Our approach employs ideas related to efficient enumeration and approximate dynamic programming (ADP).

  Interestingly, with a very minor tweak to our model described in the sequel, the Bulk Returns model reduces to the search-cost-based browsing model of Wang and Sahin (2018). An open question posed by this earlier work is whether non-trivial approximation guarantees exist for the multi-type version of their assortment problem. We answer this question in the affirmative by presented a constant factor approximation, which represents a vast improvement over the pseudo-polynomial time approach presented in Wang and Sahin (2018). To do so, we again employ a nuanced ADP-based approach, which critically relies on a novel "assortment pruning" step, where we show how to delete products from any assortment while only increasing the revenue earned by each customer type.

- Results for the Sequential Returns model: The seminal work of Wagner and Martínez-de Albéniz (2020), whose Sequential Returns model is re-hashed in Section 2.1, considers the assortment problem where the return costs are homogeneous across the products. Under this simplified variant, the authors propose a PTAS, but fail to offer hardness results of any kind. As such, whether or not this PTAS represents a best-case result remains an open questions. In Section 5, we answer this question in the negative, showing that there is an optimal polynomial-time algorithm for the assortment problem when the return costs are homogeneous. In fact, we show that this variant of the assortment problem can be formulated as a linear program, whose number of decision variables and constraints grows quadratically in the number of available products. This positive algorithmic result naturally beckons the question of whether the assortment problem under the Sequential Returns model is fundamentally harder than that of the classical MNL model. Namely, under the classical MNL model, not only does the unconstrained assortment problem admit optimal polynomial-time algorithms (Talluri and Van Ryzin, 2004), but so too does the totally unimodular (TU)-constrained variant (Sumida et al., 2021). In sharp contrast to this much celebrated positive result for MNL-based assortment optimization, in Section 3.2, we show that that when we inject TU constraints into our assortment problem, it becomes inapproximable, meaning that unless $P = NP$, one cannot hope to develop an approximation scheme with *any* positive factor guarantee.

**Comparing the two returns models - a theoretical lens.** Having crafted the Bulk Returns model so that it is parameterized identically to the Sequential Returns model, we turn our sights to comparing the two models so as to draw out managerial insights. Specifically, in Section 6.1, we first show that when the respective assortments problems are instantiated with the same set of primitives, the expected profit generated by the optimal assortment under the Sequential Returns model is at least that of its counterpart under the Bulk Returns model. The take-home insight from this result is that retailers should encourage (or force through policy) the type of sequential purchasing/return behavior portrayed by the Sequential Returns model by reducing return frictions that make ordering in bulk more convenient. Next, we make progress in understanding how the choice probabilities compare across the two models for a fixed assortment. Along this line, in Section 6.1, we first show that the overall market share is larger under the Sequential Returns model, meaning the likelihood of a no-purchase event is smaller. Digging deeper, we again consider a fixed assortment where the products are indexed in decreasing order of their intrinsic attractiveness, i.e., a parameter akin to the preference weights in the Multinomial logit (MNL) choice model. We prove that for an arbitrary assortment $S$, there always exists a threshold index $k \in S$, below which the choice probabilities of the sequential model exceed those of the bulk model, and above which the opposite is true. Put simply, this results suggests that, as compared to the Bulk Returns model, customers under the Sequential Returns model are more likely to take home a higher valued product. Generally speaking, these two latter results together suggest that customers are better off under the sequential framework, however, the assumption underlying this insight (total return disutilities/costs that grow linearly in number of returned items) are challenged within our computation experiments, as described next.

**Comparing the two returns models - an empirical lens.** In Section 6.2, we make an effort to dig deeper into the insights uncovered by our theoretical comparison of the two models by conducting an extensive set of numerical experiments, where, aided by the Durable Goods Data Set 1 (Ni et al., 2012), we create a diverse array of assortment instances through which we can empirically compare the expected profits and market shares conferred by the two returns models. Within these experiments, we allow ourselves to employ integer programming tools to uncover optimal assortments, which, in turn, opens the door for us to consider a more nuanced and intricate returns framework. Specifically, we assume that returns disutilities /costs are experienced/incurred for each return trip (i.e., the act of returning any collection of items) *and* for each returned item. With disutilities and costs no longer linear in the number of returned items, the two models might perhaps be on more equal grounds, since bulk customers will only ever require one return trip, while sequential customers might require multiple return trips. The fundamental question we seek to investigate is whether the bulk model can "catch up" to the sequential model in terms of expected profit and market share as the return trip disutility/cost increases, noting that when these terms are zero, we revert back to the original model. Ultimately, the high level insight that sequential buying behavior is best for both parties (customers and retailers) persists, as we find that both expected profits and market shares are 5-15% higher on average under the assortments recommended by the sequential model. We do, however, find that the bulk model is best when shipping costs are set to be exorbitant.

### 1.2 Related literature and outline of future sections

We begin by recapping relevant work in the operations literature that is related to product returns. We then proceed to discuss related work within the assortment optimization literature, focusing in particular on papers that include notions of consideration sets, due to their salient overlap with the choice structure imposed by our two returns models. The final paragraph of this section includes a road map for the remainder of the paper.

**Product returns.** Prior to Wagner and Martínez-de Albéniz (2020), Alptekinoğlu and Grasas (2014) were the first to consider assortment problems under a choice model that explicitly incorporates returns behavior, although the return behavior in this seminal work is exogenous, i.e., it is not influences by the offered assortment. Moreover, Alptekinoğlu and Grasas (2014) consider a setting with horizontally differentiated products that each have the same price. The earlier work of Wood (2001) was one of the first papers to incorporate returns behavior into a consumer choice model that employs a two stage buying process, where customers trial their product of interest before making a return decision. This paper studies the extent to which lenient return policies balance increased return costs with improved customer satisfaction. Su (2009) and Shulman et al. (2009) study how pricing and inventory decisions can be used to shape returns behavior. Samorani et al. (2019) study abusive returns behavior under a returns model akin to the Sequential Returns model of Wagner and Martínez-de Albéniz (2020). Finally, Singh et al. (2022) study the strategic unveiling of a new product in the face of defective product returns.

**Assortment optimization.** Our work is very closely related to the many recent developments regarding the assortment problem under various choice models that incorporate notions of a consideration sets. More specifically, the ordering phase in both returns models that we consider has close ties to existing choice models that incorporate a consideration set formation phase into the choice process, wherein customer systematically narrow down the products that interest them based on features like brand and price. Along this line, Feldman and Topaloglu (2018) and Feldman et al. (2019) incorporate implicit notion of consideration sets within classical choice models such as the MNL (Luce, 1959; McFadden, 1974) and ranking-based models (Honhon et al., 2012; Aouad et al., 2021). Namely, in these previous works, each customer type is distinguished by its own exogenously determined consideration set. On the other hand, Wang and Sahin (2018) and Derakhshan et al. (2022) explicitly model the consideration set formation process in the context of e-commerce-based product recommendation displays, thus endogenizing the consideration set formation phase. More specifically, these earlier works model the initial consideration set formation phase as a costly search process. In fact, as noted earlier, our Bulk Returns model ends up being identical (in terms of its induced choice probabilities) to the search model proposed in Wang and Sahin (2018), after replacing return disutilities with search costs, while also assuming that the return disutility is experienced for every item that is ordered, rather than just for the returned items. Moreover, this paper considers the assortment problem under an identical extension to multiple customer types, but only develops a pseudo-polynomial time approximation scheme in this setting.

**Road map of remaining sections.** The remainder of our paper is organized as follows. In Section 2, we formalize our two returns models. Next, in Section 3, we present both assortment problems and give the hardness results that we establish for each variant of the problem. Sections 4 and 5 detail the approximation schemes that we develop for the Bulk and Sequential

Returns models respectively. Finally, in Section 6, we present our comparison of the two models from theoretical and empirical vantage points.

## 2 Two Distinct Product Returns Models

In this section, we formalize the details of the Sequential and Bulk Returns models that are the central focus of our future analysis. After doing so, we conclude the section with a wrap-up discussion intended to address lingering question that might pop up after digesting the two models.

### 2.1 The Sequential Returns Model

The high-level dynamics of the Sequential Returns model can be understood to unfold over a series of sequential stages. In each stage other than first, the customer receives a product she has ordered, and then trials it for long enough so that she fully realizes her utility for keeping the product. From here, she either (i) keeps the product, (ii) returns the product and then abandons her search process, or (iii) continues her search by exchanging the product on-hand for a new product, i.e., she returns the product on-hand and orders a new product. As will be formalized shortly, the customer chooses between these three options by weighing the realized or expected utility of each. Built into this framework is the fundamental assumption that customers never re-order a product that they previously returned. To finish this high level description, we come back to the first stage, wherein the customer simply decides whether or not to initiate the search process.

**Preliminaries.** We consider a retailer who controls the availability of $n$ products indexed by the set $[n] = \{1, \dots, n\}$ [1], alongside an ever-present no-purchase, which is given index 0, and captures the option of the customer to purchase nothing. Following a random utility framework similar to that of the classical MNL model, we assume that the random utility that customers associate with product $i \in [n]$ is $U_i = v_i + \xi_i$, where $v_i$ represents the product-specific deterministic component of the utility and $\xi_1, \dots, \xi_n$ are independent logistic random variables with location-scale parameters $(0, 1)$. Additionally, it will be useful to define the "weight" of each product $i \in [n]$ as $w_i = e^{v_i}$. For the remainder of this paper, we assume that the products are indexed in decreasing order of their weight, and that each weight is unique, so that $w_1 > w_2 > \dots > w_n$ [2]. The utility of no-purchase option is fixed deterministically to 0. As will be formalized shortly, unlike the standard MNL framework, where the random utility of each offered product is assumed to be immediately realized by arriving customers, we assume that $U_i$ is only fully realized after having ordered and trialed product $i$. More specifically, we assume that the deterministic component $v_i$ is fully observable during the customer's initial search, during which she selects a single product to order. The random component $\xi_i$, on the other hand, is only realized after product $i$ has been ordered and the customer has had a chance

[1] Throughout, for integers $a, b \geq 0$, we use $[a] = \{1, 2, \dots, a\}$, $[a]_0 = \{0, 1, 2, \dots, a\}$ and $[a, b] = \{a, a+1, \dots, b\}$ all as shorthands.

[2] As will be clear once both models are fully described, the assumption on the uniqueness of the weights is merely to streamline the exposition by eliminating the need to develop tie-breaking rules for the ordering decision of consumers. All of the results continue to hold when weights can be shared across multiple products.

to familiarize herself with this product. Finally, we introduce a deterministic disutility of $f \geq 0$ that is incurred for each product that the customer returns. This disutility is meant to capture the inconvenience related to returns, e.g. repackaging, driving to the post office, etc ..., and it will play a role in both stages of the customers' decision process.

**The sequential random utility framework.** In what follows, we formalize the sequential choice behavior of each customer for a fixed assortment $S \subseteq [n]$. In essence, each customer's actions are guided by an optimal stopping problem aimed at maximizing expected utility. Formally, for arbitrary assortment $A \subseteq S$, let

$$\mathcal{U}(A) = \max \left\{ \underbrace{0}_{\text{stop search}}, \max_{i \in A} \mathbb{E}\left[ \max \left\{ \underbrace{U_i}_{\text{keep}}, \underbrace{-f + \mathcal{U}(A \setminus \{i\})}_{\text{return}} \right\} \right] \right\} \tag{1}$$

denote the maximal expected utility that a consumer can garner having yet to order any products within $A$, where the expectation is taken with respect to the random utility $U_i$. The outermost maximization trades of stopping the search process, and thus gaining no further utility, with the expected utility gained from ordering a product within $A$. For this purpose, the inner maximization over $i \in A$ determines the next product to order, if this is deemed to yield net positive utility in expectation. Conditioned that the customer orders product $i \in A$, she ultimately keeps this product (and thus ends the search process), with probability $\Pr[U_i \geq -f + \mathcal{U}(A \setminus \{i\})]$. Henceforth, we use the terms "keep" and "purchase" synonymously to capture the event that a particular product is not returned, and thus the retailer earns revenue from the sale of this product.

**Giving shape to the Sequential Returns model.** Wagner and Martínez-de Albéniz (2020) show that the search/ordering and purchasing behavior of each customer can be nailed down quite explicitly, which is somewhat surprising given the intricacies of the just-described random utility framework. The following theorem, which combines results from Lemma 1 and Theorem 1 of Wagner and Martínez-de Albéniz (2020), establishes that customers will always order products in a specific order, and also gives simple closed-form expressions for the choice probabilities, which should be interpreted as the probability that each offered product is purchased given that that the customer is yet to have ordered a single item.

THEOREM 2.1. *For fixed assortment $S \subseteq [n]$, we have that:*

*(i) Customers order products in decreasing order of weight. That is, for any subset of yet-to-be-ordered products $A \subseteq S$, the expectation in (1) is always maximized by the lowest indexed (highest weight) product within $A$.*

*(ii) Let*

$$\ell_Q = \max\{\ell \in [n] : w_\ell \geq 1 - e^{-f}\}. \tag{2}$$

*Then any product $i > \ell_Q$ will never be ordered, and hence will also never be purchased.*

*(iii) For any product $i \in S$ such that $i \leq \ell_Q$,*

$$\pi(i, S) = \frac{w_i e^{-f \cdot |S \cap [i-1]|}}{e^{-f \cdot |S \cap [\ell_Q]|} + \sum_{j \in S} w_j e^{-f \cdot |S \cap [j-1]|}}$$

*denotes the probability that product $i$ is purchased at the beginning of the customers search process.*

Part (i) of the above theorem is quite intuitive; given that the return disutility is the same for each product, it follows naturally that customers should probe (order) the products in order of their mean utilities. Part (ii) above reveals that, independent of the offered assortment, any product $i \in [n]$ that satisfies $w_i < 1 - e^{-f}$ will never be purchased. Finally, to better parse the choice probabilities given in part (iii), it helps to juxtapose them with the choice probabilities of the their well-known standard MNL model counterpart. Namely, if $f = 0$, meaning that customers experience no disutility for returning a product, then the two expressions for the choice probabilities exactly match. This occurs because, when $f = 0$, customers will surely order every product that is offered. On the other hand, when $f > 0$, product $i \in S$ can effectively be associated with an MNL-like weight of $w_i e^{-f \cdot |S \cap [i-1]|}$, which can be interpreted as the original weigth $w_i$ scaled down by the disutility of returning $|S \cap [i-1]|$ products before ultimately deciding to keep product $i$.

### 2.2 The Bulk Returns Model

Similar to the previous section, we begin with preliminary notation, noting that we are purposely redundant so that the similarities and differences of the two model are clear. From here, we formalize the choice dynamics of our two stage buying process. We note that in doing so, it is actually helpful to outline the two stages in reverse order. That is, we first describe the try-on and purchase stage for an arbitrary set of ordered products, before detailing the initial ordering decision for a fixed assortment of displayed products.

**Preliminaries.** We continue to consider an $n$-product retailing scenario with the same indexing scheme used for the Sequential Returns model. The random utility that customers associate with product $i \in [n]_0$ is $U_i = v_i + \xi_i$, where $v_i$ represents the product-specific deterministic component of the utility and $\xi_1, \ldots, \xi_n$ are independent Gumbel distributed random variables with location-scale parameters $(0, 1)$. After having read through the details of our model, it is straightforward to see that we may assume without loss of generality that $v_0 = 0$. We continue to define the "weight" of each product $i \in [n]$ as $w_i = e^{v_i}$, while also introducing the notation $w(S) = \sum_{i \in S} w_i$ for the total weight of an assortment $S \subseteq [n]$. Again, the deterministic component $v_i$ is fully observable throughout the customer's initial search, during which she selects a set of products to order. The random component $\xi_i$, on the other hand, is only realized after product $i$ has been ordered and the customer has had a chance to familiarize herself with this product. Finally, we continue to include a deterministic disutility of $f \geq 0$ that is incurred for each product that the customer returns.

**Stage 2: Try-on and keep.** A customer who orders products $\mathcal{O} \subseteq [n]$ will keep the ordered product $i \in \mathcal{O} \cup \{0\}$ whose total utility $\mathcal{U}_i(\mathcal{O}) = U_i - f \cdot (|\mathcal{O}| - 1 + \mathbb{1}_{i=0})$ is largest. The total utility $\mathcal{U}_i(\mathcal{O})$ captures the fact that if the customer keeps product $i$, she gains a utility of $U_i$, from which the disutility of returning $|\mathcal{O}| - 1 + \mathbb{1}_{i=0}$ products must be subtracted. Accordingly, the probability $\pi(i, \mathcal{O})$ that the customer keeps product $i \in \mathcal{O}$ can be computed as

$$\pi(i, \mathcal{O}) = \Pr\left[\mathcal{U}_i(\mathcal{O}) \geq \max_{j \in \mathcal{O} \cup \{0\}} \mathcal{U}_j(\mathcal{O})\right] = \frac{w_i e^{-f \cdot (|\mathcal{O}|-1)}}{e^{-f \cdot |\mathcal{O}|} + w(\mathcal{O}) \cdot e^{-f \cdot (|\mathcal{O}|-1)}} = \frac{w_i}{e^{-f} + w(\mathcal{O})},$$

where the rational expression after the second equality is precisely an MNL-like choice probability. Naturally, if $i \notin \mathcal{O}$, we have $\pi(i, \mathcal{O}) = 0$.

**Stage 1: Ordering.** Next, we take a step back and formalize the initial ordering decision. For assortment $S \subseteq [n]$, we use $\mathcal{O}(S)$ to denote customers' order decisions. Informally speaking, we assume that $\mathcal{O}(S)$ is comprised of the subset of offered products that maximize the expected utility of the product that the customer keeps. Formally, we have that

$$\mathcal{O}(S) = \operatorname*{argmax}_{\mathcal{O} \subseteq S} \mathbb{E}\left[\max_{i \in \mathcal{O} \cup \{0\}} \mathcal{U}_i(\mathcal{O})\right] = \operatorname*{argmax}_{\mathcal{O} \subseteq S} \log\left(w(\mathcal{O}) \cdot e^{-f \cdot (|\mathcal{O}|-1)} + e^{-f \cdot |\mathcal{O}|}\right) + Q,$$

where $Q$ is the Euler-Mascheroni constant. The second equality follows from the well-documented (Train, 2009; Sumida et al., 2021) expression for the expected utility that a customer obtains from the chosen alternative under the MNL model. We assume customers tie-break by choosing the largest cardinality order set. Exploiting this structure, we get the following lemma, which provides a simple and explicit way to compute $\mathcal{O}(S)$ for arbitrary offerset $S$, and also shows that it is necessarily composed of some subset of the highest weight products that are offered. This structural property of the order set implies that our tie-breaking rule is sufficient, i.e. because the weights of the products are all distinct, there cannot be multiple order sets of equal cardinality that achieve the same maximum expected utility.

Lemma 2.2. *Recalling that the products are indexed in decreasing order of weight, and fixing the offerset $S \subseteq [n]$, we have that $\mathcal{O}(S) = S \cap [\ell(S)]$ , where*

$$\ell(S) = \max\{\ell \in S \cup \{0\} : w_\ell \geq w(S \cap [\ell - 1]) \cdot (e^f - 1) + (1 - e^{-f})\}, \tag{3}$$

*with the caveat that $S \cap [-1] = \emptyset$.*

Observe that when $f = 0$, we have that $\ell_Q = \ell(S)$, which reflects the broader fact that the choice probabilities under both models reduce to those seen in a traditional MNL model when there is no disutility of return. When $f > 0$, for a fixed assortment $S$, it is straightforward to see that we have $\ell_Q \geq \ell(S)$, which implies that customers are somewhat pickier in the Bulk Return model. Specifically, since customers who practice bulk returns learn the utilities of all of the products they have ordered simultaneously, one would naturally suspect that these customers would be less likely to order lower weight products so as to hedge against the likely outcome that these items will be returned. Under the Sequential Returns model, customers only order a low-weight product with complete knowledge of how they value all higher-weight products. As such, it is not surprising that for a fixed assortment, the order set of sequential customers is nested within that of bulk customers.

**Remarks.** Having fully formalized both returns models, a few clarifying remarks are in order. First, we note that the use of distinct distributions for the error terms (logistic versus Gumbel) of the random utility specification of each model is necessary to develop closed-form expressions for the choice probabilities. That said, when the two distributions both have location-scale values of $(0, 1)$, they are near-identically bell-shaped, thus making the two random utility frameworks similar enough to warrant a meaningful comparison. Next, it is worth commenting on the assumption that, under the Bulk Returns model, the total return disutility scales linearly in the number of items returned. We justify our current set-up by noting that the return disutility parameter $f$ is meant to capture a multitude of inconveniences related to returns, beyond just re-packing and shipping back the product. Specifically, this disutility parameter incorporates the wasted time from trying on products that are ultimately returned, as well as the financial burden experienced from having to wait for refunds. That said, within the numerical experiments of Section 6.2, we introduce and study an alternative disutility framework that is perhaps closer to reality, but whose assortment problem we could only solve via IP-based tools.

Finally, we highlight the fact that, under the Sequential Returns model, the order decision is random, since customers make their keep/return decisions upon realizing the random utility of each product. On the other hand, the bulk returns order decision is based on an expected evaluation of realized utilities, and hence for a fixed assortment, the order decision is deterministic. The browsing model of Wang and Sahin (2018) similarly leads to a deterministic consideration set for a fixed set of displayed products. Following their lead, we similarly inject stochasticity into the bulk returns order decision by introducing multiple customer types, each distinguished by a distinct return disutility. We consider this multi-type extension in Section 4.3, which naturally extends the single-type version detailed above.

## 3 Assortment Optimization - Problem Formulations and Hardness Results

Having formalized the choice dynamics of the two returns models, we are now ready to formulate their corresponding assortment problems, and shed light on their computational complexity. We begin with shared notation across the two settings, before jumping into the respective formulations and various hardness results.

**Preliminaries.** For each product $i \in [n]$, let $r_i > 0$ denote its revenue and $c_i \geq 0$ denote its return cost, which represents the reverse logistical costs incurred by the retailer that are discussed at the onset of the paper. Additionally, for any assortment $S \subseteq [n]$, define $c(S) = \sum_{i \in S} c_i$ to be the cost of returning all items in $S$.

### 3.1 Bulk Returns

Recall that under the Bulk Returns model, customers peruse the set of offered products, and based on the intrinsic utilities, select a subset to order and try-out. Then, from amongst the collection of ordered products, the customer choose at most one to keep, returning the rest. As such, the expected profit from offering assortment $S$ can be expressed as

$$\begin{aligned}\mathcal{P}_B(S) &= \sum_{i\in\mathcal{O}(S)} (r_i - c(\mathcal{O}(S)\setminus\{i\}))\cdot\pi(i,\mathcal{O}(S)) - (1-\sum_{i\in\mathcal{O}(S)}\pi(i,\mathcal{O}(S)))\cdot c(\mathcal{O}(S))\\ &= \sum_{i\in\mathcal{O}(S)} (r_i + c_i)\cdot\pi(i,\mathcal{O}(S)) - c(\mathcal{O}(S)).\end{aligned}$$

The profit function above reflects the fact that if product $i \in \mathcal{O}(S)$ is purchased, then the customer will have to return all products within $\mathcal{O}(S)\setminus\{i\}$. The goal in the assortment problem is to find an assortment $S \subseteq [n]$ that maximizes expected profit, as is formally stated below

$$\max_{S\subseteq[n]} \mathcal{P}_B(S). \tag{AO-Bulk}$$

It is important to observe that AO-Bulk differs from the traditional assortment setting due to (i) the returns cost incurred by the retailer for each returned product and (ii) the link between the offered assortment $S$ and the order set $\mathcal{O}(S)$. Naturally, it is also these two novel features that bring about new technical hurdles. Namely, the introduction of the return costs makes it fairly straightforward to conceive of simple examples in which revenue-ordered assortments can perform arbitrarily poorly. Moreover, novel combinatorial difficulties are introduced by the fact that we must carefully ensure that each offered product is also ordered, since otherwise, it will not contribute to the profit.

**Hardness.** Somewhat surprisingly, we are able to show that AO-Bulk is NP-Hard even without return costs, which formally establishes that the initial ordering stage indeed complicates the assortment problem. This result is formally stated in the following theorem.

Theorem 3.1. *AO-Bulk is NP-Hard, even when $c_i = 0$ for each $i \in [n]$.*

As mentioned in Section 1.1, we will eventually show that the sequential model's assortment problem can be solved optimally in polynomial time when there are no return costs. Consequently, the above hardness result confirms that the assortment problem under the Bulk Returns model is fundamentally more difficult than its counterpart under the Sequential Returns model, which is somewhat counter-intuitive given the more nuanced order/returns behavior under the latter model.

### 3.2 Sequential Returns

Under the sequential returns model outlined in Section 2.1, customers sequentially order the products in descending order of weight $w_i$. After ordering each product, there is a brief trial period long enough so the customer fully realizes the utility she derives from the product. From here, she decides whether or not to keep the product or return it based on a comparison of the utilities derived from both options. In turn, this means that under assortment $S$, if a customer decides to keep product $i \in S$, she will have previously returned products $S \cap [i-1]$, and thus the profit garnered from this event is $r_i - c(S\cap[i-1])$. With this notion in mind, the expected profit from offering assortment $S$ can be expressed as

$$\begin{aligned}\mathcal{P}_Q(S) &= \sum_{i\in S} \pi(i, S\cap[\ell_Q])\cdot(r_i - c(S\cap[i-1])) - (1-\sum_{i\in S}\pi(i,S\cap[\ell_Q]))\cdot c(S\cap[\ell_Q]) \\ &= \sum_{i\in S}\pi(i,S\cap[\ell_Q])\cdot(r_i + c(S\cap[i,\ell_Q])) - c(S\cap[\ell_Q]).\end{aligned}$$

Since products $i > \ell_Q$ will never be purchased, we can ignore any such products in our assortment problem of interest, as is conveyed in the formulation below:

$$\max_{S\subseteq[\ell_Q]} \mathcal{P}_Q(S). \tag{AO-Seq}$$

Again, the returns cost and sequential return behavior are the features that complicate AO-Seq. In short, the fundamental novel combinatorial difficulty that arises is that the return cost for keeping product $i$ depends critically on the make-up of $S \cap [i-1]$, and thus both the choice probabilities and returns costs are nuanced functions of the offered assortment. For comparison, under the Bulk Returns model, the same is true, but in somewhat simpler form. Namely, the total return cost associated with keeping product $i$ depends only on the products offered alongside $i$, while under the Sequential Returns model, this cost is a function of the products offered alongside $i$ with lower index.

**Hardness.** Although we were unable to address the hardness of AO-Seq in its utmost generality, we are able to prove a strong separation between this assortment problem and that of the classical MNL model, for which optimal polynomial-time algorithms exist for the TU-constrained assortment problem. Generally speaking, the TU-constrained variant of the assortment problem is one in which the offered assortment is subject to operational constraints that can be encoded as a TU matrix. Examples of such constraint structures include cardinality constraints or precedence constraints, the former of which limits the total number of products that can be offered, while the latter set of constraints encodes the idea that a particular product cannot be offered unless it is paired with another product. We refer the interested reader to Sumida et al. (2021) for a more detailed discussion of practical TU constraint structures. The following hardness result formally establishes the impossibility of this constrained extension with regards to our assortment problem.

THEOREM 3.2. *Unless $P = NP$, the TU-constrained variant of AO-Seq cannot be approximated within any factor $\alpha > 0$, even when $f = 0$.*

We prove the above theorem via a reduction to the clique decision problem, which asked whether a clique of size $k$ exists within the input graph. At the core of our proof, is the realization that a clique of size $k$, by its very nature, induces a graph on $\binom{k}{2}$ edges. On the other hand, under the Sequential Returns model, the number of items returned prior to a purchase is precisely the number of offered products with smaller index (larger weight) than the item that was purchased. As such, if the offered assortment has size $k$, there are also $\binom{k}{2}$ possible pairings of purchased and returned product, which partially reveals the connection between the two problems.

## 4 Approximation Schemes for a AO-Bulk

In this section, we develop approximation schemes for assortment optimization problems under our Bulk Returns model, focusing primarily on the single customer type variant that our discussion thus far has centered around, before considering a multi-type extension at the section's tail end. Although the focus of this section is a strongly polynomial time approximation scheme (PTAS) for the single-type variant of AO-Bulk, it is the two algorithms developed for the multi-type extension that represent our strongest algorithmic contributions. We have relegated these results to the Appendix mainly only due to space limitations. Below, we present the precise profit guarantee and overall running time of our PTAS in the following theorem, whose proof unfolds over the subsequent two sections.

THEOREM 4.1. *For any $\epsilon > 0$, there is a $O(\frac{1}{\epsilon} n^{O(\frac{1}{\epsilon})})$-time algorithm that computes an assortment whose expected profit is within a $(1-\epsilon)$-fraction of optimal.*

**Technical overview.** To begin, we present a re-shaping of the assortment problem. Specifically, we show that AO-Bulk can be framed as a so-called "existential" knapsack problem. We term this knapsack problems so because its formulation requires knowledge of key features of the optimal assortment, which are not readily accessible. As such, we will never explicitly solve this knapsack problem, but instead use it to reason about the existence of a specific $\epsilon$-optimal assortment, which we refer to as the "target" assortment. Once the nature of this target assortment is formalized, our goal turns to developing an approach for its recovery.

### 4.1 The existential knapsack problem and its target assortment

In this section, we present a specially crafted knapsack problem, which is used solely for constructive purposes to establish Theorem 4.1. Re-hashing the discussion above, we note that formulating this knapsack problem requires upfront knowledge of two features of the optimal assortment, and hence its use is purely existential, meaning it is used to reason about what a near-optimal assortment might look like, rather than as means to directly identify this assortment. Specifically, we first establish an equivalency between this focal knapsack problem and AO-Bulk, which we exploit to reveal the existence of our target assortment.

**The knapsack problem.** Throughout the remainder of this section, we use $S^*$ to denote the optimal assortment to AO-Bulk, and we assume without loss of generality that $\mathcal{O}(S^*) = S^*$, meaning that we delete from $S^*$ any product that is not ordered. As alluded to above, our knapsack problem of interest wishfully assumes access to two features of the optimal assortment: $\ell(S^*)$, which corresponds to the lowest weight product offered within the optimal assortment, and the total weight $w(S^*)$. To ease notation moving forward, we will represent these two values as $\ell^*$ and $W^*$. With these two entities in-hand, we present a knapsack problem with $\ell^*$ items indexed by the set $[\ell^*]$. The reward for packing item $i \in [\ell^*]$ is $\rho_i = w_i \cdot (\frac{r_i + c_i}{e^{-f} + W^*} - \frac{c_i}{w_i})$, which results in the consumption of $w_i$ units of a $W^*$-sized knapsack. We can assume without loss of generality that $\rho_i \geq 0$ for each $i \in [n]$, since products violating this condition can be dropped while only improving the objective. Letting $x_i \in \{0, 1\}$ indicate whether item $i \in [\ell^*]$ is selected, our existential knapsack problem can be formulated as the following integer program.

$$\begin{aligned} Z^* = \max \quad & \sum_{i \in [\ell^*]} \rho_i x_i \\ \text{s.t.} \quad & \sum_{i \in [\ell^*]} w_i x_i \leq W^* \\ & x_i \in \{0,1\} \qquad\qquad \forall\, i \in [\ell^*]. \end{aligned} \tag{E-KNAP}$$

The following lemma establishes the equivalency between E-KNAP and AO-Bulk.

LEMMA 4.2. *Consider an arbitrary feasible solution $\hat{x} = (\hat{x}_1, \ldots, \hat{x}_{\ell^*})$ to E-KNAP and let $\hat{S} = \{i \in [\ell^*] : \hat{x}_i = 1\}$ denote its corresponding set of items selected. We have*

*(i)* $\mathcal{O}(\hat{S}) = \hat{S}$,

*(ii)* $\mathcal{P}_B(\hat{S}) \geq \sum_{i \in \hat{S}} \rho_i$,

*(iii) and* $Z^* = \mathcal{P}_B(S^*)$,

*noting that together (ii) and (iii) imply that the optimal solution to E-KNAP is precisely the items in $S^*$.*

**The target assortment.** In what follows, we show the existence of an $\epsilon$-optimal assortment $S_{\text{target}}$. Unfortunately, merely showing that this assortment exists is not enough to precisely identify it, since its contents critically depend on $S^*$. On the upside, we craft the target assortment so as to endow it with special structure that eventually allows for its approximate recovery. Specifically, $S_{\text{target}}$ will be the union of two disjoint sets of products. The first is defined as $S_\epsilon = S^* \cap N_\epsilon$, where $N_\epsilon = \{i \in [\ell^*] : \rho_i \geq \epsilon Z^*\}$ is the set of items with rewards of at least $\epsilon Z^*$. To create our second set of products, we focus exclusively on the set of leftover products $N_L = [\ell^*] \setminus N_\epsilon$ (the "L" subscript is for leftovers), and we introduce the notion of an item's density, defined as $\delta_i = \frac{\rho_i}{w_i} = \frac{r_i + c_i}{e^{-f} + W^*} - \frac{c_i}{w_i}$ for each product $i \in N_L$. Moreover, let $\sigma(k) \in N_L$ denote the item with the $k$-th largest density, and for $k \in [|N_L|]$, define $S_k = \{\sigma(1), \ldots, \sigma(k)\}$ to be the assortment that contains the $k$ items with largest densities. Ties can be broken arbitrarily. Adding that we define $S_0 = \emptyset$, we let $k^* = \max\{k \in [|N_L|]_0 : w(S_k) \leq W^* - w(S_\epsilon)\}$, which represents the heaviest density-ordered assortment that fits feasibly into a knapsack of size $W^* - w(S_\epsilon)$. We define $S_{\text{target}} = S_\epsilon \cup S_{k^*}$, and proceed to establish its validity as our target assortment in the following lemma.

LEMMA 4.3. *$\mathcal{P}_B(S_{\text{target}}) \geq (1 - \epsilon) \cdot \mathcal{P}_B(S^*)$.*

The sole purpose of the above lemma is to show that $S_{\text{target}}$ is good enough assortment to shoot for. In the next section, using this target assortment as our guide, we show how to uncover an assortment whose profit performance matches that of $S_{\text{target}}$.

### 4.2 Approximating the target assortment

In this section, we show how to efficiently identify an assortment $\hat{S}_{\text{target}}$, which garners an expected profit of at least $(1 - 2\epsilon) \cdot \mathcal{P}(S^*)$. Our approach for doing so is inspired by the make-up of $S_{\text{target}}$. Specifically, we set out to identify two disjoint assortments that are close replicas of $S_\epsilon$ and $S_{k^*}$. As described shortly, we will be able to exactly recover the assortment $S_\epsilon$, while subsequently doing our best to mirror the contents of $S_{k^*}$. In both cases, we employ the algorithmic tool of efficient enumeration, or "guessing". This tool is employed to recover (or approximately recover) key entities of interest related to $S^*$. To accomplish these goals via guessing, for each entity of interest, we propose a polynomial-sized collection of candidate guesses, and carry out all subsequent steps for each candidate.

**Recovering $S_\epsilon$.** In this first step, the key observation is that $|S_\epsilon| \leq \lceil \frac{1}{\epsilon} \rceil$, since each item $i \in S_\epsilon$ contributes at least $\epsilon Z^*$ to the optimal objective value of E-KNAP. As a result, we can exactly recover $S_\epsilon$ by enumerating over all assortments of cardinality at most $\lceil \frac{1}{\epsilon} \rceil$, of which there can be at most $n^{O(\frac{1}{\epsilon})}$.

**Approximating $S_{k^*}$.** In this second step, we create a distinct collection of density-ordered assortments, and choose as our approximation to $S_{k^*}$ the one that yields the highest expected profit when combined with $S_\epsilon$. Our procedure for doing so is as follows:

- Guessing $\ell^*$: We first guess $\ell^*$ by enumerating over its $n$ candidates, and subsequently drop all products indexed strictly larger than $\ell^*$.

- The approximate leftover set: To begin, we guess $\hat{Z} \in [Z^*, 2Z^*]$ by considering all candidates of the form $2^q \rho_{\max}$ for $q \in \{0, 1 \ldots, \log_2(n)\}$. Here, $\rho_{\max}$ is the highest-reward item included in the optimal solution to E-KNAP, which can be guessed by enumerating its $n$ options. From here, we define $\hat{N}_\epsilon = \{i \in [\ell^*] : \rho_i \geq \epsilon \hat{Z}\} \cup S_\epsilon$ and subsequently, $\hat{N}_L = [\ell^*] \setminus \hat{N}_\epsilon$. Note that $N_L \subseteq \hat{N}_L$.

- The approximate density-ordered assortments - preliminaries: Let $\delta_i(W) = \frac{r_i + c_i}{e^{-f} + W} - \frac{c_i}{w_i}$ denote each product's density as a function of $W$. The intent of this step is to identify a $\hat{W} \in \mathbb{R}$ such that the orderings of the densities $\delta_i(\hat{W})$ match those observed when the densities are instantiated at $W^*$. For this purpose, let $\mathcal{I} = \{W \in \mathbb{R} : \delta_i(W) = \delta_j(W) \text{ for } i, j \in [\ell^*]\}$ be the set of distinct intersection points for each pair of density functions. It is straightforward to observe that $|\mathcal{I}| = O(n^2)$, since each pair of density functions has at most one intersection. From here, we guess $W_\downarrow = \max\{W \in \mathcal{I} : W \leq W^*\}$ and $W_\uparrow = \min\{W \in \mathcal{I} : W \geq W^*\}$ by enumerating over the $O(n^2)$ possibilities for each, and subsequently define $\hat{W} = \frac{W_\uparrow + W_\downarrow}{2}$ [3]. Details and intuition regarding why such a $\hat{W}$ is needed are provided following the statement of Lemma 4.4.

- The approximate density-ordered assortments - construction: When the densities are instantiated at $\hat{W}$, let $\hat{\sigma}(k)$ denote the item whose density is $k$-th largest among $\hat{N}_L$, and for $k \in [|\hat{N}_L|]$, define $\hat{S}_k = \{\hat{\sigma}(1), \ldots, \hat{\sigma}(k)\}$ to be the approximate density-ordered assortments. Again, ties can be broken arbitrarily and we define $\hat{S}_0 = \emptyset$.

Finally, define $\kappa = \operatorname{argmax}_{k \in [|\hat{N}_L|]_0} \mathcal{P}_B(\hat{S}_k \cup S_\epsilon)$ and $\hat{S}_\kappa$ to be our approximate version of $S_{k^*}$.

[3] It suffices to choose any value for $\hat{W}$ in the interval $[W_\downarrow, W_\uparrow]$. We select the average of the two endpoints for convenience.

**The final outputted assortment.** The natural candidate for our approximation to the target assortment is $\hat{S}_{\text{target}} = S_\epsilon \cup \hat{S}_\kappa$. The following lemma shows that it is just as profitable as the original target.

LEMMA 4.4. $\mathcal{P}_B(\hat{S}_{\text{target}}) \geq (1-2\epsilon) \cdot \mathcal{P}_B(S^*)$.

Our proof of the above lemma essentially argues that $\hat{S}_{\text{target}}$ is an $\epsilon$-optimal solution to a relaxed version of E-KNAP, whose optimal solution fractionally packs the items in descending $\rho_i$-order, i.e., when the densities are instantiated at $W^*$. We overcome the fact that $W^*$ is unknown by having carefully selected $\hat{W}$ so that the ordering of the densities $\delta_i(\hat{W})$ match those observed when the densities are instantiated at $W^*$. In turn, this allows us to recovery the aforementioned greedy optimal packing, and argue for the efficacy of our proposed solution.

### 4.3 A multiple-type extension

In this section, we consider a multi-type extension of AO-Bulk. wherein for tractability, we assume the returns costs are zero for each product. We begin by describing how we augment the original single-type model, before presenting our approximation guarantee for the resulting assortment problem under the Bulk Returns model. After doing so, we describe a slight update to the bulk model that renders it identical to the browsing model of Wang and Sahin (2018), and subsequently present a new constant-factor approximation for this updated version of our assortment problem.

**The multi-type model.** We assume that there are $K$ customer types, each with a distinct disutility of return $f_k$ and an arrival probability of $\lambda_k$ such that $\sum_{k\in[K]} \lambda_k = 1$. We assume that the customer types are indexed in decreasing order of this disutility, so $f_1 \geq \ldots \geq f_k$. For tractability, it is necessary to assume that the random utility function $U_i$ is shared by each customer type, and hence the weights $\{w_i\}_{i\in[n]}$ are customer-type-independent. We note that if each type $k \in [K]$ were to associate a distinct weight of $w_{ik}$ with each product $i \in [n]$, then by setting $f_1 = \ldots = f_K = 0$, our model would reduce to the classical mixed-MNL model (McFadden and Train, 2000), whose corresponding assortment problem cannot be approximated within a factor of $O(n^{1-\epsilon})$, for any $\epsilon > 0$, unless $P = NP$ (Désir et al., 2022). Throughout the remainder of this section, we use $\ell_k(S), \mathcal{O}_k(S)$ and $\pi_k(i, S)$ to respectively denote the cut-off, order set and choice probabilities for each type $k \in [K]$ under assortment $S$. These values are all defined exactly as in the single-type setting, after replacing $f$ with $f_k$ for each customer type. Moreover, since the customers are indexed in increasing order of return disutility, it is straightforward to see that $\ell_1(S) \leq \ell_2(S) \leq \ldots \leq \ell_K(S)$, for any offered assortment $S$, i.e., the order sets are nested by type.

**The assortment problem and new technical hurdles.** For assortment $S$, let $\mathcal{R}_k(S) = \frac{1}{w_{0k}+w(\mathcal{O}_k(S))} \cdot \rho(\mathcal{O}_k(S))$ denote the expected revenue garnered from customers of type $k$, where $\rho(\mathcal{O}_k(S)) = \sum_{i\in\mathcal{O}_k(S)} r_i w_i$ and $w_{0k} = e^{-f_k}$. Accordingly, the total expected revenue garnered from offering assortment $S$ is $\mathcal{R}(S) = \sum_{k\in[K]} \lambda_k \mathcal{R}_k(S)$, and our goal in this costless version is to find the revenue-maximizing assortment. In moving to a setting with multiple customer types, we introduce new technical difficulties into the assortment problem, which arise from the fact that the retailer has to select a single assortment that balances and accounts for the individualized return behavior of each customer type. More specifically, in the single-type setting, we were able to craft an approach that ensures that each offered product will be ordered. By contrast, in the multi-type setting, each customer has a distinct order set which may, or may not, include all of the offered products. In turn, it becomes a far more complex task to understand how these order sets are shaped by the offered assortment.

**Main result - Bulk Returns model.** The following theorem, whose proof unfolds over Appendix B, gives the extent of our algorithmic findings for this multi-type setting.

THEOREM 4.5. *There is a poly*$(|\mathcal{I}|, n, K)$*-time algorithm, where* $|\mathcal{I}|$ *stands for the size of the input, that computes an assortment whose expected revenue is within a* $\Omega(\frac{1}{\log n})$*-fraction of optimal.*

Our approach for establishing the above theorem starts by identifying a partitioning of the customer types based on the total weight of their order sets. Exploiting this partitioning scheme, we show how to efficiently identify a specific subset of customer types whose order set weights are all within a factor of two of each other, and whose expected revenue contributions amount to at least a $\Omega(\frac{1}{\log n})$-factor of optimal. The former property regarding the proximity of the total weight of each order set is useful for linearizing the expressions for the choice probabilities, which in turn allows us to develop a carefully crafted approximate dynamic programs that allows us to identify an assortment that satisfies the revenue guarantee of the theorem statement.

**Main result - Browsing model.** Here, we settle an open question in Wang and Sahin (2018), which is to develop a non-trivial approximation scheme for their browsing model with multiple customer types. It turns out that the Bulk Returns model with no return costs reduces to this previously established model if the return disutility $f_k$ were to be applied to each ordered products, and not just those that are returned (in this case, $f_k$ should be thought of as a disutility of trying on each ordered item, regardless of whether they keep the item or not). In this case, the only change to the expressions for the choice probabilities of each customer type is that we now have $w_{0k} = w_0 = 1$, i.e., the no-purchase weight becomes homogeneous across types. Accordingly, for assortment $S$, the type-$k$ expected revenue becomes $\mathcal{R}_k(S) = \frac{1}{1+w(\mathcal{O}_k(S))} \cdot \rho(\mathcal{O}_k(S))$, and $\mathcal{R}(S) = \sum_{k\in[K]} \lambda_k \mathcal{R}_k(S)$ is the total expected revenue garnered from offering assortment $S$. Our goal remains to find the revenue maximizing assortment. Surprisingly, this small simplifying update to the no-purchase weight allows us to develop a more nuanced approximation scheme that garners a constant fraction of the optimal expected revenue, as stated in the following theorem, whose proof unfolds in Appendix C.

THEOREM 4.6. *There is a poly*$(|\mathcal{I}|, n, K, \frac{1}{\epsilon})$*-time algorithm, where* $|\mathcal{I}|$ *stands for the size of the input, that computes an assortment whose expected revenue revenue is within a* $(\frac{1}{7} - \epsilon)$*-fraction of optimal, for any* $\epsilon > 0$.

Our proof of Theorem 4.6 is relatively involved, and employs a handful of new ideas. In particular, we consider a multi-layered partitioning of the customer types, which again starts by grouping the customer types based on the total weight of their order sets. From here, for each group of types, we further partition the collection of products ordered by these types into three distinct sets of products. We then present ADP-based approaches to individually compete against the contributions of each of these three product sets. In the process, we introduce the notion of "pruning" an assortment, which is a simple procedure that deletes products from an assortment, while only improving the revenue of each customer type. The ability to prune any assortment is the critical algorithmic tool that allows us to accurately reason about the order sets of each type without an exact accounting of where the order set cut-off product is for each customer type.

## 5 A Linear Program for AO-Seq with Homogeneous Return Costs

In this section, we consider a variant of AO-Seq with homogeneous returns costs, meaning that $c_i = c$ for each product $i \in [n]$. Without loss of generality and for ease of notation, we assume that $\ell_Q = n$, as products indexed larger than $\ell_Q$ are never ordered, and hence can be dropped from consideration. When returns costs are homogeneous across the products, the updated profit function becomes

$$\mathcal{P}_Q(S) = \sum_{i \in S} \pi(i, S) \cdot (r_i + c \cdot |S \cap [i, n]) - c \cdot |S|.$$

Our goal remains to find an expected profit maximizing assortment $S$. Somewhat surprisingly, we show how to develop an LP with at most $O(n^2)$ decision variables and constraints to solve this variant of AO-Seq. We reiterate that this optimal polynomial-time algorithm is a strict and substantial improvement over the PTAS proposed by Wagner and Martínez-de Albéniz (2020) for this same assortment problem.

**Technical overview.** Our approach proceeds by first formulating a "'base" LP, which could, in theory, be used to directly solve our variant of AO-Seq. Unfortunately, establishing this fact rigorously turns out to be a difficult task, mainly due to rampant degeneracy issues caused by superfluous constraints that can only be identified after the optimal solution is revealed. With this in-mind, we overcome this issue by iteratively solving the base LP, deleting the just-mentioned superfluous constraints in each step until we a arrive at a non-degenerate optimal solution. From here, we show how this non-degenerate optimal solution can be used to directly reveal the optimal assortment.

### 5.1 The iterative-LP-based approach

We start by formulating the base LP, which is followed by intuition on how to interpret its decision variables and constraints. We further cement this intuition by subsequently showing that the optimal objective value of the base LP is an upper bound on the optimal profit $\mathcal{P}(S^*)$. Namely, we directly show how to construct a feasible solution to this LP from $S^*$, whose objective is precisely $\mathcal{P}(S^*)$, thus partially unveiling the connection between the base LP and AO-Seq. From here, our task turns to showing that $\mathcal{P}(S^*)$ is also an upper bound on the base LP's optimal objective, thus establishing that the two quantities are in fact equal. As hinted at above, showing this second bound using the base LP proves to be a tricky task because of degeneracy issues that arise due to superfluous constraints that read "0=0" at the optimal solution. We overcome this approach by iteratively solving the base LP until no such constraints remain, and hence it becomes easier to reason about the structure of the optimal solution.

**The Base LP.** We begin by guessing $Q^* = |S^*|$, of which there are at most $n$ candidate values. With $Q^*$ in-hand, we can charge $cQ^*$ upfront, and focus on the following assortment problem

$$\max_{S \subseteq [n]:|S|=Q^*} R_Q(S), \tag{AO-Seq-Q}$$

where

$$\begin{aligned} R_Q(S) &= \sum_{i \in S} \pi(i,S) \cdot (r_i + c \cdot |S \cap [i,n]|) \\ &= \frac{1}{1 + \sum_{i \in S} w_i e^{f \cdot (|S \cap [i,n]|)}} \cdot \sum_{i \in S} (r_i + c \cdot |S \cap [i,n]|) \cdot w_i e^{f \cdot (|S \cap [i,n]|)}. \end{aligned}$$

The second equality uses the expression for the choice probabilities under the Sequential Returns model given in Theorem 2.1, after normalizing the no-purchase weight to 1, i.e., we multiple numerator and denominator in the expression for the choice probabilities by $e^{fQ^*}$. It is straightforward to verify that AO-Seq-Q is equivalent to our original formulation. Letting $\mathcal{G} = \{(i,q) : i \in [n], q \in [i-1]_0, \}$, our base LP has decision variables $\alpha_{i,q}, \beta_{i,q}$ for each $(i,q) \in \mathcal{G}$. The $\alpha_{i,q}$-variables will eventually be shown to be associated with offering product $i$ within an assortment where $q$ products among $[i-1]$ have also been offered. The $\beta_{i,q}$-variables will be shown to be associated with not offering product $i$ in this same scenario. It is for this reason that each $i \in [n]$, is paired with $q$-values that do not exceed $i-1$, i.e., when considering product $i$, there can be at most $i-1$ offered products among $[i-1]$. Our starting LP of interest is given below:

$$\begin{aligned} Z^* = \quad & \max_{\alpha,\beta \geq 0} \sum_{(i,q) \in \mathcal{G}} \rho_{i,q} \alpha_{i,q} \\ \text{s.t.} \quad & (1) \quad \alpha_{1,0} + \beta_{1,0} = 1 - \sum_{(i,q) \in \mathcal{G}} \alpha_{i,q} w_i e^{f \cdot (Q^* - q)} \\ & (2) \quad \alpha_{i,0} + \beta_{i,0} = \beta_{i-1,0} && \forall\, i \in [2,n] \\ & (3) \quad \alpha_{i,q} + \beta_{i,q} = \alpha_{i-1,q-1} + \beta_{i-1,q} && \forall\, i \in [2,n], q \in [i-1] \\ & (4) \quad \alpha_{i,q} = 0 && \forall\, i \in [n], q \in [Q^*, n-1] \\ & (5) \quad \beta_{i,q} = 0 && \forall\, i \in [n], q \in [Q^* - n + i - 1]_0, \end{aligned} \tag{Base-LP}$$

where $\rho_{i,q} = (r_i + c \cdot (Q^* - q)) \cdot w_i e^{f \cdot (Q^* - q)}$. In Base-LP, Constraints (2) and (3) are akin to flow balance constraints; flow can "arrive" at product $i$ having offered $q$ products among products $[i-1]$ by either (i) not offering product $i-1$ and hence having offered $q$ products among $[i-2]$ (so from the variable $\beta_{i-1,q}$) or (ii) offering product $i-1$ after having offered $q-1$ products among $[i-1]$ (so from the variable $\alpha_{i-1,q-1}$). The constraints in (4) and (5) ensure that we ultimately offer exactly $Q^*$ products. The following claim is meant to further this intuition, and represent the first step in showing the equivalence of AO-Seq-Q and Base-LP.

CLAIM 5.1. *$Z^* \geq R_Q(S^*)$.*

In a nutshell, the remainder of this section is devoted to showing that $Z^* \leq R_Q(S^*)$, which together with Claim 5.1 would imply that $Z^* = R_Q(S^*)$. We do so by iteratively solving Base-LP as is described next.

**The non-degenerate LP.** Consider the following iterative approach. In the first iteration, we solve Base-LP and let $(\alpha^{(1)}, \beta^{(1)})$ denote its optimal solution. Moreover, we compute $\mathcal{G}_1 = \{(i,q) \in \mathcal{G} : \alpha^{(1)}_{i,q} + \beta^{(1)}_{i,q} = 0\}$, which constitute all pairs $(i,q)$ where constraint (2) or (3) read "0=0", and hence is superfluous. With this in mind, our second iteration considers an updated version of Base-LP, where for each $(i,q) \in \mathcal{G}_1$, we remove the associated constraint in (2) (or (3)) and add constraints enforcing $\alpha_{i,q} = \beta_{i,q} = \alpha_{i-1,q-1} = \beta_{i-1,q} = 0$ (or $\alpha_{i,0} = \beta_{i,0} = \beta_{i-1,0} = 0$). We then re-solve this new LP, whose optimal solution we denote as $(\alpha^{(2)}, \beta^{(2)})$, which then gives way to $\mathcal{G}_2 = \{(i,q) \in \mathcal{G} : \alpha^{(2)}_{i,q} + \beta^{(2)}_{i,q} = 0\}$. From here, we formulate a third LP, where for each $(i,q) \in \mathcal{G}_2$, we remove the associated constraint in (2) or (3) and add constraints enforcing the related decision variables are 0. We continue in this fashion, until we come to a particular iteration $t_{\text{final}}$, where $\mathcal{G}_{t_{\text{final}}} = \mathcal{G}_{t_{\text{final}}-1}$. We note that $t_{\text{final}} = O(n^2)$, since there are at most $O(n^2)$ constraints in (2) and (3) of Base-LP. Letting $\mathcal{G}_+ = \mathcal{G} \setminus \mathcal{G}_{t_{\text{final}}}$ denote the pairs $(i,q)$ such that $\alpha^{(t_{\text{final}})}_{i,q} + \beta^{(t_{\text{final}})}_{i,q} > 0$, the final LP that we solve can be expressed as

$$\begin{array}{lllr}
\max\limits_{\alpha,\beta\geq 0} & \multicolumn{2}{l}{\displaystyle\sum_{(i,q)\in\mathcal{G}_+} \rho_{i,q}\alpha_{i,q}} & \\
\text{s.t.} & (1) & \alpha_{1,0} + \beta_{1,0} = 1 - \displaystyle\sum_{(i,q)\in\mathcal{G}_+} \alpha_{i,q} w_i e^{f\cdot(Q^*-q)} & \\
& (2) & \alpha_{i,0} + \beta_{i,0} = \beta_{i-1,0} & \forall\, i \in [2,n] : (i,0) \in \mathcal{G}_+ \\
& (3) & \alpha_{i,q} + \beta_{i,q} = \alpha_{i-1,q-1} + \beta_{i-1,q} & \forall\, (i,q) \in \mathcal{G}_+ : q > 0 \\
& (4) & \alpha_{i,q} = 0 & \forall\, i \in [n], q \in [Q^*, n-1] \\
& (5) & \beta_{i,q} = 0 & \forall\, i \in [n], q \in [Q^* - n + i - 1]_0 \\
& (6) & \alpha_{i,q} = \beta_{i,q} = \alpha_{i-1,q-1} = \beta_{i-1,q} = 0 & \forall\, (i,q) \in \mathcal{G}_{t_{\text{final}}}.
\end{array}$$

(Non-Degen-LP)

To avoid cumbersome superscipts, we relabel the optimal solution to Non-Degen-LP as $(\alpha^*, \beta^*)$. It is straightforward to see that the optimal solution in each stage of our approach is also feasible to the previous stage's LP, since we only removed superfluous constraints moving from one stage to the next. As such, the optimal objective value is fixed at $Z^*$ during each iteration of our approach, however the optimal solution itself is likely to change. In the following lemma, we implicitly argue that by eliminating degeneracy issues caused by "0=0" constraints, we can relate the optimal solution of LP Non-Degen-LP to a specific assortment $\hat{S}$, which we show to be an optimal solution to AO-Seq-Q.

LEMMA 5.2. *Let $(\alpha^*, \beta^*)$ denote the optimal solution to Non-Degen-LP, and define $\hat{S} = \{i \in [n] : \alpha^*_{i,q} > 0\}$. We have that (i) $|\hat{S}| = Q^*$ and (ii) $R_Q(\hat{S}) = Z^*$.*

We prove the above lemma by giving explicit structure to $(\alpha^*, \beta^*)$, which immediately elucidates its connection to the assortment $\hat{S}$. Combining Claim 5.1 with Lemma 5.2, we get that $\mathcal{P}_Q(\hat{S}) = Z^* - cQ^* = \mathcal{P}_Q(S^*)$, and hence the assortment $\hat{S}$ is optimal, and directly revealed by Non-Degen-LP.

## 6 An Intricate Comparison of the Two Returns Models

The Bulk Returns model was purposefully adapted from the Sequential Returns model, endowing with an identical set of model parameters, to allow for a rigorous comparison between the two models. The primary goal of this section is to provide a thorough comparison of the two models from both theoretical and empirical perspectives.

### 6.1 A theoretical investigation

In what follows, we present a theoretical comparison of the two models along the lines of expected profit, market share, and the relative magnitude of their choice probabilities.

**Profit comparison.** Here, we set out to present a general comparison of the optimal expected profits under the two models when the returns costs are homogeneous across the products. For this purpose, let $S_B^*$ and $S_Q^*$ respectively represent the optimal assortments to AO-Bulk and AO-Seq when these two assortment problems are instantiated with the same set of weights $(w_i)_{i\in[n]}$, return disutility $f$, revenues $(r_i)_{i\in[n]}$ and refurbishing cost $c$ for each product. The following theorem establishes that the sequential optimal profit is strictly larger than its bulk counterpart.

THEOREM 6.1. *$\mathcal{P}_Q(S_Q^*) \geq \mathcal{P}_B(S_B^*)$.*

The above theorem is seemingly both surprising and inevitable, and carries with it the managerial implications discussed in Section 1.1. The proof of this result uses well-known results related to the traditional MNL assortment problem in new ways to show that $\mathcal{P}_Q(S_B^*) \geq \mathcal{P}_B(S_B^*)$. In other words, we show that the expected profit garnered by $S_B^*$ is larger under the sequential model than it is under the bulk model. To establish this fact, we begin with a preliminary result that re-hashes the just-mentioned important structural results related to the classical MNL assortment problem, before moving to the the proof of the theorem.

**Market share and choice probabilities comparison.** The analysis that follows is for a fixed assortment $S$, which satisfies $\mathcal{O}(S) = S$, noting again that products in the bulk model that are not ordered can be dropped without any changes to the underlying choice probabilities. For the purpose of our comparison, let $\pi_B(i,S)$ and $\pi_Q(i,S)$ denote the choice probabilities under the Bulk and Sequential Returns models respectively. Moreover, let $\mathrm{MS}_B(S) = \sum_{i\in S} \pi_B(i,S)$ and $\mathrm{MS}_Q(S) = \sum_{i\in S} \pi_Q(i,S)$ denote their respective market shares for fixed assortment $S$. The following lemma gives our findings.

LEMMA 6.2. *For arbitrary assortment $S \subseteq [n]$ that satisfies $\mathcal{O}(S) = S$, we have that*

*(i)* $\mathrm{MS}_Q(S) \geq \mathrm{MS}_B(S)$, *and*

*(ii) there exists $k^* \in S$ such that for $i \in S \cap [k^*]$, we have that $\pi_Q(i, S) \geq \pi_B(i, S)$, and for $i \in S \cap [k^* + 1, n]$, we have that $\pi_Q(i, S) \leq \pi_B(i, S)$.*

As discusses in Section 1.1, these two results strongly suggest that customers will generally be better off within a sequential returns framework. This insight, however, should be taken with a grain of salt, since a fundamental assumption needed to establish these theoretical properties is that the two models share a common return disutility $f$. We explore this implications of this likely asymmetry next, via an extensive set of numerical experiments.

### 6.2 An empirical investigation

In this section, we present a thorough empirical comparison of the profit and market share predicted by our two returns models. The intent of this analysis is to paint Theorem 6.1 and Lemma 6.2 with more of a practical brush. That is, we aim to go beyond an outright comparison of the two models by providing more precise profit and market share comparisons using a diverse array of test assortment instances generated from real sales data. Additionally, we seek to investigate whether the high-level insights from our theoretical comparisons persist when we augment the return cost/disutility framework of our models to add more of an air of realism. Specifically, we consider a more fine-grained model of return costs/disutilities, which incorporates a fixed cost/disutility for each return trip (i.e., when a single item or a collection of items is returned), along with the per-returned-item cost/disutility used in the current model. We give the details of this updated framework next, before describing the details of our dataset and how it is used to create realistic instances of the bulk and sequential returns assortment problems.

**The updated returns model.** In the current set-up of both models, customers experience return disutilities that are directly proportional to the number of items returned. Similarly, the retailer incurs return costs that scale in the same way. These two assumptions potentially give the sequential model a leg up when it comes to profit and market share comparisons, since the number of return trips does not factor into each customer's overall disutility, and the retailer's overall return costs. More specifically, under the bulk model, there is only ever at most a single return trip, whereas under the sequential model, there can be up to $n$ such trips. From the perspective of the customer, each return trip entails dropping the re-packaged product at a post office or return center, whereas for a retailer offering free returns, each return trip results in extra costs that could come from paying for shipping labels, for example. We therefore augment our framework by assuming that the customer experiences a return disutility of $f_s$ for each return trip, in addition to a "try-on" disutility of $f_t$ for each item that is ordered, but not kept. Similarly, we now assume that the retailer incurs a shipping cost of $c_s$ for each return trip, and a refurbishing/re-stocking cost of $c_r$ for each returned item. As described in Appendix E, the dynamics of the sequential model remain unchanged, but the bulk model must be updated slightly. In this Appendix, we derive the new expressions for the choice probabilities under the Bulk Returns model, and also give a new IP-based approach to solve its corresponding assortment problem, as our previous approach is rendered moot with this update.

**Data cleaning and parameter estimation.** We utilize the ISMS Durable Goods Data Set 1 (Ni et al., 2012) to conduct our computational experiments. This data set comprises a total of 173,262 transactions from 19,936 households over a six-year period (from December 1998 to November 2004) at a major U.S. consumer electronics retailer. Each transaction record includes detailed information such as household ID, transaction location and type, product ID, product category, and subcategory, along with purchase quantity and unit price. Our experiments focus on the sales of televisions and speakers, which are the most data rich categories. We proceed through the following three steps to derive reasonable estimates for the parameters of our model, which seed the assortment instances we create. The full details of each of three steps can be found in Appendix E, whose discussion is focused on the Television category.

- Step 1 - Data Cleaning and organization: In this first step, we uncover the most data rich categories of products, and eliminate oddly priced entries, which are likely due to large discounts or data entry errors. Each transaction is assigned to a store, whose corresponding available assortment is computed to be the set of products purchased previously at this store. Consequently, after this first step, for each transaction, we have the purchased product and the available assortment, which sets the stage for the estimation phase.

- Step 2 - Estimating the MNL-based weights: Due to a lack of detailed returns data, we cannot explicitly estimate the parameters of our models. Instead, we fit classical MNL models to the data to derive the weights in our model using an MLE-based approach that is common in the literature Train (2009).

- Step 3 - Setting return costs/disutilities: In this final step, we describe how we use the fitted weights to derive the return disutilities, while concurrently describing how the return costs are set based on what is observed in practice. For each store, we derive an overall disutility $f$ and overall cost $c$, which are respectively partitioned into $f_s$,$f_t$ and $c_s$,$c_r$ in varying proportions, as described next.

**Instance generation.** Instances are generated to represent scenarios where each product is priced within a specific store, i.e. for each store, we create an array of instances differentiated by the number of no-purchase events and the return cost/disutility parameters. The key components for these instances include:

- *Price levels:* Prices for each product are set based on the latest transaction data to reflect the most recent market conditions.

- *MNL-based weights:* Using the fitted MNL from Step 2, we assign preference weights to each product.

- *Return disutility/costs:* As detailed in Step 3, we estimate the overall return disutility $f$ and return cost $c$ for each store. In an effort to capture a wide range of scenarios, the shipping disutility values $f_s$ are selected at 25% increments, starting at 0% and going up to 100% of $f$. The try-on disutility $f_t$ is then calculated as $f_t = f - f_s$. The refurbishing cost $c_r$ is set to $c$, and the shipping cost $c_s$ is selected in 25% increments, from 0% to 100% of $c$.

Figure 1 displays the distribution of the number of products that remain after our 3-step estimation procedure and the elimination of stores with negative $f$ values. We then generate 47,160 assortment instances for the Television category by varying four no-purchase increments, three return disutility and return cost scaling factors, and the five shipping disutility and shipping cost fractions across 188 total stores (the full details of this set-up are given in Appendix E). For the Speaker category, we consider 3,330 assortment instances across 14 stores. For each instance, we are able to derive the optimal assortment under both models; for the sequential model, we use the LP developed in Section 5, while for the bulk model, we use the IP-based approach developed in Appendix E.

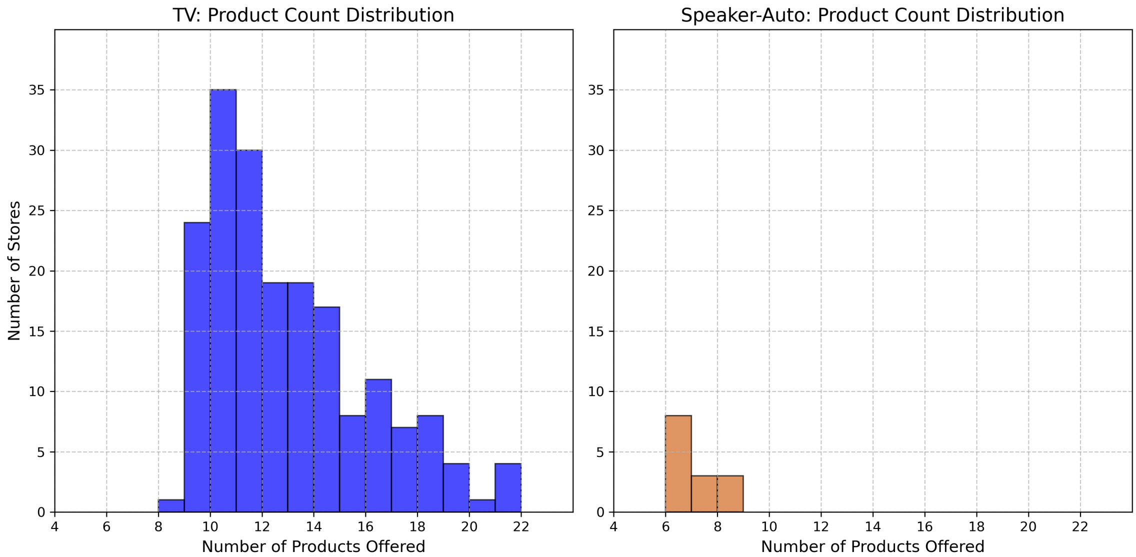
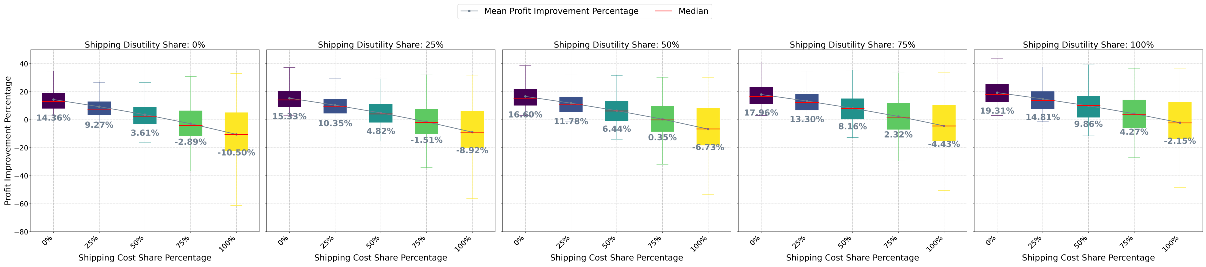


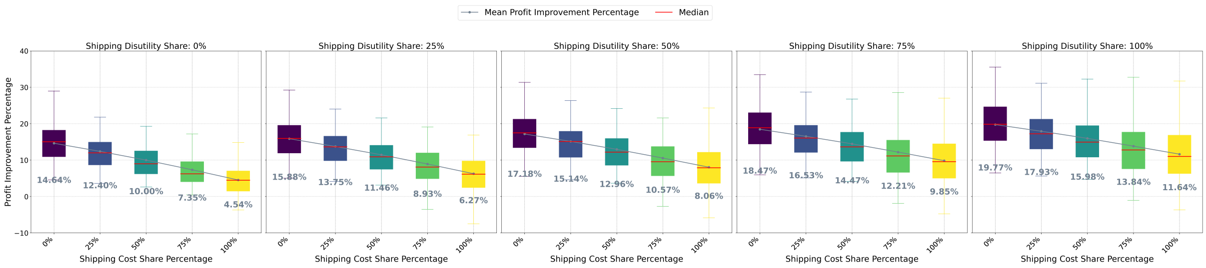

Figure 1: Distribution of number of products available at each store

**Results - profit comparison.** Recall that Theorem 6.1 established that when both the Bulk and Sequential Returns models use the same return disutility and return cost structure as in the original framework, the expected profit generated by the optimal assortment is larger under the sequential model than it is under the bulk model. Our objective here is to investigate to what extent the sequential model maintains its profit advantage under our augmented return disutility/cost framework. Figures 2a and 2b give the distributions of percentage profit improvement of the sequential model across each combination of disutility and shipping cost share. Specifically, for each problem, we compute the optimal sequential returns profit $\mathcal{P}_Q^*$ and the optimal bulk returns profit $\mathcal{P}_B^*$ and display the distribution of the percentage improvement, computed as $100 \cdot \frac{\mathcal{P}_Q^* - \mathcal{P}_B^*}{\mathcal{P}_Q^*}$ as a boxplot for each combination of disutility and shipping cost share.

With only a superficial glance at these plots, one notices that the bulk model only outperforms the sequential model on average when the shipping cost $c_s$ is set to its largest value, and this only occurs for the Television category. Moreover, the expected profit gains afforded by the sequential model are fairly substantial, generally ranging from 5-15%. Together, these two findings, along with Theorem 6.1, provide strong evidence that the sequential purchasing habits are better for bottom lines. Interestingly, we actually witness these gains magnify as the shipping disutility increases, which is somewhat surprising given that bulk customers ex-




(a) Results for the Television category



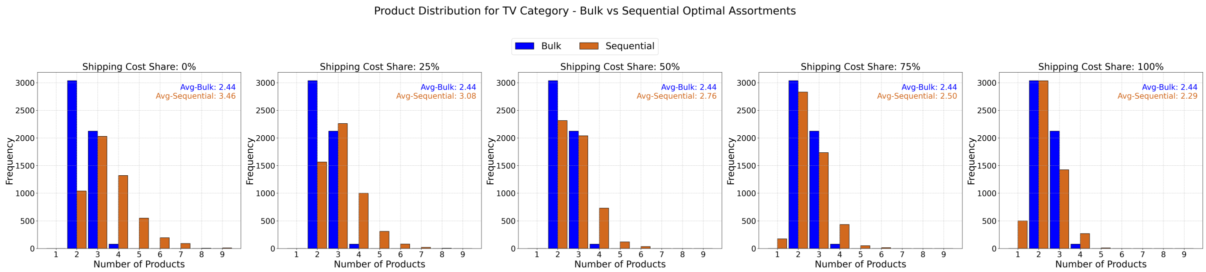

(b) Results for the Speaker category

Figure 2: Distributions of the percentage improvement in profit garnered by the sequential model for the Television and Speaker categories, shown as a function of the relative return costs and disutilities

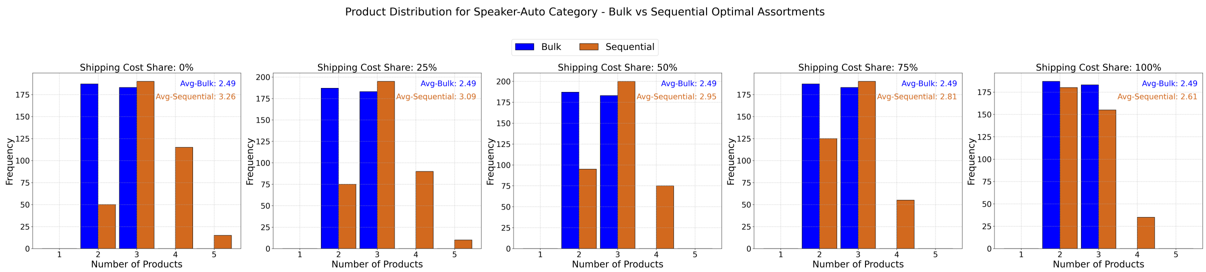
perience $f_s$ at most once, while sequential customers experience this disutility for each return event. This finding can likely be explained by the fact that, from a profit perspective, it can actually be beneficial to have a large return disutility. Specifically, when the return disutility is large, customers are unlikely to order a large number of products, since they know this will lead to numerous return events. In turn, the retailer can more easily ensure customers keep a high revenue products due to the high inconvenience of returns. This effect is likely amplified under the sequential model, wherein customers experience more total return disutility. This phenomenon is partially illustrated in Figures 3a and 3b, where we show that the distributions for the number of products included in the optimal assortments [4]. For both categories, we see that as the shipping disutility fraction increases, the optimal sequential assortment shrinks to match that of the bulk. On average, these smaller assortments include high weight, high revenue products, one of which customers are likely to keep.

**Results - market share comparison.** Finally, we investigate whether the just-discussed profit improvements conferred by sequential buying behavior come at the cost of a worse shopping experience for consumers, as measured by their likelihood to select the no-purchase option. Figures 4a and 4b show the distributions of the percentage improvement in market share garnered by the optimal sequential assortments. Here, we observe that the insights from Lemma 6.2 persist; the sequential model outperforms the bulk model with regards to market share by 5-15% again, and is only bested in the Television category when the cost share percentage is largest.

[4]The results for the bulk model do not change as we move from plot to plot, since the optimal assortment under this model is not affected by increasing $c_s$, since this cost is paid at most once.

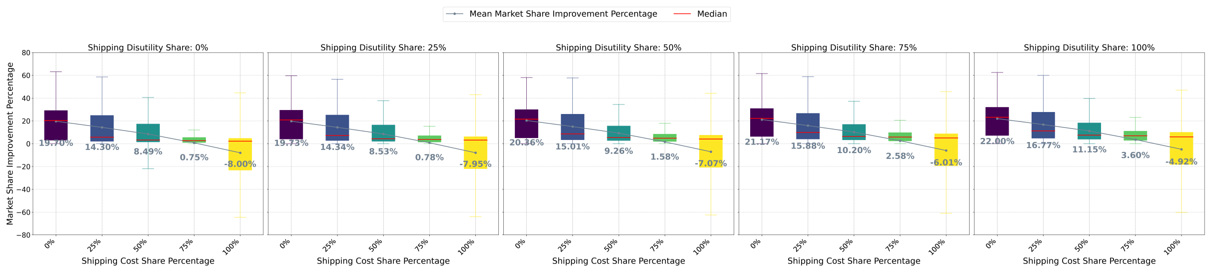



(a) Results for the Television category

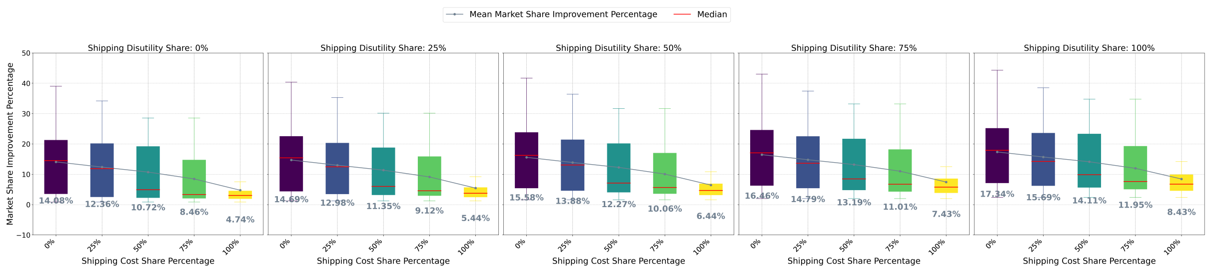



(b) Results for the Speaker category

Figure 3: Distributions of the number of products included in the optimal assortment for the Television and Speaker categories, shown as a function of the relative return costs and disutilities




(a) Results for the Television category




(b) Results for the Speaker category

Figure 4: Distributions of the percentage improvement in market share garnered by the sequential model for the Television and Speaker categories, shown as a function of the relative return costs and disutilities

## 7 Conclusion and Future Work

Our work augments the traditional assortment optimization problem by injecting notions of product returns into the problem landscape, which requires us to develop more nuanced models of customer choice that come in the form of the Sequential and Bulk Returns models. Under these models, we develop novel approximate schemes to identify provably near-optimal assortments, while concurrently juxtaposing their conferred profits and market shares to better understand what sort of return behavior should be promoted or discouraged. While we make significant progress from both algorithmic and managerial vantage points, there are numerous directions in which our work could be extended. One such direction is to pursue the development of polynomial time approximation schemes for the augmented returns framework introduced in Section 6.2. Another interesting and related extension could consider how to make assortment/pricing decisions in the presence of a so-called free shipping threshold, which dictates the minimum expenditure needed to zero-out shipping costs. In this case, a new model that potentially builds off our returns models is needed to capture how customers value the utility gains associated with reaching the free shipping threshold. How to optimally select this free shipping threshold is also a noteworthy question in this setting, which could lead to useful managerial insights.

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## A Missing Proofs

### A.1 Proof of Lemma 2.2

To start, we note that $\ell(S)$ is well-defined, since $w_0 = 1 \geq 1 - e^{-f}$. To prove the lemma, we first show that $\mathcal{O}(S) = S \cap [\ell]$ for some $\ell \in S \cup \{0\}$, meaning that the order set must be some weight-orders subset of the offered products, or the empty set. Then, we argue that $\ell(S)$ is the optimal cut-off within the weight order.

To show that that $\mathcal{O}(S) = S \cap [\ell]$ for some $\ell \in S \cup \{0\}$, we assume by way of contradiction that there exists products $i, j \in S$ such that $i < j$ (so $w_i > w_j$), and $j \in \mathcal{O}(S)$, while $i \notin \mathcal{O}(S)$. We will show that the order set $\mathcal{O}' = (\mathcal{O}(S) \setminus \{j\}) \cup \{i\}$ garners a strictly larger maximum expected utility than $\mathcal{O}(S)$. To see this, observe that

$$\begin{aligned}
\mathbb{E}\left[\max_{k\in\mathcal{O}'} \mathcal{U}_k\left(\mathcal{O}'\right)\right] &= \log\left(w\left(\mathcal{O}'\right) \cdot e^{-f\cdot(|\mathcal{O}(S)|-1)} + e^{-f\cdot|\mathcal{O}(S)|}\right) + Q \\
&> \log\left(w\left(\mathcal{O}(S)\right) \cdot e^{-f\cdot(|\mathcal{O}(S)|-1)} + e^{-f\cdot|\mathcal{O}(S)|}\right) + Q \\
&= \mathbb{E}\left[\max_{k\in\mathcal{O}(S)} \mathcal{U}_k\left(\mathcal{O}(S)\right)\right],
\end{aligned}$$

where the first equality exploits the fact that $|\mathcal{O}'| = |\mathcal{O}(S)|$, since we have just swapped products $i$ and $j$, and the lone inequality follows because $w(\mathcal{O}') > w(\mathcal{O}(S))$.

Next, we show that $\ell(S)$ is the optimal cut-off. To do so, we assume by way of contradiction that $\mathcal{O}(S) = S \cap [k]$ for some $k \in S \cup \{0\}$, such that $k \neq \ell(S)$, and then derive a contradiction by considering two cases based on the relative values of $k$ and $\ell(S)$.

- Case 1 - $k < \ell(S)$: Let $\ell_\uparrow = \min\{\ell \in S : k < \ell \leq \ell(S)\}$, and note that $\ell_\uparrow$ is well-defined since $\ell(S) \in S$. We will show that

$$\mathbb{E}\left[\max_{i\in S\cap[\ell_\uparrow]} \mathcal{U}_i\left(S \cap [\ell_\uparrow]\right)\right] \geq \mathbb{E}\left[\max_{i\in S\cap[k]} \mathcal{U}_i\left(S \cap [k]\right)\right], \tag{4}$$

  which leads to a contradiction, since either the order set $S \cap [\ell_\uparrow]$ provides a strictly larger expected maximal utility, or it provides the same expected maximal utility as $S \cap [k]$, but has larger cardinality, and hence wins the tie-break. To show (4), it suffices to show that

$$\underbrace{w\left(S \cap [\ell_\uparrow]\right) \cdot e^{-f\cdot\left(|S\cap[\ell_\uparrow]|-1\right)} + e^{-f\cdot|S\cap[\ell_\uparrow]|}}_{\text{term (i-a)}} \geq \underbrace{w\left(S \cap [k]\right) \cdot e^{-f\cdot(|S\cap[k]|-1)} + e^{-f\cdot|S\cap[k]|}}_{\text{term (ii-a)}}.$$

  Using the fact that $S \cap [\ell_\uparrow] = (S \cap [k]) \cup \{\ell_\uparrow\}$, and so $|S \cap [\ell_\uparrow]| = |S \cap [k]| + 1$ we get that

$$\text{term (i-a)} - \text{term (ii-a)} = w_{\ell_\uparrow} - w(S \cap [k]) \cdot (e^f - 1) + (1 - e^{-f}) \geq 0,$$

  thus establishing the contradiction. The inequality above follows from the fact that $\ell_\uparrow \leq \ell(S)$, along with the definition of $\ell(S)$.

- Case 2 - $k > \ell(S)$: Let $\ell_\downarrow = \max\{\ell \in S \cup \{0\} : \ell(S) \leq \ell < k\}$, and note that $\ell_\downarrow$ is well-defined since $\ell(S) \in S \cup \{0\}$. We will show that

$$\mathbb{E}\left[\max_{i\in S\cap[\ell_\downarrow]} \mathcal{U}_i\left(S \cap [\ell_\downarrow]\right)\right] > \mathbb{E}\left[\max_{i\in S\cap[k]} \mathcal{U}_i\left(S \cap [k]\right)\right], \tag{5}$$

  which leads to a contradiction, since the order set $S\cap[\ell_\downarrow]$ provides a strictly larger expected maximal utility. To show (5), it suffices to show that

$$\underbrace{w\left(S\cap[\ell_{\downarrow}]\right)\cdot e^{-f\cdot\left(|S\cap[\ell_{\downarrow}]|-1\right)}+e^{-f\cdot|S\cap[\ell_{\downarrow}]|}}_{\text{term (i-b)}}>\underbrace{w\left(S\cap[k]\right)\cdot e^{-f\cdot(|S\cap[k]|-1)}+e^{-f\cdot|S\cap[k]|}}_{\text{term (ii-b)}}.$$

Using the fact that $S\cap[k]=(S\cap[\ell_{\downarrow}])\cup\{k\}$, and so $|S\cap[\ell_{\downarrow}]|=|S\cap[k]|-1$ we get that

$$\text{term (i-b)}-\text{term (ii-b)}=-w_k+w(S\cap[\ell_{\downarrow}])\cdot(e^f-1)-(1-e^{-f})>0,$$

thus establishing the contradiction. The inequality above follows since if the reverse were true, it would contradict the definition of $\ell(S)$, since we have assumed $k>\ell(S)$.

### A.2 Proof of Theorem 3.1

Our proof relies on a reduction from the 2-partition problem, which is well-known to be NP-Hard (Karp, 1972), and whose input is a set of $n$ positive integer $a_1,\ldots,a_n$. Letting $\sum_{i\in[n]}a_i=2L$, for some integer $L\geq 1$, the goal in the partition problem is to identify a subset $I\subset[n]$ such that $\sum_{i\in I}a_i=L$. The partition problem is scale-invariant, and hence we can assume without loss of generality that $a_i\geq 2$ for each $i\in[n]$. Moreover, to match the indexing of our assortment problem, we will assume that the integers are indexed in decreasing order so that $a_1\geq a_2\geq\ldots\geq a_n$.

Given such an instance of the partition problem, we construct an instance of AO-Bulk as follows:

- We have $n+1$ products in addition to the no-purchase option, which has weight $w_0=1$. The revenue of product $i\in[n]$ is $r_i=1$, its weights is $w_i=a_i$ and its return cost is $c_i=0$. Product $n+1$ has a revenue of $r_{n+1}=1.5$, a weight of $w_{n+1}=1$ and a return cost of $c_{n+1=0}$. Note that all return costs are 0, and that product $n+1$ has the smallest weight, since we have assumed that $a_i\geq 2$ for each $i\in[n]$.
- The disutility for each returned product is $f(L)=\ln(\frac{\sqrt{L}+\sqrt{4+L}}{2\sqrt{L}})$. We make note of the fact that $e^{-f(L)}>0.5$ for $L\geq 1$, which we will exploit shortly.

**The reduction.** In Lemma A.1 and Lemma A.2 given below, we show that YES instances of the partition problem lead to an optimal assortment that garners an expected revenue of at least $\mathcal{P}^*=\frac{1.5+L}{e^{-f(L)}+L+1}$, while NO instances of the partition problem lead to an optimal assortment with revenue of no more than $\frac{1.5+(L-1)}{e^{-f(L)}+(L-1)+1}$, which is strictly smaller than $\mathcal{P}^*$, since the function $f(x)=\frac{1.5+x}{e^{-f(L)}+x+1}$ is increasing in $x$, as long as $e^{-f(L)}>0.5$, which we have just verified above in defining $f(L)$.

Lemma A.1 (Yes Instance). *Suppose there exists $I\subset[n]$ such that $\sum_{i\in I}a_i=L$, then there exists an assortment $S$ that garners an expected revenue of $\mathcal{P}(S)=\mathcal{P}^*$.*

Proof. Consider offering the assortment $S=I\cup\{n+1\}$. If we can establish that $\mathcal{O}(S)=S$, i.e. that all of the products in $S$ are ordered, then we get that

$$\mathcal{P}(S) = \frac{w(I) + r_{n+1}w_{n+1}}{e^{-f(L)} + w(I) + w_{n+1}} = \frac{L + 1.5}{e^{-f(L)} + L + 1} = \mathcal{P}^*,$$

where the second equality follows since $I$ is a valid partition. To show that $\mathcal{O}(S) = S$, note that

$$w(S \cap [n]) \cdot (e^f - 1) + (1 - e^{-f(L)}) = L \cdot (e^{f(L)} - 1) + (1 - e^{-f(L)}) = 1 = w_{n+1},$$

where the second equality follows by plugging in $f = \ln(\frac{\sqrt{L}+\sqrt{4+L}}{2\sqrt{L}})$. Via Lemma 2.2, we get that $\ell(S) = n + 1$, and so $\mathcal{O}(S) = S \cap [n+1] = S$. ■

Lemma A.2 (No Instance). *Suppose that for every $I \subset [n]$, we have $\sum_{i \in I} a_i \neq L$, then the expected revenue of the optimal assortment $S^*$ satisfies $\mathcal{P}(S^*) \leq \frac{1.5+(L-1)}{e^{-f(L)}+(L-1)+1}$.*

Proof. We consider two cases based on whether or not $\mathcal{O}(S^*)$ contains product $n + 1$.

- Case 1 - $n + 1 \in \mathcal{O}(S^*)$: We observe again that product $n + 1$ is ordered if and only if $\ell(S^*) = n + 1$, which can only occur if

$$w_{n+1} \geq w(S^* \cap [n]) \cdot (e^{f(L)}) - 1) + (1 - e^{-f(L)}).$$

  With some basic algebraic manipulations, the above inequality reduces to $w(S^* \cap [n]) \leq L$, however since we are considering instances of the partition problem in which no valid partition exists, it must the case that $w(S^* \cap [n]) \leq L - 1$. Consequently, we get that

$$\mathcal{P}(S^*) = \frac{w(S \cap [n]) + r_{n+1}w_{n+1}}{e^{-f(L)} + w(S \cap [n]) + w_{n+1}} \leq \frac{1.5 + (L-1)}{e^{-f(L)} + (L-1) + 1},$$

  as desired.

- Case 2 - $n + 1 \notin \mathcal{O}(S^*)$: In this case, we get that

$$\mathcal{P}(S^*) \leq \max_{S \subseteq [n]} \frac{w(S)}{e^{-f(L)} + w(S)} = \frac{2L}{e^{-f(L)} + 2L} \leq \frac{1.5 + (L-1)}{e^{-f(L)} + (L-1) + 1},$$

  where the last inequality can be easily verified to hold for any $L \geq 0$.

■

### A.3 Proof of Theorem 3.2

Our proof relies on a reduction from the clique decision problem, which is well-known to be NP-Complete (Karp, 1972). In this problem, the input is a graph $G = (V, E)$ on $n \geq 2$ vertices indexed by the set $[n]$, and we are interested in the question of whether or not there exists a subset of vertices $S \subseteq V$ of cardinality $K$ that are fully connected.

Given such an instance of the clique decision problem, we construct an instance of TU-constrained AO-Seq as follows. We first describe the set of products and their corresponding revenues and return costs, before formalizing the TU-constraint structure that we exploit.

**The assortment instance.** We create four distinct product classes as follows, where $\epsilon > 0$ is used as an arbitrarily small positive constant. We note that we set the return disutility $f = 0$.

- Edge products: For each edge $e \in E$, we create a product with revenue $r_e = 1$, weight $w_e = 4$ and return cost of $c_e = 0$. We will use the binary variable $x_e \in \{0, 1\}$ to indicate whether we offer each edge product $e \in E$.
- Vertex products: For each vertex $i \in [n]$, we create a product with revenue $r_i = \frac{4}{2+\epsilon}$, weight $w_i = 2 + \epsilon$ and return cost of $c_i = 0$. Abusing notation slightly, we will use the binary variable $x_i \in \{0, 1\}$ to indicate whether we offer each vertex product $i \in [n]$.
- Special vertex products: For each vertex $i \in [n]$, we create a product with revenue $r_i^S = \frac{2}{n}$, weight $w_i^S = 2$ and return cost of $c_i^S = 2$. We will use the binary variable $x_i^S \in \{0, 1\}$ to indicate whether we offer each special vertex product $i \in [n]$.
- Last product: We introduce a costly "last" product $\ell$ with revenue $r_\ell = 0$, weight $w_\ell = 1$ and return cost of $c_\ell = \frac{4K-0.5}{n}$. We will use the binary variable $z \in \{0, 1\}$ to indicate whether this last product is offered.

**The TU constraints.** We enforce three distinct sets of precedence constraints.

- An edge product cannot be offered unless both of its special vertex products are added. Formally, for each edge $e \in E$ where $e = (i, j)$, we enforce (i) $x_e \leq x_i^S$ and (ii) $x_e \leq x_j^S$.
- Each vertex product can only be added if its counterpart special product is added, formally expressed as $x_i \leq x_i^S$.
- Each special vertex products can only be added if the last product is added, which for each vertex $i \in [n]$ is expressed as $x_i^S \leq z$.

The constraint matrix that encodes these three sets of constraints clearly has exactly one +1 and one -1 in each row. Matrices of this form are well-known to be totally unimodular; see Proposition 2.6 in Chapter III.1 of Wolsey and Nemhauser (1999).

**The profit function.** First, observe that based on the prescribed weights, the products will be ordered in the following sequence: edge $\to$ vertex $\to$ special vertex $\to$ last. Moreover, since $f = 0$, we have that $\ell_Q = \ell$, and so the choice probabilities follow those observed under the traditional MNL model. In what follows, we break down the profit function under the Sequential Returns model by product class. Throughout, we use $D = 1 + \sum_{e \in E} w_e x_e + \sum_{i \in [n]} (w_i x_i + w_i^S x_i^S) + z$ (for denominator) to denote the total weight of the offered assortment, suppressing its dependence on the offered assortment, as the exact total weight will not play a role in the proof. In the individual expected profit functions that follow, we also suppress the explicit dependencies on the selected assortments of each product class, since it only complicated the notation without adding clarity.

- Edge product profit: The profit contribution of edge products is

$$\mathcal{P}_E = \frac{4E}{D},$$

where we use $E = \sum_{e \in E} x_e$ to denote the total number of edge products added.

- Vertex products: Similarly, the offered vertex products bring a profit of

$$\mathcal{P}_V = \frac{4V}{D},$$

where $V = \sum_{i \in [n]} x_i$ is the number of vertex products added.

- Special vertex products: The special vertex products are the first with returns costs, and thus their profit contribution is

$$\mathcal{P}_{SV} = \sum_{i=1}^{n} \left(\frac{2}{n} - \sum_{j=1}^{i-1} 2x_i^S\right) \cdot \frac{2x_i^S}{D} = \frac{4V_S}{nD} - \frac{4V_S \cdot (V_S - 1)}{2D},$$

where $V_S = \sum_{i \in [n]} x_i^S$ is the number of special vertex products added.

- Last product: The profit from the last product depends only on how many special vertex products were added, since it has zero revenue.

$$\mathcal{P}_\ell = \frac{-2V_S}{D} z.$$

- No purchase option. If the no-purchase item is selected, then all products were returned, and so

$$\mathcal{P}_{\text{np}} = \frac{-2V_S - z \cdot (4K - 0.5)/n}{D}.$$

In turn, the total expected profit $\mathcal{P}$ of any assortment is the sum of these five contributions.

**The reduction.** In Lemma A.3 and Lemma A.4 given below, we show that YES instances of the clique decision problem lead to an optimal assortment that garners a positive profit, while NO instances of the decision clique problem lead to an optimal assortment with a profit 0, i.e., it is optimal to offer nothing.

Lemma A.3 (Yes Instance). *Suppose there exist a clique $U \subset V$ of cardinality $K$, then there exists a feasible assortment that garners a positive expected profit.*

Proof. In this case, knowing there is a clique $U$ of cardinality $K$, we construct a feasible assortment from the four product classes as follows:

- We select the $K$ vertex and special vertex products corresponding to the clique $U$. In other words, we set $x_i^S = x_i = 1$ if $i \in U$ and $x_i^S = x_i = 0$ otherwise.

- We pick the $\frac{K \cdot (K-1)}{2}$ edge products that correspond to the edges whose endpoints are both in $U$.

- We also select the last product, so that $z = 1$.

It is straightforward to see that this solution is feasible, and so we proceed to analyzing its expected profit $\mathcal{P}$. We have that

$$
\begin{aligned}
\mathcal{P} &= \mathcal{P}_E + \mathcal{P}_V + \mathcal{P}_{SV} + \mathcal{P}_\ell + \mathcal{P}_{\text{np}} \\
&= \frac{1}{D} \cdot \left( \frac{4K \cdot (K-1)}{2} + 4K + \frac{4K}{n} - \frac{4K \cdot (K-1)}{2} - 2K - 2K - \frac{4K - 0.5}{n} \right) \\
&= \frac{1}{D} \cdot \left( \frac{4K}{n} - \frac{4K - 0.5}{n} \right) \\
&> 0.
\end{aligned}
$$

■

Lemma A.4 (No Instance). *Suppose there does not exist a clique $Q \subset V$ of cardinality $K$, then every non-empty assortment has strictly negative expected profit.*

Proof. It is straightforward to see that any feasible non-empty assortment must include the last product. We consider two cases based on whether or not the $V_S$ special vertices included in the profit-optimal non-empty assortment form a clique.

- Case 1 - The $V_S \leq K-1$ special vertices form a clique: In this case, the TU constraints ensure that this optimal assortment can include at most $V_S$ vertex products and at most $\frac{V_S \cdot (V_S - 1)}{2}$ edge products. Therefore, the optimal profit is

$$
\begin{aligned}
\mathcal{P} &= \mathcal{P}_E + \mathcal{P}_V + \mathcal{P}_{SV} + \mathcal{P}_\ell + \mathcal{P}_{\text{np}} \\
&\leq \frac{4V_S \cdot (V_S - 1)}{2} + 4V_S + \frac{4V_S}{n} - \frac{4V_S \cdot (V_S - 1)}{2} - 2V_S - 2V_S - \frac{4K - 0.5}{n} \\
&= \frac{4V_S}{n} - \frac{4K - 0.5}{n} \\
&< 0,
\end{aligned}
$$

  where the first inequality follows by setting $D$ at its lower bound of 1, and then noting that when we fix $D = 1$, our idealized profit function is increasing in the number of vertex and edge products added, and thus we set both of these terms at their respective upper bound of $V_S$ and $\frac{V_S \cdot (V_S - 1)}{2}$.

- Case 2 - The $V_S$ special vertices do not form a clique: Again, the TU constraints ensure that this optimal assortment can include at most $V_S$ vertex products and at most $\frac{V_S \cdot (V_S - 1)}{2} - 1$ edge products, since the special vertices do not form a clique. Therefore, the optimal profit is

$$
\begin{aligned}
\mathcal{P} &= \mathcal{P}_E + \mathcal{P}_V + \mathcal{P}_{SV} + \mathcal{P}_\ell + \mathcal{P}_{\text{np}} \\
&\leq 4 \cdot (\frac{V_S \cdot (V_S - 1)}{2} - 1) + 4V_S + \frac{4V_S}{n} - \frac{4V_S \cdot (V_S - 1)}{2} - 2V_S - 2V_S - \frac{4K - 0.5}{n} \\
&\leq -4 + \frac{4V_S}{n} - \frac{4K - 0.5}{n} \\
&< 0,
\end{aligned}
$$

  where the last inequality follows because $V_S \leq n$. The first inequality follows by again setting $D = 1$, and noting that when this is the case, the profit function is increasing in the number of edge and vertex products added.

**Summary.** Lemmas A.3 and A.4 together establish that, with an $\alpha > 0$ factor approximation scheme to the TU constrained version of AO-Seq, one could differentiate between YES and NO instances of clique decision problem. To do so would simply require checking whether the expected profit of the outputted assortment is positive or not.

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### A.4 Proof of Lemma 4.2

We prove the three statement sequentially.

**Proof of (i).** Define $w_{\min} = \min\{w_i : i \in \hat{S}\}$ to be smallest weight product within $\hat{S}$, and note that by construction, we must have that $w_{\min} \geq w_{\ell^*}$. Based on Lemma 2.2, it is sufficient to show that $w_{\min} \geq (w(\hat{S}) - w_{\min}) \cdot (e^f - 1) + (1 - e^{-f})$. We arrive at this just-mentioned inequality by observing that

$$w_{\min} \geq w_{\ell^*} \geq (W^* - w_{\ell^*}) \cdot (e^f - 1) + (1 - e^{-f}) \geq (w(\hat{S}) - w_{\min}) \cdot (e^f - 1) + (1 - e^{-f}),$$

where the second inequality follows by definition of $\ell^*$, and the third is a result of the fact that $w(\hat{S}) \leq W^*$ by the feasibility of $\hat{S}$ within E-KNAP.

**Proof of (ii).** Exploiting (i), along with the fact that $w(\hat{S}) \leq W^*$, we have that

$$\begin{aligned}
\mathcal{P}_B(\hat{S}) &= \sum_{i \in \hat{S}} \frac{w_i}{e^{-f} + w(\hat{S})} \cdot (r_i + c_i) - c(\hat{S}) \\
&\geq \sum_{i \in \hat{S}} \frac{w_i}{e^{-f} + W^*} \cdot (r_i + c_i) - c(\hat{S}) \\
&= \sum_{i \in \hat{S}} w_i \cdot (\frac{r_i + c_i}{e^{-f} + W^*} - \frac{c_i}{w_i}) \\
&= \sum_{i \in \hat{S}} \rho_i.
\end{aligned}$$

**Proof of (iii).** From (ii), we get that $\mathcal{P}_B(S^*) \geq Z^*$, and hence it suffices to show that $Z^* \geq \mathcal{P}(S^*)$. To do so, we construct a feasible solution to E-KNAP that earns an objective of $\mathcal{P}(S^*)$. For this purpose, consider the solution $x^* = (x_1^*, \ldots, x_{\ell^*}^*)$ where $x_i^* = 1$ if $i \in S^*$ and $x_i^* = 0$ otherwise. Clearly, this solution is feasible to E-KNAP since its total weight is precisely $W^* = w(S^*)$, and it achieves an objective of

$$\begin{aligned}
\sum_{i \in S^*} \rho_i &= \sum_{i \in S^*} w_i \cdot (\frac{r_i + c_i}{e^{-f} + W^*} - \frac{c_i}{w_i}) \\
&= \sum_{i \in S^*} \frac{w_i}{e^{-f} + w(S^*)} \cdot (r_i + c_i) - c(S^*) \\
&= \mathcal{P}(S^*).
\end{aligned}$$

### A.5 Proof of Lemma 4.3

Recalling that $k^* = \max\{k \in [|N_L|]_0 : w(S_k) \leq W^* - w(S_\epsilon)\}$, we begin by noting that if $k^* = |N_L|$, then it must be the case that $\mathcal{P}_B(S_{\text{target}}) = \mathcal{P}_B(S^*)$. To see this, if $k^* = |N_L|$, then $S_{\text{target}} = [\ell^*]$, meaning that we have a feasible solution to E-KNAP that includes every possible item. In turn, this implies that $\mathcal{P}_B(S_{\text{target}}) \geq \sum_{i \in [\ell^*]} \rho_i = Z^* = \mathcal{P}_B(S^*)$, which implies the desired equality.

Moving to the more interesting intermediate case where $k^* < |N_L|$, we will show that selecting the items in $S_{\text{target}}$ leads to an objective of at least $(1-\epsilon) \cdot \mathcal{P}_B(S^*)$ in E-KNAP. The lemma then follows immediately from property (ii) of Lemma 4.2. Our proof of this result utilizes the following intermediate claim, which re-hashes a well-known result about LP-relaxations of knapsack problems, and whose proof appears at the end of this section.

Claim A.5. *For assortment $S_{\text{UB}} = S_{k^*+1} \cup S_\epsilon$, we have that $\sum_{i \in S_{\text{UB}}} \rho_i \geq Z^*$.*

With this claim in-hand, noting that $S_{\text{target}} = S_{\text{UB}} \setminus \sigma(k^*+1)$, we get that

$$\sum_{i \in S_{\text{target}}} \rho_i = \sum_{i \in S_{\text{UB}}} \rho_i - \rho_{\sigma(k^*+1)} \geq Z^* - \rho_{\sigma(k^*+1)} \geq (1-\epsilon) \cdot Z^* = (1-\epsilon) \cdot \mathcal{P}_B(S^*),$$

where the first inequality uses Claim A.5 and the second inequality uses the fact that any item $i \in N_L$ has a reward of at most $\epsilon Z^*$ by construction.

**Proof of Claim A.5.** Consider the following LP relaxation of E-KNAP, where items within $S_\epsilon$ have already been added, and those within $N_\epsilon \setminus S_\epsilon$ have been excluded from consideration.

$$\begin{aligned} Z^*_{\text{LP}} = \max \quad & \sum_{i \in N_L} \rho_i x_i \\ \text{s.t.} \quad & \sum_{i \in N_L} w_i x_i \leq W^* - w(S_\epsilon) \\ & x_i \in [0,1] \qquad\qquad \forall\, i \in N_L. \end{aligned} \tag{6}$$

Since $S_\epsilon = S^* \cap N_\epsilon$, it is clear that

$$Z^*_{\text{LP}} + \sum_{i \in S_\epsilon} \rho_i \geq \sum_{i \in S^* \cap N_L} \rho_i + \sum_{i \in S_\epsilon} \rho_i = Z^*.$$

Consequently, to prove the claim, it suffices to show that $\sum_{i \in S_{k^*+1}} \rho_i \geq Z^*_{\text{LP}}$. To do so, we exploit the well-known property (Dantzig, 1957) of the standard knapsack problem that the optimal solution to the LP-relaxation continuously packs items in descending $\delta$-order until the knapsack is filled. In relation to (6), based on the definition of $S_{k^*}$, this means that

$$\begin{aligned} Z^*_{\text{LP}} &= \sum_{i \in S_{k^*}} \rho_i + (\underbrace{\frac{W^* - w(S_\epsilon) - w(S_{k^*})}{w_{\sigma(k^*+1)}}}_{\leq 1}) \cdot \rho_{\sigma(k^*+1)} \\ &\leq \sum_{i \in S_{k^*+1}} \rho_i, \end{aligned}$$

The first equality follows by noting that, after having packed $S_{k^*}$, there is $W^* - w(S_\epsilon) - w(S_{k^*})$ space left in the knapsack. Accordingly, in the LP-relaxation of the knapsack problem, a $\frac{W^* - w(S_\epsilon) - w(S_{k^*})}{w_{\sigma(k^*+1)}}$ -fraction of item $\sigma(k^* + 1)$ will be the lowest density item added.

### A.6 Proof of Lemma 4.4

Define $\gamma = \max\{k \in [|\hat{N}_L|]_0 : w(\hat{S}_k) \leq W^* - w(S_\epsilon)\}$, and note that by definition of $\kappa = \text{argmax}_{k \in [|\hat{N}_L|]_0} \mathcal{P}_B(\hat{S}_k \cup S_\epsilon)$, we have that $\mathcal{P}_B(\hat{S}_{\text{target}}) = \mathcal{P}_B(S_\epsilon \cup \hat{S}_\kappa) \geq \mathcal{P}_B(S_\epsilon \cup \hat{S}_\gamma)$. As such, it suffices to show that $\mathcal{P}_B(S_\epsilon \cup \hat{S}_\gamma) \geq (1 - 2\epsilon) \cdot \mathcal{P}_B(S^*)$, which we do by considering two cases based on the value $\gamma$.

**Case 1: $\gamma = |\hat{N}_L|$.** In this case, it is straightforward to once again argue that $\mathcal{P}_B(S_\epsilon \cup \hat{S}_\gamma) = \mathcal{P}_B(S^*)$ using the arguments above presented for the analogous case in the proof of Lemma 4.3, along with the observation that $N_L \subseteq \hat{N}_L$.

**Case 2: $\gamma < |\hat{N}_L|$.** In this case, consider selecting the set of items $S_\epsilon \cup \hat{S}_\gamma$ within E-KNAP. Via an almost identical argument as seen in the proof of Claim A.5, we can establish that for assortment $S_{\text{UB}} = S_{\gamma+1} \cup S_\epsilon$, we have that $\sum_{i \in S_{\text{UB}}} \rho_i \geq Z^*$. Exploiting this fact, we get that

$$\sum_{i \in S_\gamma \cup S_\epsilon} \rho_i = \sum_{i \in S_{\text{UB}}} \rho_i - \rho_{\sigma(\gamma+1)} \geq Z^* - \rho_{\sigma(\gamma+1)} \geq Z^* - \epsilon \hat{Z} \geq (1 - 2\epsilon) \cdot Z^* = (1 - 2\epsilon) \cdot \mathcal{P}_B(S^*),$$

where the second and third inequality uses the fact that any item $i \in \hat{N}_L$ has a reward of at most $\epsilon \hat{Z} \leq 2\epsilon Z^*$ by construction. From here, we get that

$$\mathcal{P}_B(\hat{S}_\gamma \cup S_\epsilon) \geq (1 - 2\epsilon) \cdot \mathcal{P}_B(S^*),$$

where the inequality follows by property (ii) of Lemma 4.2.

### A.7 Proof of Claim 5.1

To prove the claim, we construct a potential solution to Base-LP, and subsequently show that it is feasible and earns an objective of $R_Q(S^*)$. For $i \in [n]$, let $q^*(i) = |S^* \cap [i - 1]|$, and define

$$\hat{\alpha}_{i,q} = \begin{cases} \frac{1}{1 + \sum\limits_{j \in S^*} w_j e^{f \cdot (Q^* - q^*(j))}}, & \text{if } i \in S^*, q = q^*(i) \\ 0, & \text{otherwise} \end{cases}$$

and

$$\hat{\beta}_{i,q} = \begin{cases} \frac{1}{1 + \sum\limits_{j \in S^*} w_j e^{f \cdot (Q^* - q^*(j))}}, & \text{if } i \notin S^*, q = q^*(i) \\ 0, & \text{otherwise.} \end{cases}$$

**Feasibility.** We proceed to show that the constraints (1), (2), (3) and (5) of Base-LP are satisfied. To see that constraint (4) is satisfied, note that we cannot simultaneously have $i \in S^*$

and $q^*(i) = q$ for $q \geq Q^*$, since the optimal assortment includes exactly $Q^*$ products. This implies that $\hat{\alpha}_{i,q} = 0$ for each $i \in [n], q \in [Q^*, n-1]$.

- Constraint (1): This constraint reads:

$$\alpha_{1,0} + \beta_{1,0} = 1 - \sum_{(i,q)\in\mathcal{G}} \alpha_{i,q} w_i e^{f\cdot(Q^*-q)}.$$

  Observing that $|S^* \cap [0]| = 0$, we get $q^*(1) = 0$. Moreover, since we clearly must either have that $1 \in S^*$ or $1 \notin S^*$, the left-hand-side of this constraint evaluates to

$$\hat{\alpha}_{1,0} + \hat{\beta}_{1,0} = \frac{1}{1 + \sum_{j\in S^*} w_j e^{f\cdot(Q^*-q^*(j))}}.$$

  Moreover, the right-hand-side of this constraint evaluates to

$$\begin{aligned} 1 - \sum_{(i,q)\in\mathcal{G}} \hat{\alpha}_{i,q} w_i e^{f\cdot(Q^*-q^*(i))} &= 1 - \sum_{i\in S^*} \hat{\alpha}_{i,q^*(i)} w_i e^{f\cdot(Q^*-q^*(i))} \\ &= 1 - \sum_{i\in S^*} \frac{w_i e^{f\cdot(Q^*-q^*(i))}}{1 + \sum_{j\in S^*} w_j e^{f\cdot(Q^*-q^*(j))}} \\ &= \frac{1}{1 + \sum_{j\in S^*} w_j e^{f\cdot(Q^*-q^*(j))}}, \end{aligned}$$

  as desired.

- Constraint (2) (and (3)): These two constraints can be jointly expressed as

$$\alpha_{i,q} + \beta_{i,q} = \alpha_{i-1,q-1} + \beta_{i-1,q} \quad \forall\, (i,q) \in \mathcal{G}$$

  if we simply set $\alpha_{i,-1} = 0$ for any $i \in [2,n]$. We proceed in this way so that our proof can combine the two sets of constraints. For arbitrary $(i,q) \in \mathcal{G}$, we consider two cases based on the value of $q$.

  - Case 1 - $|S^* \cap [i-1]| \neq q$: This case can be rephrased as the scenario where $q^*(i) \neq q$, and hence the left-hand-side of constraint (2)/(3) evaluates to $\hat{\alpha}_{i,q} + \hat{\beta}_{i,q} = 0$. As such, our task becomes showing that the right-hand side also evaluates to 0, i.e., we set out to show that $\hat{\alpha}_{i-1,q-1} = \hat{\beta}_{i-1,q} = 0$. To do so, we consider two sub-cases based on the inclusion of product $i-1$ within $S^*$
    * Case 1a - $i-1 \in S^*$: Here, by construction, we have $\hat{\beta}_{i-1,q} = 0$. Moreover, it must be the case that $|S^* \cap [i-2]| \neq q-1$, since otherwise, we would indeed have that $|S^* \cap [i-1]| = q$, which contradicts the condition of Case 1. As a result, we must also have that $\hat{\alpha}_{i-1,q-1} = 0$, since the fact that $|S^* \cap [i-2]| \neq q-1$ implies that $q^*(i-1) \neq q-1$.
    * Case 1b - $i-1 \notin S^*$: On the hand, if $i-1 \notin S^*$, then we have that $\hat{\alpha}_{i-1,q-1} = 0$. Here, it must be that $|S^* \cap [i-2]| \neq q$, since if not and $i-1 \notin S^*$, this would

again imply $|S^* \cap [i-1]| = q$. This reasoning gives that $\hat{\beta}_{i-1,q} = 0$ using similar logic as was applied for Case 1a.

- Case 2 - $|S^* \cap [i-1]| = q$: Here, we have that $q^*(i) = q$, and hence because we either have that $i \in S^*$ or $i \notin S^*$, the left-hand-side of this constraint evaluates to

$$\hat{\alpha}_{i,q} + \hat{\beta}_{i,q} = \frac{1}{1 + \sum_{j \in S^*} w_j e^{f \cdot (Q^* - q^*(j))}}.$$

  Focusing on the right-hand-side of the constraint, we again break things down into the same two sub-cases:

  * Case 2a - $i-1 \in S^*$: If $i-1 \in S^*$, then $|S^* \cap [i-2]| = |S^* \cap [i-1]| - 1 = q-1$, and so $q^*(i-1) = q-1$. As a result, we have that $\hat{\alpha}_{i-1,q-1} = \frac{1}{1+\sum_{j\in S^*} w_i e^{f\cdot(Q^*-q^*(j))}}$, while $\hat{\beta}_{i-1,q} = 0$, and thus the constraint is satisfied.
  * Case 2b - $i-1 \notin S^*$: On the other hand, if $i-1 \notin S^*$, then $|S^* \cap [i-2]| = |S^* \cap [i-1]| = q$, and so $q^*(i-1) = q$. Consequently, we see that $\hat{\beta}_{i-1,q} = \frac{1}{1+\sum_{j\in S^*} w_i e^{f\cdot(Q^*-q^*(j))}}$, while $\hat{\alpha}_{i-1,q-1} = 0$ and again constraint (2) is satisfied.

- Constraint (5): This final set of constraints is:

$$\beta_{i,q} = 0 \quad \forall\, i \in [n], q \in [Q^* - n + i - 1]_0$$

Assume by way of contradiction that $\hat{\beta}_{i,q} > 0$ and that $q \in [Q^* - n + i - 1]_0$. Now, since $\hat{\beta}_{i,q} > 0$, it must be the case that $q = |S^* \cap [i-1]|$. Moreover, since $i \notin S^*$ and because $S^*$ adds exactly $Q^*$ products, we get hat

$$Q^* = q + |S^* \cap [i+1, n]| \le q + n - i,$$

which implies that $q \ge Q^* - n + i$, thus contradicting our initial assumption on $q$.

**Objective value.** Recalling that $q^*(i) = |S^* \cap [i-1]|$, we have that

$$\begin{aligned}
\sum_{(i,q)\in\mathcal{G}} \rho_{i,q}\alpha_{i,q} &= \sum_{i\in S^*} \rho_{i,q^*(i)}\alpha_{i,q^*(i)} \\
&= \frac{1}{1 + \sum_{j\in S^*} w_j e^{f\cdot(Q^*-q^*(j))}} \cdot \sum_{i\in S^*} (r_i + c\cdot(Q^* - q^*(i)))\cdot w_i e^{f\cdot(Q^*-q^*(i))} \\
&= \frac{1}{1 + \sum_{j\in S^*} w_j e^{f\cdot|S^*\cap[j,n]|}} \cdot \sum_{i\in S^*} (r_i + c\cdot|S^*\cap[i,n]|)\cdot w_i e^{f\cdot|S^*\cap[i,n]|} \\
&= R_Q(S^*)
\end{aligned}$$

where the third equality follows by observing that $Q^* - q^*(i) = |S^*| - |S^* \cap [i-1]| = |S^* \cap [i,n]|$ for each $i \in [n]$.

### A.8 Proof of Lemma 5.2

**Intermediate results.** Our proofs of the two properties will critically rely on the following intermediate claims, which progressively give more and more shape to the optimal solution to Non-Degen-LP. We ultimately will only directly exploit the last of these claims, but to prove this third claim first requires establishing the first two claims. The proofs of these claims can be found at the end of this section.

Claim A.6. *For each $(i, q) \in \mathcal{G}_+$, we have that $\min\{\alpha^*_{i,q}, \beta^*_{i,q}\} = 0$, i.e. exactly one of either $\alpha^*_{i,q}$ or $\beta^*_{i,q}$ must be zero.*

Claim A.7. *For each product $i \in [n]$, let $\hat{q}(i) = |\hat{S} \cap [i-1]|$. We have that $\alpha^*_{i,q} + \beta^*_{i,q} > 0$ if and only if $q = \hat{q}(i)$, meaning that $\mathcal{G}_+ = \{(i, \hat{q}(i)) : i \in [n]\}$.*

Claim A.8. *For each $(i, q) \in \mathcal{G}_+$, we have that*

$$\alpha^*_{i,q} = \begin{cases} \frac{1}{1+\sum_{j \in \hat{S}} w_j e^{f \cdot (Q^* - \hat{q}(j))}}, & \text{if } i \in \hat{S}, q = \hat{q}(i) \\ 0, & \text{otherwise} \end{cases}$$

*and*

$$\beta^*_{i,q} = \begin{cases} \frac{1}{1+\sum_{j \in \hat{S}} w_j e^{f \cdot (Q^* - \hat{q}(j))}}, & \text{if } i \notin \hat{S}, q = \hat{q}(i) \\ 0, & \text{otherwise,} \end{cases}$$

*where, again, for $i \in [n]$, we define $\hat{q}(i) = |\hat{S} \cap [i-1]|$.*

**Proof of lemma property (i).** With the above claims established, we set about showing that $|\hat{S}| = Q^*$. To do so, assume by way of contradiction that $|\hat{S}| \neq Q^*$, and define $i_\ell = \max\{i \in [n] : \beta^*_{i,\hat{q}(i)} > 0\}$. By Claim A.8, it must be that $i_\ell \notin \hat{S}$, since $\beta^*_{i_\ell,\hat{q}(i_\ell)} > 0$. Moreover, based on the definition of $\hat{S}$ and $\hat{q}(i_\ell)$, it must be that $|\hat{S}| = \hat{q}(i_\ell) + n - i_\ell$. Now, if $Q^* > |\hat{S}|$, using the just-mentioned equality concerning $|\hat{S}|$, we get that $\hat{q}(i_\ell) < Q^* - n + i_\ell$. In turn, by constraint (5) of Non-Degen-LP, we get a contradiction, since these constraints enforce that $\beta^*_{i_\ell,\hat{q}(i_\ell)} = 0$. On the other hand, if $Q^* < |\hat{S}|$, then there must exist some $i \in \hat{S}$ such that $\hat{q}(i) = Q^*$, which implies that, for such a product, we have that $\alpha^*_{i,Q^*} > 0$ by Claim A.8. This notion contradicts constraint (4) of Non-Degen-LP, which enforces that $\alpha^*_{i,Q^*} = 0$.

**Proof of lemma property (ii).** Noting that the optimal solution to Non-Degen-LP is $Z^*$, we get that

$$\begin{aligned} Z^* &= \sum_{(i,q) \in \mathcal{G}_+} \rho_{i,q} \alpha_{i,q} \\ &= \sum_{i \in \hat{S}} \rho_{i,\hat{q}(i)} \alpha_{i,\hat{q}(i)} \end{aligned}$$

$$
\begin{aligned}
&= \frac{1}{1+\sum_{j\in\hat{S}} w_j e^{f\cdot(Q^*-\hat{q}(j))}} \cdot \sum_{i\in\hat{S}} (r_i + c\cdot(Q^*-\hat{q}(i)))\cdot w_i e^{f\cdot(Q^*-\hat{q}(i))} \\
&= R_Q(\hat{S}),
\end{aligned}
$$

where the second and third equality uses Claim A.8.

**Proof of Claim A.6.** To prove the claim, we present an argument concerning the number of tight constraints (satisfied with equality) that are required at any basic feasible solution (BFS) of Non-Degen-LP. Specifically, since $\mathcal{G} = \mathcal{G}_+ \cup \mathcal{G}_{t_{\text{final}}}$, Non-Degen-LP has a total of $2\cdot|\mathcal{G}| = 2\cdot(|\mathcal{G}_+| + |\mathcal{G}_{t_{\text{final}}}|)$ total decision variables, and hence at any BFS, at least this number of tight constraints are required. We argue that the optimal solution $(\alpha^*, \beta^*)$ can only give way to a maximum of $2\cdot(|\mathcal{G}_+| + |\mathcal{G}_{t_{\text{final}}}|)$, and this maximum is also achieved only if for each $(i,q)\in\mathcal{G}_+$, we have that either $\alpha^*_{i,q} = 0$ or $\beta^*_{i,q} = 0$. We proceed to count the number of tight constraints invoked by $(\alpha^*, \beta^*)$, which come in two forms: explicit constraints as in (1) and (2)/(3), and non-basic decision variables.

- Tight constraints from (1),(2),(3): These two sets of constraints are equality constraints, so they all must be tight. There are $|\mathcal{G}_+|$ of these constraints.

- Tight constraints from non-basic decision variables: We consider each pair of decision variables $\alpha_{i,q}, \beta_{i,q}$ based on whether $(i,q)$ belongs to $\mathcal{G}_+$ or $\mathcal{G}_{t_{\text{final}}}$. By constraint (5), we have that for any $(i,q)\in\mathcal{G}_{t_{\text{final}}}$, it must be that $\alpha^*_{i,q} = \beta^*_{i,q} = 0$. This gives another $2|\mathcal{G}_{t_{\text{final}}}|$ tight constraints, bringing the total to $2|\mathcal{G}_{t_{\text{final}}}| + |\mathcal{G}_+|$, and meaning an additional $|\mathcal{G}_+|$ are needed. For pair $(i,q)\in\mathcal{G}_+$, we have that $\alpha^*_{i,q} + \beta^*_{i,q} > 0$ [5], and hence at most one of $\alpha^*_{i,q}$ and $\beta^*_{i,q}$ can be non-basic. However, since we need $|\mathcal{G}_+|$ more tight constraints, it must be that that either $\alpha^*_{i,q} = 0$ or $\beta^*_{i,q} = 0$, as desired.

**Proof of Claim A.7.** We establish the result by induction over $i$, noting that the base case of $i = 1$ is trivial. We move to considering the more general case of $i \geq 2$. Assume by way of contradiction that there exists $q \neq \hat{q}(i)$ such that $(i,q)\in\mathcal{G}_+$, meaning that $\alpha^*_{i,q} + \beta^*_{i,q} > 0$. By constraint (2)/(3) in Non-Degen-LP, this also implies that $\alpha^*_{i-1,q-1} + \beta^*_{i-1,q} > 0$, and so by the induction hypothesis, it must the case that either $\hat{q}(i-1) = q-1$ or $\hat{q}(i-1) = q$. Weconsider these two cases, showing that each leads to contradiction.

- Case 1 - $\hat{q}(i-1) = q-1$: In this case, by the induction hypothesis, it must be that $\alpha^*_{i-1,q-1} > 0$, which implies that $i-1\in\hat{S}$ based on the definition of $\hat{S}$. Consequently, we get that $\hat{q}(i) = |\hat{S}\cap[i-1]| = |\hat{S}\cap[i-2]| + 1 = q$, thus contradicting out initial assumption that $q\neq\hat{q}(i)$. The last equality uses the fact that $q-1 = \hat{q}(i-1) = |\hat{S}\cap[i-2]|$ by the condition that defines this first cse.

- Case 2 - $\hat{q}(i-1) = q$: In this case, by the induction hypothesis we have that $\beta^*_{i-1,q} > 0$, and thus $\alpha^*_{i-1,q} = 0$ by Claim A.6, and so $i-1\notin\hat{S}$ by definition of $\hat{S}$. Altogether, this

[5] It is straightforward to see that if $\alpha^*_{1,0} = \beta^*_{1,0} = 0$, then $Z^* = 0$ and the problem is trivial.

gives that $\hat{q}(i) = |\hat{S} \cap [i-1]| = |\hat{S} \cap [i-2]| = q$, once again contradicting our initial assumption on $q$.

**Proof of Claim A.8.** Let $W = \max\{\alpha^*_{1,0}, \beta^*_{1,0}\}$. Claims A.6 and A.7 together with constraint (2)/(3) of Non-Degen-LP, make it straightforward to see that for $(i,q) \in \mathcal{G}_+$, we have that $\alpha^*_{i,q} = W$ and $\beta^*_{i,q} = 0$ if $i \in \hat{S}$, and $\beta^*_{i,q} = W$ and $\alpha^*_{i,q} = 0$ if $i \notin \hat{S}$. Moreover, from constraint (1), we get that

$$\begin{aligned} W &= 1 - \sum_{(i,q) \in \mathcal{G}_+} \alpha_{i,q} w_i e^{f \cdot (Q^* - q)} \\ &= 1 - \sum_{i \in \hat{S}} \alpha_{i,q} w_i e^{f \cdot (Q^* - \hat{q}(i))} \\ &= 1 - W \cdot \sum_{i \in \hat{S}} w_i e^{f \cdot (Q^* - \hat{q}(i))}, \end{aligned}$$

which implies that

$$W = \frac{1}{1 + \sum_{j \in \hat{S}} w_j e^{f \cdot (Q^* - \hat{q}(j))}},$$

as desired. The second equality uses Claim A.7, along with the definition of $\hat{S}$.

### A.9 Proof of Theorem 6.1

Prior to jumping into the proof of the theorem, we first state a much exploited fixed point lemma related to the traditional MNL assortment problem.

**Preliminaries.** An instance of the unconstrained assortment problem under the MNL model is characterized by a collection of $n$ products, each with associated preference weight $(w_i)_{i \in [n]_0}$ (the no-purchase option is given index 0 and weight $w_0$) and revenues $(r_i)_{i \in [n]}$. The expected revenue garnered from offering assortment $S \subseteq [n]$ is given by $\mathcal{R}_{\text{MNL}}(S) = \sum_{i \in S} \frac{r_i w_i}{1 + w(S)}$, and the goal is to select the revenue maximizing assortment $S^*_{\text{MNL}}$. The following lemma is exploited in numerous papers (Rusmevichientong et al., 2010; Feldman and Topaloglu, 2015; Sumida et al., 2021) concerning MNL-based assortment problems. For the sake of completeness, we present its proof at the end of this section.

Lemma A.9. *For a given instance of the unconstrained assortment problem with weights $(w_i)_{i \in [n]_0}$ and revenues $(r_i)_{i \in [n]}$, let $Z^*_{\text{MNL}} = \mathcal{R}_{\text{MNL}}(S^*_{\text{MNL}})$ denote the optimal expected revenue. There exists a unique $Z \geq 0$ such that*

$$w_0 Z = \max_{S \subseteq [n]} \sum_{i \in S} (r_i - Z) \cdot w_i, \tag{7}$$

*and this fixed point is in fact achieved at $Z = Z^*_{\text{MNL}}$.*

We set out to prove that $\mathcal{P}_Q(S^*_B) \geq \mathcal{P}_B(S^*_B)$, which immediately establishes the theorem, since $\mathcal{P}_Q(S^*_Q) \geq \mathcal{P}_Q(S^*_B)$. To do so, we will express both profit functions as the expected revenue earned from a sale minus the expected returns costs. That is, for fixed assortment $S$, we have

$$\mathcal{P}_Q(S) = \underbrace{\sum_{i \in S} r_i \cdot \pi_Q(i,S)}_{=\mathcal{R}_Q(S)} - \underbrace{\sum_{i \in S \cup \{0\}} c(S \cap [i-1]) \cdot \pi_Q(i,S)}_{=\mathcal{C}_Q(S)}$$

and

$$\mathcal{P}_B(S) = \underbrace{\sum_{i \in S} r_i \cdot \pi_B(i,S)}_{=\mathcal{R}_B(S)} - \underbrace{\sum_{i \in S \cup \{0\}} c(S \setminus \{i\}) \cdot \pi_B(i,S)}_{=\mathcal{C}_B(S)},$$

where $\pi_Q(i,S)$ and $\pi_B(i,S)$ are used to respectively denote the choice probabilities under the Sequential and Bulk Returns models. We prove the desired result by sequentially showing that $\mathcal{R}_Q(S_B^*) \geq \mathcal{R}_B(S_B^*)$ and that $\mathcal{C}_Q(S_B^*) \leq \mathcal{C}_B(S_B^*)$, which together immediately implies that $\mathcal{P}_Q(S_B^*) \geq \mathcal{P}_B(S_B^*)$. Throughout the remainder of the proof, we assume without loss of generality that $S_B^* = \mathcal{O}(S_B^*)$, since any product not ordered by the bulk customers will also not be ordered by the sequential customers. Moreover, we remind the reader that $\ell_Q \geq \ell(S)$ for any assortment $S$, and thus all products within $S^*$ have the potential to be ordered under the Sequential Returns model.

**Revenue guarantee.** For ease of notation moving forward, let $S^* = S_B^*$ and define $Z_B^* = \mathcal{R}_B(S^*)$ and $Z_Q^* = \mathcal{R}_Q(S^*)$, noting that our goal is to show that $Z_Q^* \geq Z_B^*$. For this purpose, we consider a standard MNL assortment problem with revenues $(r_i)_{i \in [n]}$ and weights of

$$\hat{w}_i = \begin{cases} w_i e^{f \cdot (|S^* \cap [i,n]|-1)}, & \text{if } i \in S^* \\ 0, & \text{otherwise,} \end{cases}$$

where the no-purchase weight is $\hat{w}_0 = e^{-f}$. Let $\hat{Z} = \max_{S \subseteq [n]} \frac{r_i \hat{w}_i}{\hat{w}_0 + \hat{w}(S)}$ denote the optimal expected revenue to this assortment problem, where $\hat{w}(S) = \sum_{i \in S} \hat{w}_i$. We prove the revenue guarantee by sequentially showing (i) $\hat{Z} \geq Z_B^*$, when then allows us to show the desired result of (ii) $Z_Q^* \geq Z_B^*$.

- Proof of (i): Assume by way of contradiction that $\hat{Z} < Z_B^*$. Since $Z_B^* = \mathcal{R}_B(S^*) = \sum_{i \in S^*} \frac{r_i w_i}{e^{-f} + w(S^*)}$, we know that

$$e^{-f} Z_B^* = \hat{w}_0 Z_B^* = \sum_{i \in S^*} (r_i - Z_B^*) \cdot w_i \tag{8}$$

via simple algebraic manipulations. However, exploiting Lemma A.9 along with our assumption that $\hat{Z} < Z_B^*$, we get that

$$\begin{aligned} \hat{w}_0 Z_B^* &> \max_{S \subseteq [n]} (r_i - Z_B^*) \cdot \hat{w}_i \\ &\geq \sum_{i \in S^*} (r_i - Z_B^*) \cdot \hat{w}_i \\ &\geq \sum_{i \in S^*} (r_i - Z_B^*) \cdot w_i, \end{aligned}$$

which contradicts (8). The last inequality follows because $\hat{w}_i \geq w_i$ for each $i \in S^*$, combined with the following claim, which establishes that $r_i - Z_B^* \geq 0$ for each $i \in S^*$. The proof of this claim appears at the end of this section.

Claim A.10. *For each $i \in S^*$, we have that $r_i \geq Z_B^*$.*

- Proof of (ii): From (i), we have that $\hat{Z} \geq Z_B^*$, and so once again exploiting Lemma A.9, we get that

$$\begin{aligned}\hat{w}_0 Z_B^* &\leq \max_{S \subseteq [n]} (r_i - Z_B^*) \cdot \hat{w}_i \\ &= \sum_{i \in S^*} (r_i - Z_B^*) \cdot \hat{w}_i\end{aligned},$$

where the equality follows by Claim A.10, along with the fact that $\hat{w}_i = 0$ for each $i \notin S^*$. Re-arranging this inequality gives that

$$Z_B^* \leq \sum_{i \in S^*} \frac{r_i \hat{w}_i}{\hat{w}_0 + \hat{w}(S^*)} = \mathcal{R}_Q(S^*) = Z_Q^*,$$

as desired. The first equality is a result of the fact that $\pi_Q(i, S^*) = \frac{\hat{w}_i}{\hat{w}_0 + \hat{w}(S^*)}$ for each $i \in S^*$ by Theorem 2.1.

**Return cost guarantee.** Here, we show that $\mathcal{C}_Q(S^*) \leq \mathcal{C}_B(S^*)$. To begin, observe that under the Bulk Returns model with homogeneous return cost, we have that

$$\mathcal{C}_B(S) = (1 - \pi_B(0, S)) \cdot c \cdot (|S| - 1) + \pi_B(0, S) \cdot c \cdot |S| \tag{9}$$

for any assortment $S \subseteq [n]$ that satisfies $S = \mathcal{O}(S)$. To arrive at this result, note that

$$\begin{aligned}\mathcal{C}_Q(S^*) &= \sum_{i \in S^* \cup \{0\}} c \cdot \pi_Q(i, S^*) \cdot |S^* \cap [i-1]| \\ &\leq \sum_{i \in S^* \cup \{0\}} c \cdot \pi_Q(i, S^*) \cdot |S^* \setminus \{i\}| \\ &= (1 - \pi_Q(0, S^*)) \cdot c \cdot (|S^*| - 1) + \pi_Q(0, S^*) \cdot c \cdot |S^*| \\ &= (1 - \frac{e^{-f}}{e^{-f} + \hat{w}(S^*)}) \cdot c \cdot (|S^*| - 1) + \frac{e^{-f}}{e^{-f} + \hat{w}(S^*)} \cdot c \cdot |S^*| \\ &\leq (1 - \frac{e^{-f}}{e^{-f} + w(S^*)}) \cdot c \cdot (|S^*| - 1) + \frac{e^{-f}}{e^{-f} + w(S^*)} \cdot c \cdot |S^*| \\ &= (1 - \pi_B(0, S^*)) \cdot c \cdot (|S^*| - 1) + \pi_B(0, S^*) \cdot c \cdot |S^*| \\ &= \mathcal{C}_B(S^*).\end{aligned}$$

The first inequality follows because $|S^* \cap [i-1]| \leq |S^* \setminus \{i\}|$ for each $i \in S^* \cup \{0\}$, and the second inequality follows since $\hat{w}(S^*) \geq w(S^*)$.

**Proof of Lemma A.9.** We prove that the fixed point occurs at $Z = Z^*_{\text{MNL}}$, and its uniqueness follows directly from the fact that the left-hand-side of (7) is strictly increasing in $Z$, while the right-hand-side is non-increasing in $Z$. Assume by way of contradiction that the fixed point occurs at some $Z < Z^*_{\text{MNL}}$. Then we have that

$$\begin{aligned} w_0 Z^*_{\text{MNL}} &> \max_{S \subseteq [n]} (r_i - Z^*_{\text{MNL}}) \cdot w_i \\ &\geq \sum_{i \in S^*_{\text{MNL}}} (r_i - Z^*_{\text{MNL}}) \cdot w_i. \end{aligned}$$

Re-arranging this inequality give that $Z^*_{\text{MNL}} > \mathcal{R}_{\text{MNL}}(S^*_{\text{MNL}})$, which contradicts the definition of $S^*_{\text{MNL}}$. If, on the other hand, we assume that the fixed point occurs at some $Z > Z^*_{\text{MNL}}$, an identical argument can be used to establish a similar contradiction.

**Proof of Claim A.10.** To show this result, we assume by way of contradiction that there exists $i \in S^*$ such that $r_i < Z^*_B$. We proceed to show that if this is the case, then $\mathcal{P}_B(S^* \setminus \{i\}) > \mathcal{P}_B(S^*)$, thus contradicting the definition of $S^*$. To prove the result, we will show that $\mathcal{R}_B(S^* \setminus \{i\}) > \mathcal{R}_B(S^*)$ and that $\mathcal{C}_B(S^* \setminus \{i\}) \leq \mathcal{C}_B(S^*)$. Beginning with the result for the expected revenue, and noting that $S^* = \mathcal{O}(S^*)$ implies that $S^* \setminus \{i\} = \mathcal{O}(S^* \setminus \{i\})$ we have

$$\begin{aligned} Z^*_B = \mathcal{R}_B(S^*) &= \sum_{j \in S^*} \frac{r_j w_j}{e^{-f} + w(S)} \\ &= \sum_{j \in S^* \setminus \{i\}} \frac{r_j w_j}{e^{-f} + w(S \setminus \{i\})} \cdot \underbrace{\left( \frac{e^{-f} + w(S \setminus \{i\})}{e^{-f} + w(S)} \right)}_{=\lambda \in (0,1)} + r_i \cdot \underbrace{\left( \frac{w_i}{e^{-f} + w(S)} \right)}_{1-\lambda} \\ &= \lambda \mathcal{R}_B(S^* \setminus \{i\}) + (1-\lambda) \cdot r_i \\ &< \lambda \mathcal{R}_B(S^* \setminus \{i\}) + (1-\lambda) \cdot Z^*_B, \end{aligned}$$

which can be re-arranged to show that $\mathcal{R}_B(S^* \setminus \{i\}) > Z^*_B = \mathcal{R}_B(S^*)$.

Moving to the returns costs, from (9), we see that $\mathcal{C}_B(S^*) \geq |S^*| - 1$ and that $\mathcal{C}_B(S^* \setminus \{i\}) \leq |S^*| - 1$, and so $\mathcal{C}_B(S^* \setminus \{i\}) \leq \mathcal{C}_B(S^*)$.

### A.10 Proof of Lemma 6.2

We prove the two properties sequentially.

**Proof of (i).** Since the market share is is simply the demand not attributed to the no purchase option, it suffices to show that $\pi_Q(0, S) \leq \pi_B(0, S)$. For this purpose, since $\ell_Q \leq \ell(S)$, we can assume loss of generality that $|S \cap [\ell_Q]| = |S|$, and so

$$\begin{aligned} \pi_Q(0, S) &= \frac{e^{-f \cdot |S|}}{e^{-f \cdot |S|} + \sum_{i \in S} w_i e^{-f \cdot |S \cap [i-1]|}} \\ &= \frac{e^{-f}}{e^{-f} + \sum_{i \in S} w_i e^{f \cdot (|S \cap [i,n]| - 1)}} \end{aligned}$$

$$
\begin{aligned}
&\leq \frac{e^{-f}}{e^{-f} + \sum_{i \in S} w_i} \\
&= \pi_B(0, S),
\end{aligned}
$$

where the inequality follows since $f \geq 0$ and $|S \cap [i, n]| - 1 \geq 0$ for each $i \in S$.

**Proof of (ii).** For any $i \in S$, we have that

$$
\begin{aligned}
\pi_Q(i, S) &= \frac{w_i e^{f \cdot (|S \cap [i,n]| - 1)}}{e^{-f} + \sum_{j \in S} w_j e^{f \cdot (|S \cap [j,n]| - 1)}} \\
&= \frac{w_i}{e^{-f} + w(S)} \cdot \frac{w_i e^{f \cdot (|S \cap [i,n]| - 1)}}{w_i} \cdot \frac{e^{-f} + w(S)}{e^{-f} + \sum_{j \in S} w_j e^{f \cdot (|S \cap [j,n]| - 1)}} \\
&= \pi_B(i, S) \cdot e^{f \cdot (|S \cap [i,n]| - 1)} \cdot \frac{e^{-f} + w(S)}{e^{-f} + \sum_{j \in S} w_j e^{f \cdot (|S \cap [j,n]| - 1)}},
\end{aligned}
$$

and so $\pi_Q(i, S) \geq \pi_B(i, S)$ if and only if

$$
f(i, S) = e^{f \cdot (|S \cap [i,n]| - 1)} \cdot \frac{e^{-f} + w(S)}{e^{-f} + \sum_{j \in S} w_j e^{f \cdot (|S \cap [j,n]| - 1)}} \geq 1.
$$

Noting that the second term in $f(i, S)$ is upper bounded by 1 and independent of $i$, and that the first term is clearly decreasing in $i$, we see that $k^* = \max\{i \in S : f(i, S) \geq 1\}$ satisfies the desired properties. Moreover, based on property (i), we know that $k^*$ must be well-defined, since the fact that the sequential market share is larger than that of the bulk models implies that there must be at least one product $i \in S$ that satisfies $\pi_Q(i, S) \geq \pi_B(i, S)$, and hence for this particular product, we must have $f(i, S) \geq 1$.

## B Establishing Theorem 4.5

We begin with some preliminaries and an overview of our approach, before explicitly laying out the approximation scheme and establishing the theorem.

**Preliminaries part 1.** We start by detailing a partitioning of the customer types by the total weight of their order sets under the optimal assortment $S^*$. Specifically, letting $w_{\max}$ denote the highest weight product offered within $S^*$ (note that this product is ordered by all customers), define

$$
\mathcal{K}_q = \{k \in [K] : w(\mathcal{O}_k(S^*)) \in [w_{\max} \cdot 2^{q-1}, w_{\max} \cdot 2^q)\}
$$

to be the set of customer types whose order set has total weight in the interval $[w_{\max} \cdot 2^{q-1}, w_{\max} \cdot 2^q)$. We will refer to $\mathcal{K}_q$ as the set of "class-$q$ types". For $Q = \log_2 n$, it is straightforward to see that $\mathcal{K}_1, \ldots, \mathcal{K}_Q$ are disjoint, and that they form a partition of the $K$ customer types. Moreover, as can be seen from Figure 5, each set of class-$q$ types will be comprised of customer types whose indices are consecutive. Given this partitioning, we can express the optimal expected revenue as $\mathcal{R}(S^*) = \sum_{q \in [Q]} R(q, S^*)$, where

$$R(q, S^*) = \sum_{k \in \mathcal{K}_q} \frac{\lambda_k}{w_{0k} + w(\mathcal{O}_k(S^*))} \cdot \rho(\mathcal{O}_k(S^*))$$

is the total expected revenue contribution coming from the class-$q$ types. Finally, define $q^* = \text{argmax}_{q \in [Q]} R(q, S^*)$ to be the class of customer types that earns the largest expected revenue, breaking ties arbitrarily, and let $\mathcal{K}_{q^*} = [k_{\min}, k_{\max}]$, where $k_{\min}$ and $k_{\max}$ are the smallest and largest indexed class-$q^*$ types. For the remainder of this section, we will use the shorthand $R_{q^*}$ to refer to $R(q^*, S^*)$. Finally, for the remainder of this section, we use $\ell_k^*$ as a shorthand for $\ell_k(S^*)$ to ease notation moving forward, recalling that $\ell_k(S^*)$ is defined to be the lowest weight product ordered type $k \in [K]$ under the optimal assortment $S^*$.

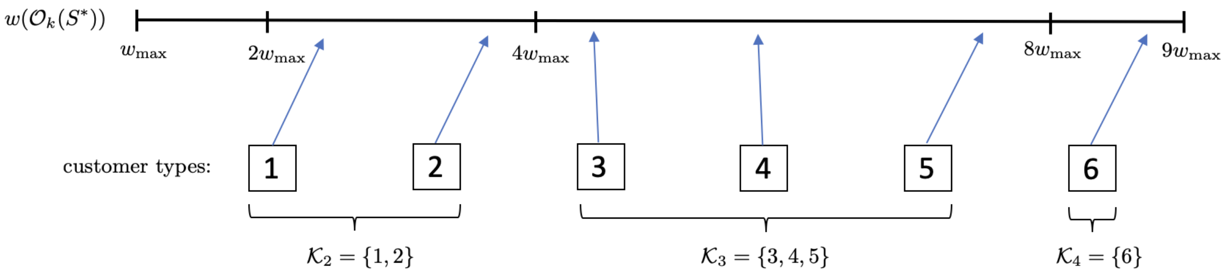


Figure 5: A depiction of the partition of the customer types when in a setting where $K = 6$. In this example, if $q^* = 3$, then $k_{\min} = 3$ and $k_{\max} = 5$.

**Technical overview.** The crux of our approach lies in the observation that $R_{q^*} \geq \frac{1}{Q}\mathcal{R}(S^*) = \Omega(\frac{1}{\log n}) \cdot \mathcal{R}(S^*)$, and so if we can produce an assortment that garners a constant fraction of $R_{q^*}$, then we will have met the revenue guarantee of Theorem 4.5. To uncover such an assortment, we observe that $R_{q^*}$ can be decomposed into the revenue contribution from product within $[\ell^*_{k_{\min}}]$ and from products within $[\ell^*_{k_{\min}} + 1, \ell^*_{k_{\max}}]$. Accordingly, the larger of these two contributions must be at least $\frac{1}{2} R_{q^*}$. With this in-mind, we develop two distinct approximate dynamic programming (ADP) approaches, which respectively compete against the revenue contributions from the two distinct sets of products mentioned above within an $\epsilon$-factor. Ultimately, our approach outputs the better of the two assortments returned by the two dynamic programs, which is guaranteed to garner an expected revenue of at least $\frac{1-\epsilon}{2} \cdot R_{q^*} \geq \frac{1-\epsilon}{2Q} \cdot \mathcal{R}(S^*) = \Omega(\frac{1}{\log n}) \cdot \mathcal{R}(S^*)$.

### B.1 The approximation scheme

The individual steps of our approximation scheme are detailed below, with the addition of two interludes that provide additional notation, definitions and important intermediate results. We note that in what appears below, since we have assumed that the return costs are zero, we will recycle the notation $c(S)$ for a particular assortment $S$, giving it an alternative meaning than the total return cost.

**Step 1: Guessing order cut-offs.** We first guess $w_{\max}$, $\ell^*_{k_{\min}}$ and $\ell^*_{k_{\max}}$ by enumerating over the $n$ possibilities for each.

**Step 2: Guessing $\mathcal{K}_{q^*}$ and $q^*$.** Next, we guess $k_{\min}$ and $k_{\max}$ by enumerating over the $K$ possibilities for each. Note that these two guesses together fully reveal $\mathcal{K}_{q^*}$. Finally, we also guess $q^*$ by enumerating over the $O(\log n)$ options, and with $w_{\max}$ already in-hand from the first guessing step, we let $W_{q^*} = w_{\max} \cdot 2^{q^*-1}$.

**Interlude: Decomposing $R_{q^*}$.** With the guesses from Steps 1 and 2 in-hand, define disjoint product sets $N_1 = [\ell^*_{k_{\min}}]$ and $N_2 = [\ell^*_{k_{\min}} + 1, \ell^*_{k_{\max}}]$, and observe that

$$\begin{aligned} R_{q^*} &= \sum_{k \in \mathcal{K}_{q^*}} \frac{\lambda_k}{w_{0k} + w(\mathcal{O}_k(S^*))} \cdot \rho(\mathcal{O}_k(S^*)) \\ &= \sum_{k \in \mathcal{K}_{q^*}} \frac{\lambda_k}{w_{0k} + w(\mathcal{O}_k(S^*))} \cdot \left(\rho(\mathcal{O}_k(S^*) \cap N_1) + \rho(\mathcal{O}_k(S^*) \cap N_2)\right), \end{aligned}$$

where the second equality follows since $N_1$ and $N_2$ are disjoint, combined with the fact that $\mathcal{O}_k(S^*) \subseteq N_1 \cup N_2$ for any $k \in \mathcal{K}_{q^*}$ by construction. Letting

$$\begin{aligned} R^{N_1}_{q^*} &= \sum_{k \in \mathcal{K}_{q^*}} \frac{\lambda_k}{w_{0k} + w(\mathcal{O}_k(S^*))} \cdot \rho(\mathcal{O}_k(S^*) \cap N_1) \\ R^{N_2}_{q^*} &= \sum_{k \in \mathcal{K}_{q^*}} \frac{\lambda_k}{w_{0k} + w(\mathcal{O}_k(S^*))} \cdot \rho(\mathcal{O}_k(S^*) \cap N_2) \end{aligned}$$

respectively denote the expected revenue contribution from the product sets $N_1$ and $N_2$, it must be the case that $\max\{R^{N_1}_{q^*}, R^{N_2}_{q^*}\} \geq \frac{1}{2} R_{q^*}$, since $R_{q^*} = R^{N_1}_{q^*} + R^{N_2}_{q^*}$. In other words, the expected revenue contribution coming either from products within $N_1$ or $N_2$ must exceed $\frac{1}{2} R_{q^*}$. In the proceeding steps, we set about individually competing against the contribution from these two sets of products across the class-$q^*$ types.

**Step 3: The approximate dynamic program for $N_1$.** We start by rounding each product's $\rho$-value down to the the nearest integer multiple of $\frac{\epsilon}{n} \hat{\rho}_{N_1}$, where $\hat{\rho}_{N_1}$ satisfies $\hat{\rho}_{N_1} \leq \rho(S^* \cap N_1) \leq 2\hat{\rho}_{N_1}$ (See Appendix B.5 for how to uncover such a $\hat{\rho}_{N_1}$). We let $(\hat{\rho}_i)_{i \in [n]}$ denote these rounded $\rho$-values, and for assortment $S \subseteq [n]$, we use $\hat{\rho}(S) = \sum_{i \in S} \hat{\rho}_i$ to denote the total $\hat{\rho}$-value. From here, we define a dynamic program that proceeds sequentially over the products in $N_1$, and decides for each whether to offer it or not, with the intention to minimize the total weight of the offered products while hitting a target total $\hat{\rho}$-value. For this purpose, the state space of our dynamic program is comprised of the following two parameters:

- The current product $i \in N_1$, for which we will proceed to make an offer or not decision. We will enforce that product $\ell^*_{k_{\min}}$ is offered to simplify future proofs, however this requirement is not necessary.

- A target total $\hat{\rho}$-value to accumulate across products $i, \dots, \ell^*_{k_{\min}}$. We will restrict this parameter to the grid
$$\mathcal{G}_{N_1} = \{q \cdot \frac{\epsilon}{n} \hat{\rho}_{N_1} : q = 0, \dots, \lceil 2\frac{n}{\epsilon} \rceil\}.$$

The value functions $\mathcal{V}_{N_1}(i, \rho)$ represent the minimum weight assortment from among products $i, \dots, \ell^*_{k_{\min}}$ needed to accrue a total $\hat{\rho}$-value of at least $\rho$. Formally, we have

$$\mathcal{V}_{N_1}(i, \rho) = \begin{cases} \min\left\{w_i + \mathcal{V}_{N_1}\left(i+1, [\rho - \hat{\rho}_i]^+\right), \mathcal{V}_{N_1}(i+1, \rho)\right\}, & \text{if } i < \ell^*_{k_{\min}} \\ w_i + \mathcal{V}_{N_1}\left(i+1, [\rho - \hat{\rho}_i]^+\right), & \text{if } i = \ell^*_{k_{\min}}. \end{cases} \tag{10}$$

with base cases of
$$\mathcal{V}_{N_1}(\ell^*_{k_{\min}} + 1, \rho) = \begin{cases} 0, & \text{if } \rho = 0 \\ \infty, & \text{otherwise.} \end{cases} \tag{11}$$

It is only useful to describe how we use this dynamic program to help us identify a final assortment after having presented the alternative dynamic program to compete against $N_2$. This discussion appears below in the paragraph labeled "outputted assortment".

**Interlude: Preliminaries part 2.** Before detailing our approximate dynamic program to select the items from $N_2$, it necessary for us to introduce some additional notation and intermediate claims. Specifically, for $i \in N_2$, let $\mathcal{K}^{(i)} = \{k \in \mathcal{K}_{q^*} : w_i \geq W_{q^*} \cdot (e^{f_k} - 1) + (1 - e^{-f_k})\}$, which can be thought of as the set of class-$q^*$ types that will surely order product $i$ if the offered assortment has total no larger than $W_{q^*}$. Note that for each $i \in N_2$, the set $\mathcal{K}^{(i)}$ can be fully computed upfront, since we have already guessed $\mathcal{K}_{q^*}$ and $q^*$. Furthermore, for each product $i \in N_2$, let $c_i = \sum_{k \in \mathcal{K}^{(i)}} \frac{\lambda_k}{w_{0k} + W_{q^*}} \cdot \rho_i$ be its "contribution", and define $c(S) = \sum_{i \in S} c_i$ to be the total contribution of assortment $S \subseteq N_2$. The following claim relates the total contribution of a particular set of low-weight assortments to their expected revenue from class-$q^*$ types, thus partially illustrated why these two newly defined quantities ($\mathcal{K}^{(i)}$ and $c_i$) are important.

Claim B.1. *For any assortment $S \subseteq N_2$ whose total weight satisfies $w(S) \leq W_{q^*}$, we have $R(q^*, S) \geq c(S)$.*

Proof. We have

$$\begin{aligned} R(q^*, S) &= \sum_{k \in \mathcal{K}_{q^*}} \frac{\lambda_k}{w_{0k} + w(\mathcal{O}_k(S))} \cdot \rho(\mathcal{O}_k(S)) \\ &= \sum_{i \in S} \sum_{\substack{k \in \mathcal{K}_{q^*}: \\ i \in \mathcal{O}_k(S)}} \frac{\lambda_k}{w_{0k} + w(\mathcal{O}_k(S))} \cdot \rho_i \\ &\geq \sum_{i \in S} \sum_{k \in \mathcal{K}^{(i)}} \frac{\lambda_k}{w_{0k} + w(\mathcal{O}_k(S))} \cdot \rho_i \\ &\geq \sum_{i \in S} \sum_{k \in \mathcal{K}^{(i)}} \frac{\lambda_k}{w_{0k} + W_{q^*}} \cdot \rho_i \\ &= c(S). \end{aligned}$$

The first inequality follows from the observation that, if $k \in \mathcal{K}^{(i)}$, then by Lemma 2.2, we must have that $i \in \mathcal{O}_k(S)$ for any $i \in S$, since we have assumed that $w(S) \leq W_{q^*}$. ■

With this result in-hand, we set about constructing an approximate dynamic program that ultimately yields an assortment whose total weight does not exceed $W_{q^*}$, and whose total contribution is at least $(1-\epsilon) \cdot R^{N_2}_{q^*}$ . Leveraging Claim B.1, we immediately get that the expected revenue of this assortment must exceed $(1-\epsilon) \cdot R^{N_2}_{q^*}$.

**Step 4: The approximate dynamic program for $N_2$.** We start by rounding the contribution $c_i$ of each product $i \in N_2$ down to the nearest integer multiple of $\frac{\epsilon}{n}\hat{\mathcal{C}}$, where $\hat{\mathcal{C}}$ satisfies $\hat{\mathcal{C}} \leq R^{N_2}_{q^*} \leq 2\hat{\mathcal{C}}$. In Appendix B.5, we describe how to uncover such a $\hat{\mathcal{C}}$ in polynomial time. We use $\hat{c}_i$ to denote the rounded contribution of product $i \in N_2$, and the function $\hat{c}(S) = \sum_{i \in S} \hat{c}_i$ to denote the total rounded contribution of assortment $S \subseteq N_2$. From here, we define a dynamic program that proceeds sequentially over the products in $N_2$, and decides for each whether to offer it or not, with the intention to minimize the total weight of the offered products while hitting a target total $\hat{c}$-value. For this purpose, the state space of our dynamic program is comprised of the following two parameters:

- The current product $i \in N_2$, for which we will proceed to make an offer or not decision. We will enforce that product $\ell^*_{k_{\max}}$ is offered.
- A target total $\hat{c}$-value to accumulate across products $i, \dots, \ell^*_{k_{\max}}$. We will restrict this parameter to the grid
$$\mathcal{G}_{N_2} = \{q \cdot \frac{\epsilon}{n}\hat{\mathcal{C}} : q = 0, \dots, \lceil 2\frac{n}{\epsilon} \rceil\}.$$

The value functions $\mathcal{V}_{N_2}(i, c)$ represent the minimum weight assortment from among products $i, \dots, \ell^*_{k_{\max}}$ needed to accrue a total $\hat{c}$-value of at least $c$. Formally, we have

$$\mathcal{V}_{N_2}(i,c) = \begin{cases} \min\left\{w_i + \mathcal{V}_{N_2}\left(i+1, [c-\hat{c}_i]^+\right), \mathcal{V}_{N_2}(i+1,c)\right\}, & \text{if } i < \ell^*_{k_{\max}} \\ w_i + \mathcal{V}_{N_2}\left(i+1, [c-\hat{c}_i]^+\right), & \text{if } i = \ell^*_{k_{\max}}. \end{cases} \tag{12}$$

with base cases of

$$\mathcal{V}_{N_2}(\ell^*_{k_{\max}}+1, c) = \begin{cases} 0, & \text{if } c = 0 \\ \infty, & \text{otherwise.} \end{cases} \tag{13}$$

**The outputted assortment.** Let $S_{N_1}(\rho)$ denote the assortment derived from following the dynamic program given in (10)-(11) starting from the state $(1, \rho)$, and define $S_{N_2}(c)$ to be the assortment derived from following the dynamic program given in (12)-(13) starting from the state $(\ell^*_{k_{\min}}+1, c)$. Our approximation scheme outputs the assortment

$$\hat{S} = \underset{S \in \mathcal{S}_{N_1} \cup \mathcal{S}_{N_2}}{\operatorname{argmax}} \; \mathcal{R}(S), \tag{14}$$

where $\mathcal{S}_{N_1} = \{S_{N_1}(\rho)\}_{\rho \in \mathcal{G}_{N_1}}$ and $\mathcal{S}_2 = \{S_{N_2}(c)\}_{c \in \mathcal{G}_{N_2}}$.

## B.2 The Analysis

We first identify assortments $\bar{S}_1 \in \mathcal{S}_{N_1}$ and $\bar{S}_2 \in \mathcal{S}_{N_2}$ that respectively garner an expected revenue of of at least $(1-\epsilon) \cdot R_{q^*}^{N_1}$ and $(1-\epsilon) \cdot R_{q^*}^{N_2}$, for any $\epsilon > 0$. From here, it is relatively straightforward to combine these results to show that the expected revenue of $\hat{S}$ is within a $\Omega(\log n)$-factor of $\mathcal{R}(S^*)$. We conclude the analysis by arguing that $\hat{S}$ can indeed be obtained in a running time that is polynomial in the input.

**Competing against $R_{q^*}^{N_1}$.** Let $\bar{S}_1$ denote the assortment returned by following the dynamic program in (10)-(11) from the initial state $(1, \hat{\rho}(S^* \cap N_1))$. The following claim, whose proof can be found in Appendix B.2, establishes important properties of $\bar{S}_1$, which will subsequently aid in relating its expected revenue to $R_{q^*}^{N_1}$.

Claim B.2. *For any $k \in \mathcal{K}_{q^*}$, we have*

*(i)* $w(\bar{S}_1) \leq w(\mathcal{O}_k(S^*))$,

*(ii)* $\mathcal{O}_k(\bar{S}_1) = \bar{S}_1$,

*(iii) and* $\rho(\mathcal{O}_k(\bar{S}_1)) \geq (1-\epsilon) \cdot \rho(\mathcal{O}_k(S^*) \cap N_1)$.

An immediate consequence of Claim B.2 is that

$$\begin{aligned}
\mathcal{R}(\bar{S}_1) \geq \sum_{k \in \mathcal{K}_{q^*}} R_k(\bar{S}_1) &= \sum_{k \in \mathcal{K}_{q^*}} \frac{\lambda_k}{w_{0k} + w(\mathcal{O}_k(\bar{S}_1))} \cdot \rho(\mathcal{O}_k(\bar{S}_1)) \\
&= \sum_{k \in \mathcal{K}_{q^*}} \frac{\lambda_k}{w_{0k} + w(\bar{S}_1)} \cdot \rho(\bar{S}_1) \quad \text{(By (ii) of Claim B.2)} \\
&\geq \sum_{k \in \mathcal{K}_{q^*}} \frac{(1-\epsilon) \cdot \lambda_k}{w_{0k} + w(\mathcal{O}_k(S^*))} \cdot \rho(\mathcal{O}_k(S^*) \cap N_1) \quad \text{(By (i),(iii) of Claim B.2)} \\
&= (1-\epsilon) \cdot R_{q^*}^{N_1}.
\end{aligned}$$

**Competing against $R_{q^*}^{N_2}$.** Let $\bar{S}_2$ to be the assortment returned by following the dynamic program in (12)-(13) from the initial state $(\ell^*_{k_{\min}} + 1, \hat{c}(S^* \cap N_2)$. We first establish the following claim, whose proof can be found in Appendix B.3, which allows us to show that $\mathcal{R}(\bar{S}_2)$ is within an $\epsilon$-factor of $R_{q^*}^{N_2}$.

Claim B.3. *We have*

*(i)* $\bar{S}_2 \subseteq N_2$,

*(ii)* $w(\bar{S}_2) \leq W_{q^*}$,

*(iii) and* $c(\bar{S}_2) \geq (1-\epsilon) \cdot R_{q^*}^{N_2}$.

Noting the three properties of Claim B.3, we can directly exploit Claim B.1 to get that $R_{q^*}(\bar{S}_2) \geq (1-\epsilon) \cdot R_{q^*}^{N_2}$.

**The final revenue guarantee.** To wrap things up, observe that

$$\mathcal{R}(\hat{S}) \geq \max\{\mathcal{R}(\bar{S}_1), \mathcal{R}(\bar{S}_2)\} \geq (1-\epsilon) \cdot \max\{R_{q^*}(\bar{S}_1), R_{q^*}(\bar{S}_2)\} \geq \frac{1-\epsilon}{2} \cdot R_{q^*},$$

which implies that $\mathcal{R}(\hat{S}) = \Omega(\frac{1}{\log n}) \cdot \mathcal{R}(S^*)$. The first inequality above follows since we clearly have that $\bar{S}_1 \in \mathcal{S}_{N_1}$ and $\bar{S}_2 \in \mathcal{S}_{N_2}$.

**Overall running time.** We analyze the running times of the individual steps of our approach, and argue that the result is a $poly(|\mathcal{I}|, n, K, \frac{1}{\epsilon})$-time algorithm.

- The guessing in step 1 can be carried out in $O(n^3)$, since we are guessing three quantities that each can take on $n$ possible values, while the second guessing step can be executed in $O(K^2 \cdot \log n)$, since there are $K$ possibilities for $k_{\min}$ and $k_{\max}$, and $O(\log n)$ possibilities for $q^*$.

- The running time to solve the two dynamic programs is $O((\frac{n}{\epsilon})^{O(1)})$, since both have a total of $O(\frac{n^2}{\epsilon})$ states, whose value functions can each be computed in $O(1)$. The running time to build $\mathcal{S}_{N_2}$ just adds an addition multiplicative dependence on $|\mathcal{I}|$ that is needed to recover $\hat{\mathcal{C}}$, as described in Section B.5. Computing $\hat{S}$ can be be done by enumerating over all assortments in $\mathcal{S}_{N_1}$ and $\mathcal{S}_{N_2}$, of which there are $O((\frac{n}{\epsilon})^{O(1)})$.

### B.3 Proof of Claim B.2

We prove the three properties sequentially.

**Proof of (i).** For any $k \in \mathcal{K}_{q^*}$, we have

$$w(\mathcal{O}_k(S^*)) = w(S^* \cap [\ell_k^*]) \geq w(S^* \cap [\ell_{k_{\min}}^*]) = w(S^* \cap N_1) \geq w(\bar{S}_1).$$

The first inequality follows because $k \geq k_{\min}$ for any $k \in \mathcal{K}_{q^*}$, and the last inequality follows from the fact that $\bar{S}_1$ is precisely the lowest weight assortment comprised of products from $N_1$ whose total $\hat{\rho}$-value exceeds $\hat{\rho}(S^* \cap N_1)$. Clearly the assortment $S^* \cap N_1$ achieves said total $\hat{\rho}$-value, so it must be that $w(S^* \cap N_1) \geq w(\bar{S}_1)$.

**Proof of (ii).** Recalling that $\ell_{k_{\min}}^* \in \bar{S}_1$, it suffices to show that customer type $k_{\min}$ will order $\ell_{k_{\min}}^*$, since this customer type has the largest return disutility among all those in $\mathcal{K}_{q^*}$ and $\ell_{k_{\min}}^*$ is the smallest weight product in $N_1$. To show this result, observe that

$$\begin{aligned}
w_{\ell_{k_{\min}}^*} &\geq w(S^* \cap [\ell_{k_{\min}}^* - 1]) \cdot (e^{f_{k_{\min}}} - 1) + (1 - e^{-f_{k_{\min}}}) \\
&= w\left((S^* \setminus \{\ell_{k_{\min}}^*\}) \cap N_1\right) \cdot (e^{f_{k_{\min}}} - 1) + (1 - e^{-f_{k_{\min}}}) \\
&\geq w\left((\bar{S}_1 \setminus \{\ell_{k_{\min}}^*\})\right) \cdot (e^{f_{k_{\min}}} - 1) + (1 - e^{-f_{k_{\min}}}) \\
&= w(\bar{S}_1 \cap [\ell_{k_{\min}}^* - 1]) \cdot (e^{f_{k_{\min}}} - 1) + (1 - e^{-f_{k_{\min}}}),
\end{aligned}$$

which, by Lemma 2.2, implies that $\ell_{k_{\min}}(\bar{S}_1) = \ell_{k_{\min}}^*$. The first inequality above follows by definition of the order set for customer type $k_{\min}$, and the second inequality follows because both $\bar{S}_1$ and $S^*$ include product $\ell^*_{k_{\min}}$, along with the fact that $w(S^* \cap N_1) \geq w(\bar{S}_1)$, as established by property (i) of the claim.

**Proof of (iii).** Initially using property (ii) of the current claim, we get that get that

$$\rho(\mathcal{O}_k(\bar{S}_1)) = \rho(\bar{S}_1) \geq \hat{\rho}(\bar{S}_1) \geq \hat{\rho}(S^* \cap N_1) \geq (1-\epsilon) \cdot \rho(S^* \cap N_1) = (1-\epsilon) \cdot \rho(\mathcal{O}_k(S^*) \cap N_1)$$

for any $k \in \mathcal{K}_{q^*}$. The second inequality follows by noting that, via property (i), we have established that $\mathcal{V}_{N_1}(1, \hat{\rho}(S^* \cap N_1)) < \infty$, and hence $\hat{\rho}(\bar{S}_1) \geq \hat{\rho}(S^* \cap N_1)$. The third inequality uses the fact that $\hat{\rho}(S^* \cap N_1) \geq \rho(S^* \cap N_1) - \epsilon \hat{\rho}_{N_1} \geq (1-\epsilon) \cdot \rho(S^* \cap N_1)$. The final inequality follows by noting that for each $k \in \mathcal{K}_{q^*}$, we have that $\ell^*_k \geq \ell^*_{k_{\min}}$, and so $\mathcal{O}_k(S^*) \cap N_1 = S^* \cap N_1$.

### B.4 Proof of Claim B.3

We prove the three properties sequentially.

**Proof of (i).** The fact that $\bar{S}_2 \subseteq N_2$ is trivial, since the dynamic program in (12)-(13) only considers adding products in $N_2$.

**Proof of (ii).** Given that $\mathcal{V}_{N_2}(\ell^*_{k_{\min}} + 1, \hat{c}(S^* \cap N_2)) = w(\bar{S}_2)$ is the minimum weight needed to garner a total rounded contribution of $\hat{c}(S^* \cap N_2)$, we trivially get that this value function is upper bounded by $w(S^* \cap N_2)$, since the assortment $S^* \cap N_2$ has a total rounded contribution of $\hat{c}(S^* \cap N_2)$. Consequently, to prove this second property, it suffices to show that $w(S^* \cap N_2) \leq W_{q^*}$. To see this observe that

$$\begin{aligned} w(S^* \cap N_2) &= w(S^* \cap [\ell^*_{k_{\max}}]) - w(S^* \cap [\ell^*_{k_{\min}}]) \\ &\leq w_{\max} \cdot 2^{q^*} - w_{\max} \cdot 2^{q^*-1} \\ &\leq w_{\max} \cdot 2^{q^*-1} \\ &= W_{q^*}, \end{aligned}$$

where the first and second inequality follows definition of $\mathcal{K}_{q^*}$.

**Proof of (ii).** We start by observing that

$$c(\bar{S}_2) \geq \hat{c}(\bar{S}_2) \geq \hat{c}(S^* \cap N_2) \geq c(S^* \cap N_2) - \frac{\epsilon}{n} \hat{\mathcal{C}} \cdot |S^* \cap N_2| \geq c(S^* \cap N_2) - \epsilon R^{N_2}_{q^*}.$$

To complete the proof, we will show that $c(S^* \cap N_2) \geq R^{N_2}_{q^*}$, which implies that $c(\bar{S}_2) \geq (1-\epsilon) \cdot R^{N_2}_{q^*}$. To see this final inequality, note that

$$\begin{aligned} c(S^* \cap N_2) &= \sum_{i \in S^* \cap N_2} \sum_{k \in \mathcal{K}^{(i)}} \frac{\lambda_k}{w_{0k} + W_{q^*}} \cdot \rho_i \\ &\geq \sum_{i \in S^* \cap N_2} \sum_{\substack{k \in \mathcal{K}_{q^*}: \\ i \in \mathcal{O}_k(S^*) \cap N_2}} \frac{\lambda_k}{w_{0k} + W_{q^*}} \cdot \rho_i \end{aligned}$$

$$
\begin{aligned}
&= \sum_{k \in \mathcal{K}_{q^*}} \sum_{i \in \mathcal{O}_k(S^*) \cap N_2} \frac{\lambda_k}{w_{0k} + W_{q^*}} \cdot \rho_i \\
&= \sum_{k \in \mathcal{K}_{q^*}} \frac{\lambda_k}{w_{0k} + W_{q^*}} \cdot \left( \sum_{i \in \mathcal{O}_k(S^*) \cap N_2} \rho_i \right) \\
&= \sum_{k \in \mathcal{K}_{q^*}} \frac{\lambda_k}{w_{0k} + W_{q^*}} \cdot \rho(\mathcal{O}_k(S^*) \cap N_2) \\
&\geq \sum_{k \in \mathcal{K}_{q^*}} \frac{\lambda_k}{w_{0k} + w(\mathcal{O}_k(S^*))} \cdot \rho(\mathcal{O}_k(S^*) \cap N_2) \\
&= R_{q^*}^{N_2}.
\end{aligned}
$$

The first inequality comes from the fact that if $i \in \mathcal{O}_k(S^*) \cap N_2$ for $k \in \mathcal{K}_{q^*}$, it must be the case that

$$
\begin{aligned}
w_i &\geq w(S^* \cap [\ell_k^* - 1]) \cdot (e^{f_{k_{\min}}} - 1) + (1 - e^{-f_{k_{\min}}}) \\
&\geq w(S^* \cap [\ell_{k_{\min}}^*]) \cdot (e^{f_{k_{\min}}} - 1) + (1 - e^{-f_{k_{\min}}}) \\
&\geq W_{q^*} \cdot (e^{f_{k_{\min}}} - 1) + (1 - e^{-f_{k_{\min}}}),
\end{aligned}
$$

and so $k \in \mathcal{K}^{(i)}$. The second and third inequality follow from the fact that for any $k \in \mathcal{K}_{q^*}$, we have that $k > k_{\min}$ and that $w(\mathcal{O}_k(S^*)) \geq W_{q^*}$ by construction.

### B.5 The Anchor Values for Rounding

This section is devoted to describing how of each of the "anchor" values used in the rounding procedures are uncovered via efficient enumeration in polynomial time.

**Computing $\hat{\rho}_{N_1}$.** We can uncover $\hat{\rho}_{N_1}$ by enumerating over the following set of candidates $\{\rho^*_{\max,N_1} \cdot 2^q : q = 0, \ldots, \log(n)\}$, where $\rho^*_{\max,N_1} = \max_{i \in (S^* \cap N_1)} \rho_i$ is the maximum $\rho$-value across all products offered in $S^* \cap N_1$. Since $\rho^*_{\max,N_1} \leq \rho(S^* \cap N_1) \leq n\rho^*_{\max,N_1}$, it must be the case that one of the just-proposed candidates satisfied the requirements of $\hat{\rho}_{N_1}$.

**Computing $\hat{\mathcal{C}}$.** We can uncover $\hat{\mathcal{C}}$ by enumerating over the following set of candidates $\{\left(\frac{\lambda_{\min}\rho_{\min}}{1+2W_{q^*}}\right) \cdot 2^q : q = 0, \ldots, \log\left(\frac{r_{\max}\cdot(1+2W_{q^*})}{\lambda_{\min}\rho_{\min}}\right)\}$, where $\lambda_{\min} = \min_{i \in [n]} \lambda_i$, $\rho_{\min} = \min_{i \in [n]} \rho_i$, and $r_{\max} = \max_{i \in [n]} r_i$ is the maximum revenue across all products. Since $\frac{\lambda_{\min}\rho_{\min}}{1+2W_{q^*}} \leq R_{q^*}^{N_2} \leq r_{\max}$, it must be the case that one of the just-proposed candidates satisfied the requirements of $\hat{\mathcal{C}}$.

## C Establishing Theorem 4.6

The proof of this theorem unfolds over the five subsections that follow, the first two of which build intuition, introduce additional notation and establish preliminary results, while the latter three are devoted to explicit details of our approximation scheme. All proofs from this section appear in Appendix D.

### C.1 The multi-layered partition

In this section, we present a two layered partitioning of the customer types and products based on features of the order sets under the optimal assortment $S^*$. Specifically, we first partition the $K$ customer types based on the total weight of their order sets under $S^*$, grouping customer types so that within each group, the total weight of each order set is within a factor of two. From here, we partition the products in $S^*$ into three distinct sets based again on the accumulation of the order set weight for each group of customer types. For future reference, we present a depiction of these partitions in Figure 6, which builds off Figure 5.

Two important remarks are in order. First, since the above two partitions fundamentally depend on $S^*$, their respective constructions are purely existential constructs that help us craft our approximation scheme. Along this line, after having formalized the two partitions, the last paragraph of this section formalizes how we intend to proceed. Namely, having partitioned $S^*$ into three sub-assortments, we set out to develop approximate-dynamic-programming-based approaches that separately compete against the revenue contribution of each of the three sub-assortments.

**Partitioning the customer types.** To start, let $w_{\max} = \max_{i \in S^*} w_i$ denote the highest weight (and therefore lowest indexed) product offered within $S^*$. We can assume without loss of generality that all customer types will order this product, since any customer type that does not do so will order nothing, and hence generate zero expected revenue. As such, for each $k \in [K]$, we have that $\mathcal{O}_k(S^*) \in [w_{\max}, nw_{\max}]$. Define

$$\mathcal{K}_q = \{k \in [K] : w(\mathcal{O}_k(S^*)) \in [w_{\max} \cdot 2^{q-1}, w_{\max} \cdot 2^q)\}$$

to be the customer types that make-up order weight class $q$ (weight class $q$, for short). Observe that for $Q = \lceil \log_2(n) \rceil$, we have that $\biguplus_{q \in [Q]} \mathcal{K}_q = [K]$, i.e. the weight classes $\mathcal{K}_1, \ldots, \mathcal{K}_Q$ form a valid partition of the customer types. Finally, let $\mathcal{Q} = \{q \in [Q] : \mathcal{K}_q \neq \emptyset\}$ denote the weight classes that house at least one customer type. For $t \in [|\mathcal{Q}|]$, we will use $q_t \in \mathcal{Q}$ to denote the $t$-th largest indexed weight class among those in $\mathcal{Q}$, e.g., $q_1 = \min \mathcal{Q}$. Moreover, for each non-empty weight class $q \in \mathcal{Q}$, let $k^*_{q,\min} = \min \mathcal{K}_q$ and $k^*_{q,\max} = \max \mathcal{K}_q$, noting that $\mathcal{K}_q = [k^*_{q,\min}, k^*_{q,\max}]$, since the order sets are nested by customer type index. Moreover, for $t \in [|\mathcal{Q}| - 1]$, we have that $k^*_{q_t,\max} = k^*_{q_{t+1},\min} - 1$, as illustrated in Figure 6.

**Partitioning the optimal assortment - Part I.** For each $t \in [|\mathcal{Q}|]$, let $S^*_{q_t} \subseteq S^*$ denote the products ordered by any customer type within $\mathcal{K}_{q_t}$, that are not ordered by customer types who are a part of lower indexed weight classes $\mathcal{K}_{q_1}, \ldots, \mathcal{K}_{q_{t-1}}$. Formally, we have that

$$S^*_{q_t} \quad = \quad \mathcal{O}_{k^*_{q_t,\max}}(S^*) \setminus \mathcal{O}_{k^*_{q_{t-1},\max}}(S^*),$$

as depicted in Figure 6. It is again straightforward to observe that $S^* = \biguplus_{q \in \mathcal{Q}} S^*_q$, i.e., the sets $S^*_q$ partition $S^*$.

**Partitioning the optimal assortment - Part II.** Next, for each $q \in \mathcal{Q}$, we partition $S^*_q$ into three disjoint sub-assortments. To do so, define product

$$i^*_q = \max\{i \in S^*_q : w(S^* \cap [i]) < w_{\max} \cdot 2^{q-1}\}$$

along with product $j^*_q = i^*_q + 1$. It is important to observe that all products within $S^* \cap [j^*_q]$ will be ordered by all customer within $\mathcal{K}_q$. We partition $S^*_q$ into the following three assortments:

$$S^*_{q,1} = S^*_q \cap [i^*_q], \quad S^*_{q,2} = \{j^*_q\}, \quad S^*_{q,3} = S^*_q \setminus (S^*_{q,1} \cup S^*_{q,2}).$$

Finally, for $p \in \{1, 2, 3\}$, define $S^*_p = \biguplus_{q \in \mathcal{Q}} S^*_{q,p}$, noting that $S^* = \biguplus_{p=1}^{3} S^*_p$.

**Remark.** To allow for parallel reasoning in describing how we compete against the three assortments, we will assume that $S^*_{q,p}$ is well-defined and never empty. Having read through our approximation schemes, it will be clear that these assumptions are innocuous, and are simply enacted to ease the exposition of our approaches. One implication of this assumption is that we can assume $q_1 > 1$, since otherwise, it would be the case that $S^*_{1,1} = \emptyset$ based on the fact that all order sets have weight at least $w_{\max}$.

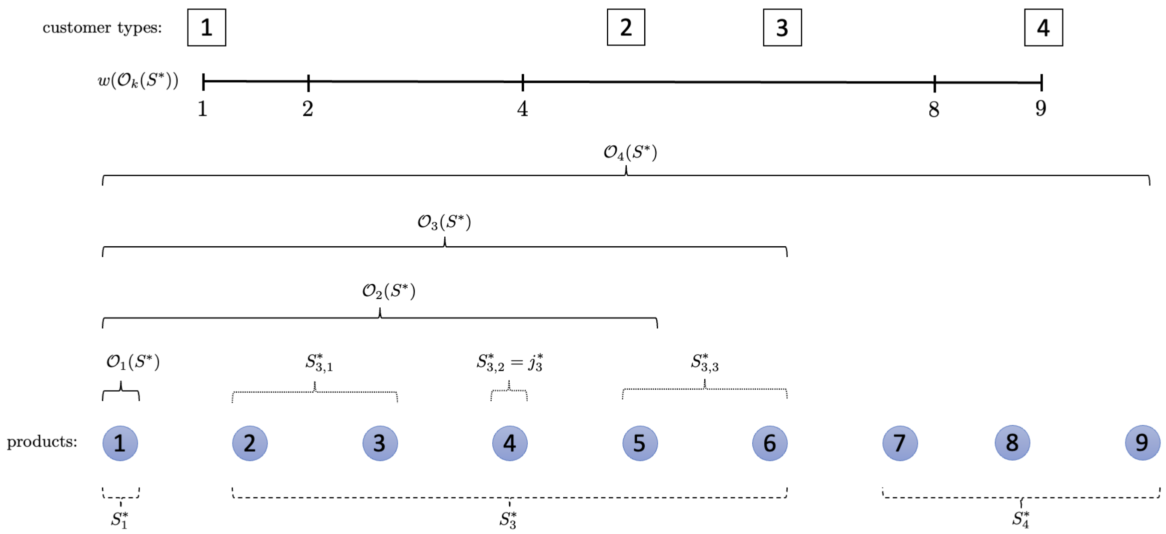


Figure 6: A depiction of the multi-layered partition when each of the nine products has a weight of 1. The location of the squares on the number line denotes the total weight of the order set for each customer type. Accordingly, we have that $\mathcal{K}_1 = \{1\}, \mathcal{K}_3 = \{2, 3\}$ and $\mathcal{K}_4 = \{4\}$.

**Optimal expected revenue breakdown.** For $p \in \{1, 2, 3\}$, let

$$R_p(S^*) = \sum_{q \in \mathcal{Q}} \sum_{k \in \mathcal{K}_q} \lambda_k \cdot R_{k,p}(S^*),$$

capture the expected revenue contribution of $S^*_p$ towards the optimal expected revenue $\mathcal{R}(S^*)$, where

$$R_{k,p}(S^*) = \frac{1}{1 + w(\mathcal{O}_k(S^*))} \cdot \rho(\mathcal{O}_k(S^*) \cap S^*_{q,p}).$$

denotes the part of this "contribution" attributed to customer type $k$. Observing that $\mathcal{R}(S^*) = \sum_{p=1}^{3} R_p(S^*)$, we have that

$$\max_{p\in\{1,2,3\}} R_p(S^*) \geq \frac{1}{3} \cdot \mathcal{R}(S^*).$$

As such, the remaining sections are devoted to developing separate approximation schemes that identify five assortments that compete within a constant factor of the above-defined contribution of $S_1^*, S_2^*$ and $S_3^*$ respectively. That said, before presenting our approach, we first make a few simple remarks, which are then followed by a section comprised of intermediate claims that are critical to the development of our approximation schemes, and also help to start to build intuition.

**Remarks.** It is important note that $R_p(S^*)$ is likely different from $\mathcal{R}(S_p^*)$, since in the latter term, we have potentially dropped products from $S^*$ when considering the expected revenue of $S_p^*$, which in turn could affect the order sets of each customer type. Moreover, we note that it is straightforward to see that $\mathcal{R}_k(S^*) \geq R_{k,p}(S^*)$ based on their respective definitions, noting that the former term is the customer type $k$ expected revenue. Table 1 summarizes all of the most important pieces of newly developed notation for quick reference.

| | | |
|---|---|---|
| $\mathcal{K}_q$ | = | Customer types whose total order weight satisfy $w(\mathcal{O}_k(S^*)) \in [w_{\max} \cdot 2^{q-1}, w_{\max} \cdot 2^q]$. |
| $\mathcal{Q}$ | = | The set of non-empty weight classes, |
| $k^*_{q,\min}$ ($k^*_{q,\max}$) | = | The lowest (highest) indexed customer class within $\mathcal{K}_q$ |
| $q_t$ | = | The $t$-th largest indexed weight class within $\mathcal{Q}$. |
| $S_q^*$ | = | The products ordered by types $\mathcal{K}_q$ under $S^*$, which are not ordered by lower indexed classes |
| $S^*_{q,p}$ | = | One of three assortments that partition $S_q^*$. |
| $R_p(S^*)$ | = | The expected revenue earned from products $S_p^*$ under $S^*$. |
| $R_{k,p}(S^*)$ | = | The expected revenue earned from products $S_p^*$ under $S^*$ due to customer type $k$. |

Table 1: Most important pieces of notation for the approximation scheme of this section.

### C.2 Intermediate results and preliminary guessing steps

To ease the development of our approximation schemes, it will be helpful to establish a number of preliminary results, which consist of various claims and initial algorithmic steps that get the ball rolling with regards to the development of our approximation schemes. In particular, the notion of a "pruned" assortment will play a pivotal role in our analysis that follows.

**Initial guessing steps.** First, we use efficient enumeration to exactly recover $\mathcal{Q}$, which are the non-empty weight groups. Since $Q = O(\log n)$, we can do so by enumerating over the all $2^Q = O(n)$ options for $\mathcal{Q}$. Additionally, we also guess $w_{\max} = \max_{i\in S^*} w_i$, of which there are once again only $O(n)$ options. We assume that $w_1 = w_{\max}$, i.e. product 1 is the highest weight product included in the optimal assortment. This assumption is without loss of generality, since if it fails to hold, we can delete products with weight larger than $w_{\max}$ and re-index the products accordingly.

**Total weight upper bounds.** For each $q \in \mathcal{Q}$ and $p \in \{1, 2, 3\}$, we establish a basic claim that reveals upper bounds on the total weight of each assortment $S^*_{q,p}$. We eventually exploit this result in two related ways within our ADP-based approach. First, these upper bounds will help us ensure that we have an accurate accounting of our accumulated expected revenue within the dynamic programs that we develop. Second, these bounds will allow us to pinpoint the order sets of specific customer classes, which is also critical for arguing that we accumulate expected revenue accurately. The exact nature of this first claim is presented below.

CLAIM C.1. *For each $t \in [|\mathcal{Q}|]$, we have*

*(i)* $w(S^*_{q_t,1}) \leq w_{\max} \cdot (2^{q_t - 1} - 2^{q_{t-1} - 1})$,

*(ii)* $w(S^*_{q_t,2}) \leq w_{\max}$,

*(iii) and* $w(S^*_{q_t,3}) \leq w_{\max} \cdot 2^{q_t - 1}$,

*where we define* $q_0 = -\infty$

**Assortment pruning.** In what follows, we show how to produce the pruned version of any assortment $S \subseteq [n]$ by carefully removing products from $S$. We denote the resulting assortment $\mathcal{P}(S)$, and naturally we have that $\mathcal{P}(S) \subseteq S$ [6]. In a nutshell, the act of pruning an assortment involves carefully removing products from $S$ in a manner that ensures that the expected revenue garnered from each customer type increases, while also endowing the pruned assortment with other helpful qualities. Specifically, consider an arbitrary assortment $S \subseteq [n]$, and initialize the step 1 pruned assortment as $P_1 = S$. We prune the assortment $S$ through $n-1$ sequential steps indexed $1, \dots, n-1$, where in step $t \in [n-1]$, we compute the standard MNL expected revenue of the assortment $P_t \cap [t]$, which we denote as $\mathcal{R}_{\text{MNL}}(P_t \cap [t]) = \frac{\rho(P_t \cap [t])}{1 + w(P_t \cap [t])}$. Next, we compute the set of products $D_t = \{i \in P_t \cap [t+1, n] : r_i < \mathcal{R}_{\text{MNL}}(P_t \cap [t])\}$ and move to step $t+1$ with $P_{t+1} = P_t \setminus D_t$, i.e the step $t+1$ pruned assortment is the step $t$ pruned assortment after removing products within $D_t$. Note that if $t \notin P_t$, then $P_{t-1} \cap [t-1] = P_t \cap [t]$, and hence the step $t$ pruning process will delete nothing, i.e. $D_t = \emptyset$. The result after the $n-1$ pruning steps is the pruned assortment $\mathcal{P}(S) = P_n$. The following lemma gives important features of this pruned assortment, whose implications for our approximation schemes are discussed proceeding the lemma statement.

LEMMA C.2. *For any assortment $S \subseteq [n]$, customer type $k \in [K]$, and product $\ell \in [\ell_k(S)]$, we have that*

$$\mathcal{R}_k(S \cap [\ell]) \leq \mathcal{R}_k(\mathcal{P}(S)).$$

With regards to the above lemma, first note that when $\ell = \ell_k(S)$, we get that $S \cap [\ell] = \mathcal{O}_k(S)$, and so the lemma gives that $\mathcal{R}_k(S) \leq \mathcal{R}_k(\mathcal{P}(S))$, meaning that the pruned assortment has improved the customer type $k$ expected revenue. Moreover, since this result holds for arbitrary $k \in [K]$, we get that $\mathcal{R}(\mathcal{P}(S)) \geq \mathcal{R}(S)$. Finally, we note that the fact that the above lemma holds for any $\ell \in [\ell_k(S)]$ will be useful within the accounting of our ADP-based approaches.

[6] Note that we do not consider profit in this problem, and so this notational overlap should not cause any confusion, as we will only use $\mathcal{P}(S)$ to denote the pruned version of $S$

More specifically, the accounting within each dynamic program will assume that the products within our estimated version of $S^*_{q,p}$ are the last products ordered by the customer types of $\mathcal{K}_q$ (our DP will also attempt to uncover $\mathcal{K}_q$ for each $q \in \mathcal{Q}$). In reality, however, because our accounting of the total added weight will not be exact, it may be the case that the order sets of customer types within $\mathcal{K}_q$ "spillover" to include products from assortments targeted at higher indexed weight groups. We will leverage Lemma C.2 to ensure that any such spillover can only improve revenues, after pruning the assortment returned by our dynamic program.

**Outline and final guarantee.** In the three sections that follow, we separately show how to compete against $R_p(S^*)$ for each $p \in \{1, 2, 3\}$. We start with $p = 2$, as this is the simplest case, since the assortments $S^*_{q,2}$ are singletons. This approach then sets the stage for the slightly more nuanced ADP-based approach used to compete against the other two revenue contributions. The sections that follow should be thought of as independent of each other, and hence we recycle a handful of terms to avoid overly cumbersome notation. Ultimately, it is straightforward to see that taking the highest revenue assortment out of the five produced by the approximation schemes guarantees at least a $(\frac{1}{7} - \epsilon)$-fraction of the optimal expected revenue.

### C.3 Competing against $R_2(S^*)$

The goal in this section is to identify an assortment that closely approximates $S^*_2$, and thus achieves an expected revenue of nearly $R_2(S^*)$. In what follows, we formalize our ADP-based approach to accomplish this goal.

**State space.** Each component of the state space is defined below, where in doing so, we exploit the fact that we have already guessed $\mathcal{Q}$. We also note that for ease of exposition with regards to handling base cases in the dynamic program, we introduce a "dummy" customer type $K + 1$ with $\lambda_{K+1} = f_{K+1} = 0$.

- The index of a weight group $t \in [|\mathcal{Q}| + 1]$.
- An interval of customer types $[k_{\min}, k_{\max}]$, such that $k_{\min}, k_{\max} \in [K + 1]$.
- A product $j \in [n]_0$.
- A rounded accumulated total $\rho$-value, which we restrict to the set of exponentially spaced grid points
$$G_\delta = \{\rho_{\min} \cdot (1 + \delta)^y : y = 0, \dots, \lceil \log_{1+\delta}(\frac{n\rho_{\max}}{\rho_{\min}})\rceil\} \cup \{0\},$$
where we set $\delta = \frac{\epsilon}{Q}$, $\rho_{\min} = \min_{i\in[n]} \rho_i$ and $\rho_{\max} = \max_{i\in[n]} \rho_i$. Abusing notation slightly, we use the operator $G_\delta()$ to round its input down to the nearest grid point in $G_\delta$.

We use

$$\mathcal{U} = \{(t, [k_{\min}, k_{\max}], j, \rho) : \begin{array}{c} t \in [|\mathcal{Q}| + 1] \\ k_{\min}, k_{\max} \in [K + 1] : k_{\min} \leq k_{\max} \\ j \in [n]_0 \end{array},$$

$$\rho \in G_\delta\}$$

to denote the universe of possible states. The exact interpretation of each state variable is explained in the paragraph where we formally define the value functions.

**Action space.** For state $U = (t, [k_{\min}, k_{\max}], j, \rho) \in \mathcal{U}$, we define

$$\begin{aligned} \mathcal{F}(U) = \{(k'_{\max}, j') : \quad & k'_{\max} \in [K+1] : k'_{\max} \geq k_{\max} + 1 \\ & j' \in [j+1, n] : w_{j'} \geq (t-1) \cdot w_{\max} \cdot (e^{f_{k_{\min}}} - 1) + (1 - e^{-f_{k_{\min}}})\} \end{aligned}$$

to denote the feasible set of actions. When action $(k'_{\max}, j') \in \mathcal{F}(U)$ in taken in state $U = (t, [k_{\min}, k_{\max}], j, \rho) \in \mathcal{U}$, we transition to state

$$U' = (t+1, [k_{\max}+1, k'_{\max}], j', G_\delta(\rho + \rho_{j'})),$$

noting that the interval of customer types transitions from $[k_{\min}, k_{\max}]$ to $[k_{\max}+1, k'_{\max}]$. It is straightforward to see that $U' \in \mathcal{U}$.

**Value functions.** For state $U = (t, [k_{\min}, k_{\max}], j, \rho) \in \mathcal{U}$, the value functions $V(U)$ will represent what we establish to be a lower bound on the maximal expected revenue that it is possible to accrue when:

- we only have $|\mathcal{Q}| - t + 1$ products left to add to our assortment,
- for the state $U$ at-hand, we will only potentially accrue revenue from customer types $[k_{\min}, k_{\max}]$,
- the last product selected was $j$,
- the total accumulated rounded $\rho$-value from the first $t-1$ products added is $\rho$.

We can recursively define the value functions as

$$\begin{aligned} V(U) \quad = \quad & \max_{(k'_{\max}, j')} \left\{ \sum_{k \in [k_{\min}, k_{\max}]} \lambda_k \cdot \frac{\rho + \rho_{j'}}{1 + t w_{\max}} + V\left(U'\right) \right\} \\ & s.t. \quad (k'_{\max}, j'). \in \mathcal{F}(U) \end{aligned} \tag{15}$$

As for base cases, we have that any state $U = (|\mathcal{Q}|+1, [k_{\min}, k_{\max}], j, \rho) \in \mathcal{U}$, we set $V(U) = 0$, and if for any state $U \in \mathcal{U}$ we have that $\mathcal{F}(U) = \emptyset$, then we set $V(U) = -\infty$. These base cases will persist for all dynamic programs that we develop, and hence we only state them once here.

**The initial state.** Our dynamic program will start from the initial state $U^{(1)} \in \mathcal{U}$, defined as

$$U^{(1)} = (1, [k^{(1)}_{\min}, k^{(1)}_{\max}], \underbrace{j^{(0)}}_{=0}, \underbrace{\rho^{(0)}}_{=0}),$$

where $k_{\min}^{(1)} = k_{q_1,\min}^*$ and $k_{\max}^{(1)} = k_{q_1,\max}^*$, whose values can both be guessed by enumerating over the $O(K^2)$ possibilities. It is straightforward to see that $U^{(1)} \in \mathcal{U}$.

**Outputted assortment.** Starting from the just-defined initial state $U^{(1)}$, we can recursively define the sequence of states visited by the optimal policy. Namely, for $t \in [|\mathcal{Q}|]$, define the action taken by the optimal policy in state $U^{(t)}$ as $(k_{\max}^{(t+1)}, j^{(t)})$, i.e., this tuple is the argmax of (15) when in state $U^{(t)}$. Accordingly, the optimal policy progresses over states $U^{(1)}, U^{(2)}, \ldots, U^{(|\mathcal{Q}|+1)}$, where for $t \in [2, |\mathcal{Q}|+1]$, we have that

$$U^{(t)} = (t, [k_{\max}^{(t-1)} + 1, k_{\max}^{(t)}], j^{(t-1)}, G_\delta(\rho^{(t-1)} + \rho_{j^{(t-1)}})).$$

The assortment that we output and proceed to analyze is $S_2 = \mathcal{P}(\biguplus_{t=1}^{|\mathcal{Q}|}\{j^{(t)}\})$, which is the pruned version of the assortment that consists of the product added in each stage.

**Performance guarantee.** The following lemma shows that the expected revenue garnered by $S_2$ is at least the revenue contribution $R_2(S^*)$, with a arbitrarily small fudge-factor.

Lemma C.3. *For any $\epsilon > 0$, we have that*

*(i)* $V(U^{(1)}) \geq (1-\epsilon) \cdot R_2(S^*)$

*(ii)* $\mathcal{R}(S_2) \geq V(U^{(1)})$,

We note that, together, the two properties immediately imply that $\mathcal{R}(S_2) \geq (1-\epsilon) \cdot R_2(S^*)$, as desired. Finally, we argue that the running time needed to uncover $S_2$ is polynomial in the input and $\frac{1}{\epsilon}$.

**Running time.** To establish a running time to compute $S_2$ requires computing $|\mathcal{U}|$, i.e., the number of states in our dynamic program. Also, we must show that the maximization in (15) can be computed efficiently. Going component by component, we see that

$$|\mathcal{U}| = O(|\mathcal{Q}| \cdot K^2 n \cdot |G_\delta|) = O((Kn)^{O(1)} \cdot \frac{|\mathcal{I}|}{\epsilon}).$$

Solving the maximization can be done by enumeration over all options, of which there are at most $O(Kn)$, and thus the running time to compute $S_2$ is indeed $\mathrm{poly}(|\mathcal{I}|, n, k, \frac{1}{\epsilon})$.

### C.4 Competing against $R_1(S^*)$

Recall that to allow for parallel reasoning in developing our approximation scheme, we assume that each assortment $S_{q,p}^*$ is non-empty. Note that, in order to ensure that $S_{q_t,1}^* \neq \emptyset$ for each $t \in [|\mathcal{Q}|]$, we need to assume that $q_1 > 1$, which also implies that that product 1 is contained within $S_{q_t,1}^*$. In the scenario where $q_1 = 1$, then $S_{q_t,1}^* = \emptyset$ (since every order set will have weight at least $w_{\max}$), and we can just ignore this first weight group within our dynamic program, i.e. this corner case can easily br handled. Moving forward, to compete against $R_1(S^*)$, we start by breaking down $R_1(S^*)$ into the contributions from product 1 (recall that $w_1 = w_{\max}$), and products $S_1^* \setminus \{1\}$. Along this line, note that

$$\begin{aligned} R_1(S^*) &= \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in \mathcal{K}_{q_t}} \frac{\lambda_k}{1 + w(\mathcal{O}_k(S^*))} \cdot \rho(\mathcal{O}_k(S^*) \cap S^*_{q_t,1}) \\ &= \underbrace{\sum_{t \in [|\mathcal{Q}|]} \sum_{k \in \mathcal{K}_{q_t}} \frac{\lambda_k}{1 + w(\mathcal{O}_k(S^*))} \cdot \rho_1}_{R_1^{\diamond}(S^*)} \\ &\qquad + \underbrace{\sum_{t \in [|\mathcal{Q}|]} \sum_{k \in \mathcal{K}_{q_t}} \frac{\lambda_k}{1 + w(\mathcal{O}_k(S^*))} \cdot \left(\rho(\mathcal{O}_k(S^*) \cap S^*_{q_t,1}) - \rho_1\right)}_{R_1^{\bullet}(S^*)} . \end{aligned}$$

In what follows, we show how to compete against these two values, noting that singling out product 1 is necessary for future accounting arguments when it comes to reasoning about order sets. Specifically, we will compete against $R_1^{\diamond}(S^*)$ by offering product 1 by itself, and we will compete again $R_1^{\bullet}(S^*)$ using a near-identical ADP framework to the one seen in Appendix C.3 to identify an assortment $S_1$ whose expected revenue can be made arbitrarily close to $R_1^{\bullet}(S^*)$. The primary difference between the approach presented in this section versus the last, is that the sub-assortments selected from each class that go into building $S_1$ are not necessarily singletons, as was the case when building $S_2$. As such, we will need a more nuanced approach to select the assortment from each weight group. which will rely on the solution to a specific knapsack problem.

**State space.** Again, for ease of exposition with regards to handling base cases in the dynamic program, we introduce a "dummy" customer type $K+1$ with $\lambda_{K+1} = f_{K+1} = 0$, and dummy product $n+1$ with $w_{n+1} = r_{n+1} = 0$. The individual components of the state space are given below, with the lone update from the previous section that we replace the previous offered products $j$ with an interval of product $[j_{\min}, j_{\max}]$

- The index of a weight group $t \in [|\mathcal{Q}| + 1]$.
- An interval of customer types $[k_{\min}, k_{\max}]$, such that $k_{\min}, k_{\max} \in [K+1]$.
- An interval of products $[j_{\min}, j_{\max}]$, such that $j_{\min}, j_{\max} \in [2, n+1]$. We do not consider product 1 within this approach, as it plays no role in competing again $R_1^{\bullet}(S^*)$.
- A rounded accumulated total $\rho$-value, which we restrict to the set of exponentially spaced grid points
$$G_\delta = \{\rho_{\min} \cdot (1+\delta)^y : y = 0, \ldots, \lceil \log_{1+\delta}(\frac{n\rho_{\max}}{\rho_{\min}}) \rceil\},$$
where we set $\delta = \frac{\epsilon}{Q}$, and $\rho_{\min} = \min_{i \in [n]} \rho_i$ and $\rho_{\max} = \max_{i \in [n]} \rho_i$.

We use

$$\mathcal{U} = \{(t, [k_{\min}, k_{\max}], [j_{\min}, j_{\max}], \rho) : \quad \begin{array}{c} t \in [|\mathcal{Q}| + 1] \\ k_{\min}, k_{\max} \in [K+1] : k_{\min} \le k_{\max} \end{array} ,$$

$$
\begin{aligned}
&j_{\min}, j_{\max} \in [2, n+1] : j_{\min} \leq j_{\max} \\
&\rho \in G_\delta\}
\end{aligned}
$$

to denote the universe of possible states. The exact interpretation of each state variable is explained in the paragraph where we formally define the value functions.

**Action space.** For state $U = (t, [k_{\min}, k_{\max}], [j_{\min}, j_{\max}], \rho) \in \mathcal{U}$, we define

$$
\begin{aligned}
\mathcal{F}(U) = \{(k'_{\max}, j') : \quad & k'_{\max} \in [K+1] : k'_{\max} \geq k_{\max} + 1 \\
& j'_{\max} \in [2, n+1] : j'_{\max} \geq j_{\max} + 1\}
\end{aligned}
$$

to denote the feasible set of actions. When action $(k'_{\max}, j'_{\max},) \in \mathcal{F}(U)$ in taken in state $U = (t, [k_{\min}, k_{\max}], [j_{\min}, j_{\max}], \rho) \in \mathcal{U}$, we transition to state

$$
U' = \left(t+1, [k_{\max}+1, k'_{\max}], [j_{\max}+1, j'_{\max}], G_\delta\left(\rho + \rho\left(\text{KNAP}\left(U\right)\right)\right)\right),
$$

where $\rho(\text{KNAP}(U))$ is defined next.

**Knapsack problem - (•).** Finally,we proceed to define the knapsack problem through which we actually make assortment decisions in each stage of our dynamic program. For this purpose, define the set of products

$$
N(U) = \{i \in [j_{\min}, j_{\max}] : w_i \geq W_t \cdot (e^{f_{k_{\min}}} - 1) + (1 - e^{-f_{k_{\min}}})\},
$$

where $W_t = (2^{q_t - 1} - 1) \cdot w_{\max}$. This set denotes a subset of products within the interval $[j_{\min}, j_{\max}]$, whose weights meet a specific lower bound. Note that if $i \in N(U)$, then we will have that

$$
w_i \geq W_t \cdot (e^{f_k} - 1) + (1 - e^{-f_k})
$$

for any $k \in [k_{\min}, k_{\max}]$ as this bound is decreasing in $k$. Consider the following knapsack problem, formulated as an integer program, over the set of products $N(U)$:

$$
\begin{aligned}
Z(U) = \max \quad & \sum_{i \in N(U)} \rho_i x_i \\
\text{s.t.} \quad & \sum_{i \in N(U)} w_i x_i \leq w_{\max} \cdot (2^{q_t - 1} - \max\{1, 2^{q_{t-1} - 1}\}) \\
& x_i \in \{0, 1\} \qquad \forall\, i \in N(U),
\end{aligned}
\tag{KNAP-(•)}
$$

recalling that we have defined $q_0 = -\infty$. As we will eventually show, the combination of the definition of $N(U)$ along with the weight constraint in KNAP-(•) will ensure that, within our dynamic program, we do not misstep when it comes to understanding the order sets of each customer type. Moving forward, we assume blackbox access to an FPTAS for KNAP-(•) (see, e.g., (Vazirani, 2013, Chap. 8))), which produces a feasible assortment $\text{KNAP}(U) \subseteq N(U)$ such that $\rho(\text{KNAP}(U)) \geq (1 - \epsilon) \cdot Z(U)$, for any $\epsilon > 0$.

**Value functions.** For state $U = (t, [k_{\min}, k_{\max}], [j_{\min}, j_{\max}], \rho) \in \mathcal{U}$, the value functions $V(U)$ will represent a lower bound on the optimal expected revenue when

- we only gain revenue from products in weight groups $\{t, t+1, \ldots, |\mathcal{Q}|\}$. Also, in the current state, we only
- potentially accrue revenue from classes $[k_{\min}, k_{\max}]$,
- and products within $[j_{\min}, j_{\max}]$.
- Finally, the total accumulated rounded $\rho$-value from the products added in previous stages is $\rho$.

We can recursively define the value functions as

$$\begin{aligned} V(U) \quad &= \quad \sum_{k \in [k_{\min}, k_{\max}]} \lambda_k \cdot \frac{\rho + \rho(\mathrm{KNAP}(U))}{1 + W_t} + \max_{(k'_{\max}, j'_{\max})} V\left(U'\right) \\ &s.t. \quad (k'_{\max}, j'_{\max}). \in \mathcal{F}(U) \end{aligned} \tag{16}$$

**The initial state.** Our dynamic program will start from the initial state $U^{(1)} \in \mathcal{U}$, defined as

$$U^{(1)} = (1, [k^{(1)}_{\min}, k^{(1)}_{\max}], [j^{(1)}_{\min}, j^{(1)}_{\max}], \underbrace{\rho^{(0)}}_{=0}),$$

where $k^{(1)}_{\min} = k^*_{q_1,\min}$ and $k^{(1)}_{\max} = k^*_{q_1,\max}$ and $j^{(1)}_{\min} = 2$ and $j^{(1)}_{\max} = \max S^*_{q_1,1}$, the latter of which can be recovered by enumerating over its $O(n)$ possible identities.

**Outputted assortment.** Starting from the just-defined initial state $U^{(1)}$, we can recursively define the sequence of states visited by the optimal policy. Namely, for $t \in [|\mathcal{Q}|]$, define the actions taken by the optimal policy in state $U^{(t)}$ as $(k^{(t+1)}_{\max}, j^{(t+1)}_{\max}, )$ and $\mathrm{KNAP}(U^{(t)})$ respectively. The former tuple is the argmax of (16) when in state $U^{(t)}$, and the latter assortment is the near-optimal knapsack solution, which we denote as $\mathrm{KNAP}^{(t)}$ as a shorthand. Accordingly, the optimal policy progresses over states $U^{(1)}, U^{(2)}, \ldots, U^{(|\mathcal{Q}|+1)}$, where for $t \in [2, |\mathcal{Q}|+1]$, we have that

$$U^{(t)} = \left(t, [k^{(t-1)}_{\max} + 1, k^{(t)}_{\max}], [j^{(t-1)}_{\max} + 1, j^{(t)}_{\max}], G_\delta\left(\rho^{(t-1)} + \rho\left(\mathrm{KNAP}^{(t)}\right)\right)\right).$$

The assortment that we output and proceed to analyze is $S_1 = \mathcal{P}(\biguplus_{t=1}^{|\mathcal{Q}|} \mathrm{KNAP}^{(t)})$, i.e., the pruned union of all the items selected in each near-optimal solution to KNAP-($\bullet$).

**Performance guarantee.** The following lemma shows that the expected revenue garnered by $S_1$ can be made to be arbitrarily close to $R_1(S^*)$

LEMMA C.4. *For any $\epsilon > 0$, we have that*

*(i)* $V(U^{(1)}) \geq (1-\epsilon)^2 \cdot R^\bullet_1(S^*)$

*(ii)* $\mathcal{R}(S_1) \geq V(U^{(1)})$,

We note that, since $\mathcal{R}(\{1\}) \geq R^\diamond_1(S^*)$, we get that $\max\{\mathcal{R}(\{1\}), \mathcal{R}(S_1)\} \geq (\frac{1}{2} - \epsilon) \cdot R_1(S^*)$.

**Running time.** To establish a running time to compute $S_1$ requires again computing $|\mathcal{U}|$, and also showing that the maximization in 16 can be computed efficiently. Going component by component, we see that

$$|\mathcal{U}| = O(|\mathcal{Q}| \cdot K^2 n^2 \cdot |G_\delta|) = O((Kn)^{O(1)} \cdot \frac{|\mathcal{I}|}{\epsilon}).$$

Solving the maximization can be be done be enumeration over all options, of which there are at most $O(Kn)$, and thus the running time to compute $S_1$ is indeed $\text{poly}(|\mathcal{I}|, n, k, \frac{1}{\epsilon})$. Of course, adding in the running time to solve the knapsack problems does not change the overall polynomial running time of our approach since we just need to solve this knapsack problem once for each state.

### C.5 Competing against $R_3(S^*)$

To compete against $R_3(S^*)$ requires our most nuanced approach, due to the fact that there may be distinct customer types $k, k' \in \mathcal{K}_q$ such that $\mathcal{O}_k(S^*) \cap S^*_{q,3} \neq \mathcal{O}_{k'}(S^*) \cap S^*_{q,3}$ (the same is not true when we look at sub-assortment 1 and 2, meaning that class $q$ customers all have the same overlap in order sets when it comes to $S^*_{q,1}$ and $S^*_{q,2}$). To handle this issue, we start by breaking down $R_3(S^*)$ into two separate contributions, and proceed to develop distinct ADP-based approaches to compete against each part. Along this line, note that

$$\begin{aligned}
R_3(S^*) &= \sum_{t\in[|\mathcal{Q}|]} \sum_{k\in\mathcal{K}_{q_t}} \frac{\lambda_k}{1+w(\mathcal{O}_k(S^*))} \cdot \rho(\mathcal{O}_k(S^*) \cap S^*_3) \\
&= \sum_{t\in[|\mathcal{Q}|]} \sum_{k\in\mathcal{K}_{q_t}} \frac{\lambda_k}{1+w(\mathcal{O}_k(S^*))} \cdot \left(\rho(\biguplus_{\tau\in[t-1]} S^*_{q_\tau,3}) + \rho(\mathcal{O}_k(S^*) \cap S^*_{q_t,3})\right) \\
&= \underbrace{\sum_{t\in[|\mathcal{Q}|]} \sum_{k\in\mathcal{K}_{q_t}} \frac{\lambda_k}{1+w(\mathcal{O}_k(S^*))} \cdot \rho(\biguplus_{\tau\in[t-1]} S^*_{q_\tau,3})}_{R_3^{\triangleleft}(S^*)} \\
&\qquad + \sum_{t\in[|\mathcal{Q}|]} \underbrace{\sum_{k\in\mathcal{K}_{q_t}} \frac{\lambda_k}{1+w(\mathcal{O}_k(S^*))} \cdot \rho(\mathcal{O}_k(S^*) \cap S^*_{q_t,3})}_{R^{\star}_{t,3}(S^*)},
\end{aligned}$$

where the the second equality follows by the fact that the order sets are nested by customer type index. We also define $R^{\star}_3(S^*) = \sum_{t\in[|\mathcal{Q}|]} R^{\star}_{t,3}(S^*)$. We use superscript $\triangleleft$ for the first term, since the total $\rho$-value comes from weight groups with index less than $t$ (i.e., to the left of $t$), and we use $\star$ for the second because the $\rho$-value comes form the weight group $t$ at-hand. In what follows, we show how to compete against these two values, noting that the final running time to recover the two assortments using our ADP-based approaches can be shown to polynomial in the input using similar reasoning as seen in previous sections.

### C.5.1 ADP for $R_3^{\triangleleft}(S^*)$

The approach we develop in this section very much mirrors the approach presented in the last section to compete again $R_1^{\bullet}(S^*)$. The primary difference will be the knapsack problem employed to make the decisions in each state, and the fact that we do not omit product 1 from the analysis.

**State space.** Again, for ease of exposition with regards to handling base cases in the dynamic program, we introduce a "dummy" customer type $K+1$ with $\lambda_{K+1} = f_{K+1} = 0$, and dummy product $n+1$ with $w_{n+1} = \infty$ and $r_{n+1} = 0$. The individual components of the state space are given below:

- The index of a weight group $t \in [|\mathcal{Q}|+1]$.
- An interval of customer types $[k_{\min}, k_{\max}]$, such that $k_{\min}, k_{\max} \in [K+1]$.
- An interval of products $[j_{\min}, j_{\max}]$, such that $j_{\min}, j_{\max} \in [n+1]$. Note that unlike the previous section, we include product 1 as candidates for these two components.
- A rounded accumulated total $\rho$-value, which we restrict to the set of exponentially spaced grid points
$$G_\delta = \{\rho_{\min} \cdot (1+\delta)^y : y = 0, \dots, \lceil \log_{1+\delta}(\frac{n\rho_{\max}}{\rho_{\min}})\rceil\},$$
where we set $\delta = \frac{\epsilon}{Q}$, and $\rho_{\min} = \min_{i\in[n]} \rho_i$ and $\rho_{\max} = \max_{i\in[n]} \rho_i$.

We use

$$\mathcal{U} = \left\{(t, [k_{\min}, k_{\max}], [j_{\min}, j_{\max}], \rho) : \begin{array}{c} t \in [|\mathcal{Q}|+1] \\ k_{\min}, k_{\max} \in [K+1] : k_{\min} \le k_{\max} \\ j_{\min}, j_{\max} \in [n+1] : j_{\min} \le j_{\max} \\ \rho \in G_\delta \end{array}\right\},$$

to denote the universe of possible states.

**Action space.** For state $U = (t, [k_{\min}, k_{\max}], [j_{\min}, j_{\max}], \rho) \in \mathcal{U}$, we define

$$\mathcal{F}(U) = \left\{(k'_{\max}, j') : \begin{array}{l} k'_{\max} \in [K+1] : k'_{\max} \ge k_{\max} + 1 \\ j'_{\max} \in [n+1] : j'_{\max} \ge j_{\max} + 1 \end{array}\right\}$$

to denote the feasible set of actions. When action $(k'_{\max}, j'_{\max},) \in \mathcal{F}(U)$ in taken in state $U = (t, [k_{\min}, k_{\max}], [j_{\min}, j_{\max}], \rho) \in \mathcal{U}$, we transition to state

$$U' = \left(t+1, [k_{\max}+1, k'_{\max}], [j_{\max}+1, j'_{\max}], G_\delta\left(\rho + \rho\left(\text{KNAP}\left(U\right)\right)\right)\right),$$

where $\rho(\text{KNAP}(U))$ is defined next.

**Knapsack problem - ($\triangleleft$).** Throughout this section, for each $t \in [|\mathcal{Q}|]$, we set $W_t = w_{\max} \cdot 2^{q_t - 1}$, and define

$$N(U) = \{i \in [j_{\min}, j_{\max}] : w_i \geq (W_{t+1} - w_{\max}) \cdot (e^{f_{k_{\max}+1}} - 1) + (1 - e^{-f_{k_{\max}+1}})\}.$$

From here, consider the following knapsack problem, formulated as an integer program, over this set of products:

$$\begin{aligned} Z(U) = \max \quad & \sum_{i \in N(U)} \rho_i x_i \\ \text{s.t.} \quad & \sum_{i \in N(U)} w_i x_i \leq W_t \\ & x_i \in \{0,1\} \qquad \forall i \in N(U), \end{aligned} \tag{KNAP-($\triangleleft$)}$$

where we again assume blackbox access to an FPTAS for KNAP-($\triangleleft$), which produces a feasible assortment $\text{KNAP}(U) \subseteq N(U)$ such that $\rho(\text{KNAP}(U)) \geq (1-\epsilon) \cdot Z(U)$, for any $\epsilon > 0$.

**Value functions.** These value functions will mirror those in (16), as seen below:

$$\begin{aligned} V(U) \quad = \quad & \sum_{k \in [k_{\min}, k_{\max}]} \lambda_k \cdot \frac{\rho}{1 + W_t} + \max_{(k'_{\max}, j'_{\max})} V\left(U'\right) \\ s.t. \quad & (k'_{\max}, j'_{\max}), \in \mathcal{F}(U), \end{aligned} \tag{17}$$

with the lone difference being that we do not accrue the $\rho$-value from the current state's knapsack solution in the immediate reward, and hence the make-up of this approximate knapsack solution only influences the subsequent state $U'$.

**The initial state.** Our dynamic program will start from the initial state $U^{(1)} \in \mathcal{U}$, defined as

$$U^{(1)} = (1, [k^{(1)}_{\min}, k^{(1)}_{\max}], [j^{(1)}_{\min}, j^{(1)}_{\max}], \underbrace{\rho^{(0)}}_{=0}),$$

where $k^{(1)}_{\min} = k^*_{q1,\min}$ and $k^{(1)}_{\max} = k^*_{q1,\max}$ and $j^{(1)}_{\min} = 1$ and $j^{(1)}_{\max} = \max S^*_{q1,3}$, the latter of which can be recovered by enumerating over their $O(n)$ possible identities.

**Outputted assortment.** The assortment that we output and proceed to analyze is $S^{\triangleleft}_3 = \mathcal{P}(\biguplus_{t=1}^{|\mathcal{Q}|} \text{KNAP}^{(t)})$, i.e., the pruned union of all the items selected in each near-optimal solution to KNAP-($\triangleleft$), whose shorthand is $\text{KNAP}^{(t)} = \text{KNAP}(U^{(t)})$.

**Performance guarantee.** The following lemma shows that the expected revenue garnered by $S^{\triangleleft}_3$ is within a constant factor of $R^{\triangleleft}_3(S^*)$.

LEMMA C.5. *For any $\epsilon > 0$, we have that*

*(i)* $V(U^{(1)}) \geq (1-\epsilon)^2 \cdot R^{\triangleleft}_3(S^*)$

*(ii)* $\mathcal{R}(S_3^{\lhd}) \geq V(U^{(1)})$,

We note that, together, the two properties immediately imply that $\mathcal{R}(S_3^{\lhd}) \geq (1-\epsilon)^2 \cdot R_3^{\lhd}(S^*)$.

### C.5.2 ADP for $R_3^{\star}(S^*)$

Our final ADP-based approach is presented below, which has some slight differences with the previous approaches.

**State space.** Again, for ease of exposition with regards to handling base cases in the dynamic program, we introduce a "dummy" customer type $K+1$ with $\lambda_{K+1} = f_{K+1} = 0$, and dummy product $n+1$ with $w_{n+1} = \infty$ and $r_{n+1} = 0$. The individual components of the state space are given below:

- The index of a weight group $t \in [|\mathcal{Q}|+1]$.
- An interval of customer types $[k_{\min}, k_{\max}]$, such that $k_{\min}, k_{\max} \in [K+1]$.
- An interval of products $[j_{\min}, j_{\max}]$, such that $j_{\min}, j_{\max} \in [n+1]$. Note that unlike the previous section, we include product 1 as candidates for these two components.

We use

$$\mathcal{U} = \{(t, [k_{\min}, k_{\max}], [j_{\min}, j_{\max}]) : \begin{array}{c} t \in [|\mathcal{Q}|+1] \\ k_{\min}, k_{\max} \in [K+1] : k_{\min} \leq k_{\max} \\ j_{\min}, j_{\max} \in [n+1] : j_{\min} \leq j_{\max}\} \end{array},$$

to denote the universe of possible states.

**Action space.** For state $U = (t, [k_{\min}, k_{\max}], [j_{\min}, j_{\max}]) \in \mathcal{U}$, we define

$$\mathcal{F}(U) = \{(k'_{\max}, j') : \begin{array}{l} k'_{\max} \in [K+1] : k'_{\max} \geq k_{\max} + 1 \\ j'_{\max} \in [n+1] : j'_{\max} \geq j_{\max} + 1\} \end{array}$$

to denote the feasible set of actions. When action $(k'_{\max}, j'_{\max},) \in \mathcal{F}(U)$ in taken in state $U = (t, [k_{\min}, k_{\max}], [j_{\min}, j_{\max}]) \in \mathcal{U}$, we transition to state

$$U' = \left(t+1, [k_{\max}+1, k'_{\max}], [j_{\max}+1, j'_{\max}]\right).$$

**Knapsack problem - ($\star$).** For our final knapsack problem, in each state $U \in \mathcal{U}$ we define the following set of customer types for each product $i \in [j_{\min}, j_{\max}]$:

$$\mathcal{K}_i(U) = \{k \in [k_{\min}, k_{\max}] : w_i \geq W_t \cdot (e^{f_k} - 1) + (1 - e^{-f_k})\},$$

where $W_t = w_{\max} \cdot 2^{q_t - 1}$. With regards to $\mathcal{K}_i(U)$ note that (i) it will consist of some subset of the highest indexed types within the interval $[k_{\min}, k_{\max}]$, and (ii) for $i, i' \in [j_{\min}, j_{\max}]$ such that $i < i'$, we have that $\mathcal{K}_{i'}(U) \subseteq \mathcal{K}_i(U)$. From here, for each $i \in [j_{\min}, j_{\max}]$ we define

$$p_i(U) = \sum_{k \in \mathcal{K}_i(U)} \frac{\lambda_k}{1 + W_t} \cdot \rho_i,$$

and we consider the following knapsack problem, formulated as an integer program:

$$\begin{aligned} Z(U) = \max \quad & \sum_{i \in [j_{\min}, j_{\max}]} p_i(U) x_i \\ \text{s.t.} \quad & \sum_{i \in [j_{\min}, j_{\max}]} w_i x_i \le \min\{w_{\max}, \frac{W_t}{2}\} \\ & x_i \in \{0,1\} \qquad\qquad \forall\, i \in [j_{\min}, j_{\max}], \end{aligned} \tag{KNAP-($\star$)}$$

where we again assume blackbox access to an FPTAS for KNAP-($\star$), which produces a feasible assortment $\text{KNAP}(U) \subseteq [j_{\min}, j_{\max}]$ such that $\sum_{i \in \text{KNAP}(U)} p_i(U) \ge (1-\epsilon) \cdot Z(U)$, for any $\epsilon > 0$.

**Value functions.** These value functions will mirror those in (16), as seen below:

$$\begin{aligned} V(U) \quad &= \quad \sum_{i \in \text{KNAP}(U)} p_i(U) + \max_{(k'_{\max}, j'_{\max})} V(U') \\ & s.t. \quad (k'_{\max}, j'_{\max}), \in \mathcal{F}(U), \end{aligned} \tag{18}$$

with the lone difference being that we do not accrue the $\rho$-value from the current state's knapsack solution in the immediate reward, and hence the make-up of this approximate knapsack solution only influences the subsequent state $U'$.

**The initial state.** Our dynamic program will start from the initial state $U^{(1)} \in \mathcal{U}$, defined as

$$U^{(1)} = (1, [k^{(1)}_{\min}, k^{(1)}_{\max}], [j^{(1)}_{\min}, j^{(1)}_{\max}]),$$

where $k^{(1)}_{\min} = k^*_{q_1,\min}$ and $k^{(1)}_{\max} = k^*_{q_1,\max}$ and $j^{(1)}_{\min} = 1$ and $j^{(1)}_{\max} = \max S^*_{q_1,3}$.

**Outputted assortment.** The assortment that we output and proceed to analyze is $S^\star_3 = \mathcal{P}(\biguplus_{t=1}^{|Q|} \text{KNAP}^{(t)})$, i.e., the pruned union of all the items selected in each near-optimal solution to KNAP-($\star$), whose shorthand is $\text{KNAP}^{(t)} = \text{KNAP}(U^{(t)})$.

**Performance guarantee.** The following lemma shows that the expected revenue garnered by $S^\star_3$ is within a constant factor of $R^\star_3(S^*)$.

Lemma C.6. *For any $\epsilon > 0$, we have that*

*(i)* $V(U^{(1)}) \ge \frac{1}{3}(1-\epsilon) \cdot R^\star_3(S^*)$

*(ii)* $\mathcal{R}(S^\star_3) \ge V(U^{(1)})$,

Finally,w e note that, that $\max\{\mathcal{R}(S^\triangleleft_3), \mathcal{R}(S^\star_3)\} \ge (\frac{1}{4} - \epsilon) \cdot R_3(S^*)$, based on Lemmas C.5 and C.6

## D Theorem 4.6 - Proofs

This section contains the missing proofs from Appendix C.

### D.1 Proof of Claim C.1

We prove the three properties sequentially. Each proof uses basic fundamental properties of how we create the weight groups.

**Proof of (i).** Recalling that $S^*_{q_t,1} = S^*_{q_t} \cap [i^*_{q_t}]$, for $t = 1$, we have that

$$\begin{aligned} w(S^*_{q_t,1}) &= w(\mathcal{O}_{k^*_{q_1,\min}}(S^*) \cap [i^*_{q_t}]) \\ &\leq w_{\max} \cdot 2^{q_1 - 1} \\ &= w_{\max} \cdot 2^{q_1-1} - w_{\max} \cdot 2^{q_0 - 1}, \end{aligned}$$

where the last inequality follows because we have defined $q_0 = -\infty$. For $t > 1$,we have

$$\begin{aligned} w(S^*_{q_t,1}) &= w(\mathcal{O}_{k^*_{q_t,\min}}(S^*) \cap [i^*_{q_t}]) - w(\mathcal{O}_{k^*_{q_{t-1},\max}}(S^*)) \\ &\leq w_{\max} \cdot 2^{q_t - 1} - w(\mathcal{O}_{k^*_{q_{t-1},\max}}(S^*)) \\ &= w_{\max} \cdot 2^{q_t-1} - w_{\max} \cdot 2^{q_{t-1} - 1}, \end{aligned}$$

as desired. The first inequality follows by definition of $i^*_{q_t}$, and the second follows by definition of weight group $q_{t-1}$, i.e. the total order weight of any customer class in this group must be at least $w_{\max} \cdot 2^{q_{t-1}-1}$.

**Proof of (ii).** This second property follows directly from the fact $|S^*_{q,2}| \leq 1$, and that all products within $S^*$ have weights that do not exceed $w_{\max}$.

**Proof of (iii).** If $S^*_{q_t,3} = \emptyset$, then the claim holds trivially. Otherwise, it must be the case that $j^*_{q_t} \in S^*_{q_t}$, and so

$$\begin{aligned} w(S^*_{q_t,3}) &= w(\mathcal{O}_{k^*_{q_t,\max}}(S^*)) - w(S^* \cap [j^*_{q_t}]) \\ &\leq w_{\max} \cdot 2^{q_t} - w_{\max} \cdot 2^{q_t - 1} \\ &= w_{\max} \cdot 2^{q_t-1}, \end{aligned}$$

where the lone inequality uses the fact the definition of the weight group $\mathcal{K}_{q_t}$, along with the fact that the definition of $i^*_{q_t}$ together with the assumption that $j^*_{q_t} \in S^*_{q_t}$ implies that $w(S^* \cap [j^*_{q_t}]) \geq w_{\max} \cdot 2^{q_t-1}$.

### D.2 Proof of Lemma C.2

Before jumping into the proof of the lemma, we first present a well-known structural property of the MNL model's revenue function, before moving to a handful of intermediate claims, which we then exploit to establish the lemma.

**MNL facts.** We start with a well-known fact of the traditional MNL revenue function, which states that the updated revenue that results from adding product $i \notin S$ to assortment $S$ is some convex combination of the expected revenue of $S$ and the revenue $r_i$.

FACT D.1. *For any assortment $S \subset [n]$ and product $i \notin S$, we have that*

$$\mathcal{R}_{\mathrm{MNL}}(S \cup \{i\}) = \alpha_S \cdot \mathcal{R}_{\mathrm{MNL}}(S) + (1 - \alpha_s) \cdot r_i,$$

*where $\alpha_S = \frac{w(S)}{1+w(S)+w_i} \in [0, 1]$.*

There are two natural corollaries of this fact, which we present below. Both corollaries follow by simple iterative applications of Fact D.1.

COROLLARY D.2. *For any two disjoint assortments $S, S' \subseteq [n]$ such that $\mathcal{R}_{\mathrm{MNL}}(S) \leq \min_{i \in S'} r_i$, we have that*

$$\mathcal{R}_{\mathrm{MNL}}(S \cup S') \geq \mathcal{R}_{\mathrm{MNL}}(S).$$

COROLLARY D.3. *Consider two disjoint assortments $S, S' \subseteq [n]$ and $\alpha = \frac{w(S')}{1+w(S \cup S')} \in [0, 1]$. If $\mathcal{R}_{\mathrm{MNL}}(S) \geq \max_{i \in S'} r_i$, we have that*

$$\mathcal{R}_{\mathrm{MNL}}(S \cup S') \leq \alpha \cdot \mathcal{R}_{\mathrm{MNL}}(S) + (1 - \alpha) \cdot \max_{i \in S'} r_i.$$

*On the other hand, if $\mathcal{R}_{\mathrm{MNL}}(S) \leq \min_{i \in S'} r_i$, then*

$$\mathcal{R}_{\mathrm{MNL}}(S \cup S') \geq \alpha \cdot \mathcal{R}_{\mathrm{MNL}}(S) + (1 - \alpha) \cdot \min_{i \in S'} r_i.$$

$\alpha = \frac{w(S')}{1+w(S \cup S')} \in [0, 1]$.

**Intermediate claims.** Here, we present a number of intermediate claims regarding the pruning process, which together will allow for a very succinct proof of the lemma. The proofs of each claim are presented at the end of this section.

CLAIM D.4. *For each step $t \in [n]$, we have that $\mathcal{P}(S) \cap [t] = P_t \cap [t]$.*

CLAIM D.5. *For each step $t \in [n]$, we have that $\mathcal{R}_{\mathrm{MNL}}(P_t \cap [t]) \geq \mathcal{R}_{\mathrm{MNL}}(P_{t-1} \cap [t-1])$.*

CLAIM D.6. *For steps $t, \tau \in [n]$ that satisfy $\tau \leq t$, we have that*

$$\mathcal{R}_{\mathrm{MNL}}(P_\tau \cap [t]) \geq \mathcal{R}_{\mathrm{MNL}}(P_{\tau-1} \cap [t]).$$

CLAIM D.7. *For each step $t \in [n]$, we have that*

$$\min_{i \in \mathcal{P}(S) \cap [t+1,n]} r_i \geq \mathcal{R}_{\mathrm{MNL}}(P_t \cap [t]).$$

**Proof of the lemma.** To prove the lemma, we establish that

$$\mathcal{R}_k(S \cap [\ell]) \leq \mathcal{R}_{\mathrm{MNL}}(\mathcal{P}(S) \cap [\ell_k(S)]) \leq \mathcal{R}_k(\mathcal{P}(S)). \tag{19}$$

To see the first inequality in (19), observe that

$$\begin{aligned}\mathcal{R}_k(S \cap [\ell]) &= \mathcal{R}_{\text{MNL}}(S \cap [\ell]) \\ &= \mathcal{R}_{\text{MNL}}(P_1 \cap [\ell]) \\ &\leq \mathcal{R}_{\text{MNL}}(P_\ell \cap [\ell]) \\ &\leq \mathcal{R}_{\text{MNL}}(P_{\ell_k(S)} \cap [\ell_k(S)]) \\ &= \mathcal{R}_{\text{MNL}}(\mathcal{P}(S) \cap [\ell_k(S)]),\end{aligned}$$

where the second equality follows because $P_1 = S$, the first inequality follows by Claim D.6, and the second inequality by Claim D.5. The last equality uses Claim D.4.

To establish the second inequality in (19), observe that since $\mathcal{P}(S) \subseteq S$, we must have that $\ell_k(\mathcal{P}(S)) \geq \ell_k(S)$. Define $\mathcal{O}_{\text{rest}} = \mathcal{O}_k(\mathcal{P}(S)) \cap [\ell_k(S)+1, n]$ to be the product indexed larger than $\ell_k(S)$ that ordered by customer $k$ after we prune, noting that $\mathcal{O}_k(\mathcal{P}(S)) = (\mathcal{P}(S) \cap [\ell_k(S)]) \cup \mathcal{O}_{\text{rest}}$. From here, we have that

$$\begin{aligned}\mathcal{R}_k(\mathcal{P}(S)) &= \mathcal{R}_{\text{MNL}}(\mathcal{O}_k(\mathcal{P}(S))) \\ &= \mathcal{R}_{\text{MNL}}\left((\mathcal{P}(S) \cap [\ell_k(S)]) \cup \mathcal{O}_{\text{rest}}\right) \\ &\geq \alpha \cdot \mathcal{R}_{\text{MNL}}\left((\mathcal{P}(S) \cap [\ell_k(S)])\right) + (1-\alpha) \cdot \min_{i \in \mathcal{O}_{\text{rest}}} r_i \\ &\geq \alpha \cdot \mathcal{R}_{\text{MNL}}\left(\mathcal{P}(S) \cap [\ell_k(S)]\right) + (1-\alpha) \cdot \min_{i \in \mathcal{P}(S) \cap [\ell_k(S)+1, n]} r_i \\ &\geq \alpha \cdot \mathcal{R}_{\text{MNL}}\left(\mathcal{P}(S) \cap [\ell_k(S)]\right) + (1-\alpha) \cdot \mathcal{R}_{\text{MNL}}\left(P_{\ell_k(S)} \cap [\ell_k(S)]\right) \\ &= \alpha \cdot \mathcal{R}_{\text{MNL}}\left(\mathcal{P}(S) \cap [\ell_k(S)]\right) + (1-\alpha) \cdot \mathcal{R}_{\text{MNL}}\left(\mathcal{P}(S) \cap [\ell_k(S)]\right) \\ &= \mathcal{R}_{\text{MNL}}\left(\mathcal{P}(S) \cap [\ell_k(S)]\right),\end{aligned}$$

as desired. The first inequality uses Corollary D.3, the second uses the fact that $\mathcal{O}_{\text{rest}} \subseteq \mathcal{P}(S) \cap [\ell_k(S)+1, n]$, and the last inequality uses Claim D.7. The second to last inequality uses Claim D.4.

**Proof of Claim D.4.** We show the result by induction over $t$. To establish the base case of $t = 1$, we consider two cases based on whether or not $1 \in P_1$.

- Case 1 - $1 \notin P_1$: In this case, it is straightforward to see that we have that $\mathcal{P}(S) \cap [1] = \emptyset$ and $P_1 \cap [t] = \emptyset$.
- Case 2 - $1 \in P_1$: Here, since $D_1 \subseteq [2, n]$, this first product will never be deleted in the pruning process. As a result, we have that $\mathcal{P}(S) \cap [1] = \{1\} = P_1 \cap [1]$.

We now show the result for general step $t > 1$ assuming the result holds for steps $1, \ldots, t-1$. To do so, we again consider two cases based on inclusion of product $t$ within $P_t$.

- Case 1 - $t \notin P_t$: In this case, we have that

$$P_t \cap [t] = P_{t-1} \cap [t-1] = \mathcal{P}(S) \cap [t-1] = \mathcal{P}(S) \cap [t],$$

where the first inequality exploits the fact that if $t \notin P_t$, then we have that $P_t = P_{t-1}$ in the pruning process, and the second equality uses the induction hypothesis. The last equality following from the observation that the pruning process never adds products, so that if $t \notin P_t$, then it must be that $t \notin \mathcal{P}(S)$.

- Case 2 - $t \in P_t$: In this second case, we have that

$$P_t \cap [t] = (P_{t-1} \cap [t-1]) \cup \{t\} = (\mathcal{P}(S) \cap [t-1]) \cup \{t\} = \mathcal{P}(S) \cap [t],$$

where the second equality uses the induction hypothesis, and the last equality follows because $t \in \mathcal{P}(S)$. To see this last inclusion, observe that $\mathcal{P}(S) = P_t \setminus (\biguplus_{\tau \in [t,n-1]} D_\tau)$. Furthermore, since $D_\tau \subseteq [t+1, n]$ for any $\tau \in [t, n-1]$, and since $t \in P_t$, we must have that $t \in \mathcal{P}(S)$.

**Proof of Claim D.5.** To prove the claim, we again consider two cases based on whether or not $t \in P_t$.

- Case 1 - $t \notin P_t$: In this case, we have that $P_t \cap [t] = P_{t-1} \cap [t-1]$ and so the claim holds trivially.

- Case 2 - $t \in P_t$: Here, if $t \in P_t$, it must have survived the stage $t-1$ pruning and so $r_t \geq \mathcal{R}_{\text{MNL}}(P_{t-1} \cap [t-1])$. As such, we have that

$$\begin{aligned}
\mathcal{R}_{\text{MNL}}(P_t \cap [t]) &= \mathcal{R}_{\text{MNL}}((P_{t-1} \cap [t-1]) \cup \{t\}) \\
&= \alpha \cdot \mathcal{R}_{\text{MNL}}(P_{t-1} \cap [t-1]) + (1-\alpha) \cdot r_t \\
&\geq \mathcal{R}_{\text{MNL}}(P_{t-1} \cap [t-1]),
\end{aligned}$$

where the second equality used Fact D.1 and the inequality follows since $r_t \geq \mathcal{R}_{\text{MNL}}(P_{t-1} \cap [t-1])$.

**Proof of Claim D.6.** Since $P_\tau = P_{\tau-1} \setminus D_{\tau-1}$, and because $D_{\tau-1} \subseteq [\tau, n]$ we have that

$$P_\tau \cap [t] = (P_{\tau-1} \cap [\tau-1]) \cup \left( \underbrace{(P_{\tau-1} \cap [\tau, t]) \setminus D_{\tau-1}}_{\text{products in } [\tau,t] \text{ not deleted in stage } \tau-1 \text{ pruning}} \right) \tag{20}$$

$$P_{\tau-1} \cap [t] = (P_\tau \cap [t]) \cup ( \underbrace{D_{\tau-1} \cap [\tau, t]}_{\text{products in } [\tau,t] \text{ deleted in stage } \tau-1 \text{ pruning}} ) \tag{21}$$

By definition of $D_{\tau-1}$, if $i \in (P_{\tau-1} \cap [\tau, t]) \setminus D_{\tau-1}$, then its revenue must satisfy $r_i \geq \mathcal{R}_{\text{MNL}}(P_{\tau-1} \cap [t-1])$, and thus combining (20) with Corollary D.2, we get that

$$\mathcal{R}_{\text{MNL}}(P_\tau \cap [t]) \geq \mathcal{R}_{\text{MNL}}(P_{\tau-1} \cap [\tau-1]). \tag{22}$$

From here, now exploiting (21), we get that

$$\mathcal{R}_{\text{MNL}}(P_{\tau-1} \cap [t]) = \mathcal{R}_{\text{MNL}}((P_\tau \cap [t]) \cup (D_{\tau-1} \cap [\tau, t]))$$

$$\begin{aligned} &\le \alpha \cdot \mathcal{R}_{\text{MNL}}(P_\tau \cap [t]) + (1-\alpha) \cdot \mathcal{R}_{\text{MNL}}(P_{\tau-1} \cap [\tau - 1]) \\ &\le \mathcal{R}_{\text{MNL}}(P_\tau \cap [t]), \end{aligned}$$

where the first inequality combines the definition of $D_{\tau-1}$ with Corollary D.3 and the last inequality uses (22).

**Proof of Claim D.7.** Assume by way of contradiction that there exists $i \in \mathcal{P}(S) \cap [t+1, n]$ such that $r_i < \mathcal{R}_{\text{MNL}}(P_t \cap [t])$. In this case, since we have that $\mathcal{P}(S) \subseteq P_t$, it must be that $i \in P_t \cap [t+1, n]$. In turn, this would then imply that $i \in D_t$, and thus would be pruned in step $t$, thus contradicting the fact that $i \in \mathcal{P}(S)$.

### D.3 Proof of Lemma C.3 - property (i)

We prove the result by constructing a feasible sequence of states that leads to an accumulated reward of at least $(1-\epsilon) \cdot R_2(S^*)$. This sequence will effectively mirror the decisions made by $S_2^* = \biguplus_{t \in [|\mathcal{Q}|]} j_{q_t}^*$. Specifically, for each $t \in [|\mathcal{Q}|]$, we take actions $(\hat{k}_{\max}^{(t+1)}, \hat{j}^{(t)})$, where

$$\hat{j}^{(t)} = j_{q_t}^*,$$

and

$$\hat{k}_{\max}^{(t+1)} = \begin{cases} k_{q_{t+1},\max}^*, & \text{if } t < |\mathcal{Q}| \\ K+1, & \text{if } t = |\mathcal{Q}|. \end{cases}$$

Starting from state $\hat{U}^{(1)} = U^{(1)}$, this sequence of actions will lead to progressing over states $\hat{U}^{(1)}, \hat{U}^{(2)}, \ldots, \hat{U}^{(|\mathcal{Q}|+1)}$, where for $t \in [2, |\mathcal{Q}|]$, we have that

$$\hat{U}^{(t)} = (t, [k_{q_t,\min}^*, k_{q_t,\max}^*], j_{q_{t-1}}^*, \hat{\rho}^{(t)}),$$

and for the terminal state, we have

$$\hat{U}^{(|\mathcal{Q}|+1)} = (|Q|+1, [K+1, K+1], j_{q_{|\mathcal{Q}|}}^*, \hat{\rho}^{(|\mathcal{Q}|+1)}),$$

where $\hat{\rho}^{(t)} = G_\delta(\hat{\rho}^{(t-1)} + \rho_{j_{q_{t-1}}^*})$. We first show that this sequence of state/action pairs is feasible, and then proceed to argue that it accrues sufficient reward to establish property (i).

**Feasibility.** Here, we will show that $\hat{U}^{(t)} \in \mathcal{U}$ for each $t \in [|\mathcal{Q}|+1]$, which requires showing that $(\hat{k}_{\max}^{(t+1)}, \hat{j}^{(t)}) \in \mathcal{F}(\hat{U}^{(t)})$ for each $t \in [|\mathcal{Q}|]$. To do so, we consider arbitrary $t \in [|\mathcal{Q}|]$, and sequentially argue the feasibility of each of the two decisions.

- $\hat{k}_{\max}^{(t+1)}$: Feasibility in this case boils down to showing that $\hat{k}_{\max}^{(t+1)} > \hat{k}_{\max}^{(t)}$. If $t = |\mathcal{Q}|$, we have that $\hat{k}_{\max}^{(t+1)} = K+1 > k_{q_{|\mathcal{Q}|},\max}^* = \hat{k}_{\max}^{(t)}$, and hence the inequality holds. To see that the inequality holds when $t < |\mathcal{Q}|$, note that $\hat{k}_{\max}^{(t+1)} = k_{q_{t+1},\max}^* > k_{q_t,\max}^* = \hat{k}_{\max}^{(t)}$, where the strict inequality is a result of the definition of the weight groups.

- $\hat{\jmath}^{(t)}$: In this case, we need to show that $\hat{\jmath}^{(t)} > \hat{\jmath}^{(t-1)}$. To see this, note that $\hat{\jmath}^{(t)} = j^*_{q_t} > j^*_{q_{t-1}} = \hat{\jmath}^{(t-1)}$, where strict inequality again follows by definition of the weight classes.

**Accumulated reward.** The total reward from following the the above-described sequence of actions is

$$\hat{V} = \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in [k^*_{q_t,\min}, k^*_{q_t,\max}]} \lambda_k \cdot \frac{\hat{\rho}^{(t)} + \rho_{\hat{\jmath}^{(t)}}}{1 + t w_{\max}}.$$

We proceed to show that $\hat{V} \geq (1-\epsilon) \cdot R_2(S^*)$, which implies property (i), since the feasibility argument above gives that $V(U^{(1)}) \geq \hat{V}$. Along this line, we have that

$$\begin{aligned} \hat{V} &= \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in \mathcal{K}_{q_t}} \lambda_k \cdot \frac{\hat{\rho}^{(t)} + \rho_{\hat{\jmath}^{(t)}}}{1 + t w_{\max}} \\ &\geq \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in \mathcal{K}_{q_t}} \lambda_k \cdot \frac{(1-\epsilon) \cdot \sum_{\tau \in [t]} \rho_{j^*_{q_t}}}{1 + t w_{\max}} \end{aligned} \tag{23}$$

where the equality following from the observation that $\mathcal{K}_{q_t} = [k^*_{q_t,\min}, k^*_{q_t,\max}]$. The inequality can be shown via a straightforward inductive argument over $t \in [|\mathcal{Q}|]$, which establishes that

$$\hat{\rho}^{(t)} \geq (1-\delta)^{t-1} \cdot \sum_{\tau \in [t-1]} \rho_{j^*_{q_t}}$$

for every $t \in [|\mathcal{Q}|]$, which implies that

$$\hat{\rho}^{(t)} \geq (1 - \frac{\epsilon}{|\mathcal{Q}|})^{|\mathcal{Q}|} \cdot \sum_{\tau \in [t-1]} \rho_{j^*_{q_t}} \geq (1-\epsilon) \cdot \sum_{\tau \in [t-1]} \rho_{j^*_{q_t}}$$

Continuing from here, we have

$$\begin{aligned} (23) &\geq \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in \mathcal{K}_{q_t}} \lambda_k \cdot \frac{(1-\epsilon) \cdot \sum_{\tau \in [t]} \rho_{j^*_{q_t}}}{1 + 2^{q_t - 1} \cdot w_{\max}} \\ &\geq \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in \mathcal{K}_{q_t}} \lambda_k \cdot \frac{(1-\epsilon) \cdot \sum_{\tau \in [t]} \rho_{j^*_{q_t}}}{1 + w(\mathcal{O}_k(S^*))} \\ &= (1-\epsilon) \cdot R_2(S^*), \end{aligned}$$

as desired. The first inequality follows because $q_t \geq t$, and thus $t \leq 2^{q_t - 1}$. The second inequality uses the definition of the weight group $q_t$, for which the order weight of all customer types is at least $2^{q_t-1} \cdot w_{\max}$ by construction.

### D.4 Proof of Lemma C.3 - property (ii)

**Intermediate claims.** We begin with a simple intermediate claim, whose proof appear at the end of this section, and for which we start by delevoping additional notation. For each $t \in [|\mathcal{Q}|]$, define $A_{t,2} = \biguplus_{\tau \in [t]} \{j^{(t)}\}$ to be the assortment we have selected after the first $t$ stages of the dynamic program, where we use the shorthand $A_2 = \biguplus_{\tau \in [|\mathcal{Q}|]} \{j^{(\tau)}\}$ for the final assortment.

Informally speaking, the first claim establishes that each of these products will be ordered by the customer types within $[k^{(t)}_{\min}, k^{(t)}_{\max}]$. This claim will allow us to directly exploit the affects of pruning.

Claim D.8. *For any $t \in [|\mathcal{Q}|]$ and $k \in [k^{(t)}_{\min}, k^{(t)}_{\max}]$, we have that $A_{t,2} \subseteq \mathcal{O}_k(A_2)$.*

**Proof of property (ii).** With the above two intermediate claims in-hand, we set out to show that $\mathcal{R}(S_2) \geq V(U^{(1)})$. We have

$$
\begin{aligned}
\mathcal{R}(S_2) &= \sum_{t\in[|\mathcal{Q}|]} \sum_{k\in\mathcal{K}_{q_t}} \lambda_k \cdot \mathcal{R}_k(\mathcal{P}(A_2)) \\
&\geq \sum_{t\in[|\mathcal{Q}|]} \sum_{k\in\mathcal{K}_{q_t}} \lambda_k \cdot \mathcal{R}_k(A_2 \cap [t]) \\
&= \sum_{t\in[|\mathcal{Q}|]} \sum_{k\in\mathcal{K}_{q_t}} \lambda_k \cdot \mathcal{R}_k(A_{t,2}) \\
&= \sum_{t\in[|\mathcal{Q}|]} \sum_{k\in\mathcal{K}_{q_t}} \lambda_k \cdot \frac{\rho(A_{t,2})}{1+w(A_{t,2})} \\
&\geq \sum_{t\in[|\mathcal{Q}|]} \sum_{k\in\mathcal{K}_{q_t}} \lambda_k \cdot \frac{\rho(A_{t,2})}{1+tw_{\max}} \\
&\geq \sum_{t\in[|\mathcal{Q}|]} \sum_{k\in[k^{(t)}_{\min},k^{(t)}_{\min}]} \lambda_k \cdot \frac{\rho(A_{t,2})}{1+tw_{\max}} \\
&\geq V(U^{(1)}).
\end{aligned}
$$

where the first inequality uses Claim D.8, which gives that $t \leq \ell_k(A_2)$, to employ Lemma C.2. The second inequality uses the fact that $w(A_{t,2}) \leq tw_{\max}$ since it includes at most $t$ products, and the third inequality follows from observing that the customer type intervals $[k^{(t)}_{\min}, k^{(t)}_{\max}]$ are disjoint and might not partition $[K]$ The last inequality follows because the total accumulated $\rho$ values within our dynamic program are rounded down, so we have that $\rho(A_{t,2}) \geq \rho^{(t)} + \rho_{j^{(t)}}$ for any $t \in [|\mathcal{Q}|]$.

**Proof of Claim D.8.** To prove the result, we consider arbitrary $\tau \in [t]$, and we show that product $j^{(\tau)}$ must be ordered by all customer types within $[k^{(t)}_{\min}, k^{(t)}_{\max}]$. Specifically, we have that

$$
\begin{aligned}
w_{j^{(\tau)}} \geq w_{j^{(t)}} &\geq (t-1)\cdot w_{\max}\cdot(e^{f_{k_{\min}}}-1)+(1-e^{-f_{k_{\min}}}) \\
&\geq w(A_2\cap[j^{(t)}-1])\cdot(e^{f_{k_{\min}}}-1)+(1-e^{-f_{k_{\min}}})
\end{aligned}
$$

where the second inequality follows by the feasibility of $j^{(t)}$, and the third inequality follows since product $j^{(t)}$ is the $t$-th product added to $S_2$, and all products have weight no more than $w_{\max}$. Note that by Lemma 2.2, this implies that $j^{(\tau)} \in \mathcal{O}_k(A_2)$ for any $k \in [k^{(t)}_{\min}, k^{(t)}_{\max}]$, since the last expression in the chain of inequalities is decreasing in $k$.

### D.5 Proof of Lemma C.4 - property (i)

We prove the result by constructing a feasible sequence of states that leads to an accumulated reward of at least $(1-\epsilon)^2 \cdot R_1(S^*)$. This sequence will effectively mirror the decisions made by $S_1^* \setminus \{1\}$ within each weight group. Specifically, for each $t \in [|\mathcal{Q}|]$, we take actions $(\hat{k}_{\max}^{(t+1)}, \hat{j}_{\max}^{(t+1)})$, where

$$\hat{j}_{\max}^{(t+1)} = \begin{cases} \max S^*_{q_{t+1},1}, & \text{if } t < |\mathcal{Q}| \\ n+1, & \text{if } t = |\mathcal{Q}|, \end{cases}$$

and

$$\hat{k}_{\max}^{(t+1)} = \begin{cases} k^*_{q_{t+1},\max}, & \text{if } t < |\mathcal{Q}| \\ K+1, & \text{if } t = |\mathcal{Q}|. \end{cases}$$

Starting from state $\hat{U}^{(1)} = U^{(1)}$, this sequence of actions will lead to progressing over states $\hat{U}^{(1)}, \hat{U}^{(2)}, \ldots, \hat{U}^{(|\mathcal{Q}|+1)}$, where for $t \in [2, |\mathcal{Q}|]$, we have that

$$\hat{U}^{(t)} = (t, [k^*_{q_t,\min}, k^*_{q_t,\max}], [\max S^*_{q_{t-1},1} + 1, \max S^*_{q_t,1}], \hat{\rho}^{(t)}),$$

and for the terminal state we have

$$\hat{U}^{(|Q|+1)} = (|Q|+1, [K+1, K+1], [n+1, n+1], \hat{\rho}^{(|Q|+1)}),$$

where $\hat{\rho}^{(t)} = G_\delta(\hat{\rho}^{(t-1)} + \rho(\text{KNAP}^{(t)}))$. We first show that this sequence of state/action pairs is feasible, and then proceed to argue that it accrues sufficient reward to establish property (i).

**Feasibility.** Here, we will show that $\hat{U}^{(t)} \in \mathcal{U}$ for each $t \in [|\mathcal{Q}|+1]$, which requires showing that $(\hat{k}_{\max}^{(t+1)}, \hat{j}_{\max}^{(t+1)}) \in \mathcal{F}(\hat{U}^{(t)})$ for each $t \in [|\mathcal{Q}|]$. To do so, we consider arbitrary $t \in [|\mathcal{Q}|]$, and sequentially argue the feasibility of each of the two decisions.

- $\hat{k}_{\max}^{(t+1)}$: Follows using identical arguments as those made in the previous section.
- $\hat{j}_{\max}^{(t+1)}$: In this case, we just need to show that $\hat{j}_{\max}^{(t+1)} > \hat{j}_{\max}^{(t)}$, which follows trivially from their respective definitions given above.

**Accumulated reward.** For ease of notation moving forward, we slightly abuse notation, using $\text{KNAP}^{(t)}$ to denote the blackbox knapsack FPTAS output in state $\hat{U}^{(t)}$. The total reward from following the the above-described sequence of actions is

$$\hat{V} = \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in [k^*_{q_t,\min}, k^*_{q_t,\max}]} \lambda_k \cdot \frac{\hat{\rho}^{(t)} + \rho(\text{KNAP}^{(t)}))}{1 + W_t}.$$

We proceed to show that $\hat{V} \geq (1-\epsilon)^2 \cdot R_2(S^*)$, which implies property (i), since the feasibility argument above implies that $V(U^{(1)}) \geq \hat{V}$. Along this line, we will exploit the following intermediate claim, whose proof can be found at the end of this section.

Claim D.9. *For every $t \in [|\mathcal{Q}|]$, we have $\rho(\text{KNAP}^{(t)}) \geq (1-\epsilon) \cdot \rho(S^*_{q_t,1} \setminus \{1\})$.*

Exploiting the above claim, along with the fact that $\mathcal{K}_{q_t} = [k^*_{q_t,\min}, k^*_{q_t,\max}]$, we get that

$$\begin{aligned}
\hat{V} &\geq \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in \mathcal{K}_{q_t}} \lambda_k \cdot \frac{(1-\epsilon)^2 \cdot \sum_{\tau \in [t]} \rho(S^*_{q_\tau,1} \setminus \{1\})}{1 + W_t} \\
&\geq \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in \mathcal{K}_{q_t}} \lambda_k \cdot \frac{(1-\epsilon)^2 \cdot \sum_{\tau \in [t]} \rho(S^*_{q_\tau,1} \setminus \{1\})}{1 + w(\mathcal{O}_k(S^*))} \\
&= (1-\epsilon)^2 \cdot R_1(S^*),
\end{aligned}$$

as desired. The extra factor of $(1-\epsilon)$ in the first inequality comes out of the fact a simple inductive argument over $t$ can be again used to show that

$$\hat{\rho}^{(t)} \geq (1-\delta)^{t-1} \cdot \sum_{\tau \in [t-1]} \rho(\mathrm{KNAP}^{(\tau)})$$

which implies that

$$\hat{\rho}^{(t)} \geq (1 - \frac{\epsilon}{|\mathcal{Q}|})^{|\mathcal{Q}|} \cdot \sum_{\tau \in [t-1]} \rho(\mathrm{KNAP}^{(\tau)}) \geq (1-\epsilon) \cdot \sum_{\tau \in [t-1]} \rho(\mathrm{KNAP}^{(\tau)}).$$

The second inequality uses the definition of the weight group $q_t$, for which the order weight of all customer types is at least $2^{q_t - 1} \cdot w_{\max} \geq W_t$.

**Proof of Claim D.9.** Proving this claim boils down to showing that, for any $t \in [|\mathcal{Q}|]$, the assortment $S^*_{q_t,1} \setminus \{1\}$ is feasible to KNAP-($\bullet$) when instantiated at state $\hat{U}^{(t)}$. By property (i) of Claim C.1, for any $t \in [2, |\mathcal{Q}|]$, we know that

$$w(S^*_{q_t,1}) = w(S^*_{q_t,1} \setminus \{1\}) \leq w_{\max} \cdot (2^{q_t-1} - 2^{q_{t-1}-1}) \leq w_{\max} \cdot (2^{q_t-1} - \max\{1, 2^{q_{t-1}-1}\}),$$

where the first equality follows because $1 \in S^*_{q_1,1}$, and the second inequality follows since $q_t > 1$. As a result, the upper bound on the total weight is satisfied for for $t \in [2, |\mathcal{Q}|]$. On the other hand, for $t = 1$, we have

$$w(S^*_{q_1,1} \setminus \{1\}) \leq w_{\max} \cdot (2^{q_t-1}) - w_{\max} = w_{\max} \cdot (2^{q_t-1} - \max\{1, 2^{q_{t-1}-1}\}),$$

and hence the knapsack constraint is again satisfied. From here, the final step is showing that $S^*_{q_t,1} \setminus \{1\} \subseteq N(\hat{U}^{(t)})$. To do so, we more generally show (i) $S^*_{q_t,1} \subseteq [\max S^*_{q_{t-1},1} + 1, \max S^*_{q_t,1}]$, and (ii) for each $i \in S^*_{q_t,1}$, we have that $w_i \geq W_t \cdot (e^{f_{k^*_{q_t,\min}}-1}) + (1 - e^{-f_{k^*_{q_t,\min}}})$.

- Proof of (i): Noting that $\max S^*_{q_{t-1},1} + 1 \leq \min S^*_{q_t,1}$ by construction, the condition then holds trivially.
- Proof of (ii): Recalling that

$$i^*_{q_t} = \mathrm{argmax}\{i \in S^*_q : w(\mathcal{O}_{k^*_{q_t,\min}}(S^*) \cap [i]) < w_{\max} \cdot 2^{q_t-1}\}$$

and that $S^*_{q_t,1} = S^*_{q_t} \cap [i^*_{q_t}]$, which implies that $i^*_{q_t}$ is the lowest weight product within $S^*_{q_t,1}$.

As such, for any $i \in S^*_{q_t,1}$, we have that

$$\begin{aligned} w_i \geq w_{i^*_{q_t}} &\geq w(\mathcal{O}_{k^*_{q_t,\min}}(S^*) \cap [i_{q_t} - 1]) \cdot (e^{f_{k^*_{q_t,\min}}-1}) + (1 - e^{-f_{k^*_{q_t,\min}}}) \\ &\geq W_t \cdot (e^{f_{k^*_{q_t,\min}}-1}) + (1 - e^{-f_{k^*_{q_t,\min}}}), \end{aligned}$$

where the first inequality follows since type $k^*_{q_t,\min}$ orders product $i^*_{q_t}$, and the last inequality follows by the definition of the product $i^*_{q_t}$, which allows us to argue that $w(\mathcal{O}_{k^*_{q_t,\min}}(S^*) \cap [i_{q_t} - 1]) \geq w_{\max} \cdot 2^{q_t-1} - w_{i^*_{q_t}} \geq w_{\max} \cdot (2^{q_t-1} - 1) = W_t$.

### D.6 Proof of Lemma C.4 - property (ii)

**Intermediate claims.** We again begin with a helpful intermediate claim, whose proofs appear at the end of this section. For each $t \in [|\mathcal{Q}|]$, define $A_{t,1} = \biguplus_{\tau \in [t]} \mathrm{KNAP}^{(\tau)}$ to be the cumulative assortment we have selected after the first $t$ stages of the dynamic program and $A_1 = \biguplus_{\tau \in [|\mathcal{Q}|]} \mathrm{KNAP}^{(\tau)}$.

Claim D.10. *For any $t \in [|\mathcal{Q}|]$ and $k \in [k^{(t)}_{\min}, k^{(t)}_{\max}]$, we have that $A_{t,1} \subseteq \mathcal{O}_k(A_1)$.*

**Proof of property (ii).** With the above intermediate claim in-hand, we set out to show that $\mathcal{R}(S_1) \geq V(U^{(1)})$. Note that

$$\begin{aligned} \mathcal{R}(S_1) &= \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in \mathcal{K}_{q_t}} \lambda_k \cdot \mathcal{R}_k(\mathcal{P}(A_1)) \\ &\geq \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in \mathcal{K}_{q_t}} \lambda_k \cdot \mathcal{R}_k(A_1 \cap [\max A_{t,1}]) \\ &= \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in \mathcal{K}_{q_t}} \lambda_k \cdot \mathcal{R}_k(A_{t,1}) \\ &= \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in \mathcal{K}_{q_t}} \lambda_k \cdot \frac{\rho(A_{t,1})}{1 + w(A_{t,1})} \\ &\geq \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in \mathcal{K}_{q_t}} \lambda_k \cdot \frac{\rho(A_{t,1})}{1 + W_t} \\ &\geq \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in [k^{(t)}_{\min}, k^{(t)}_{\max}]} \lambda_k \cdot \frac{\rho(A_{t,1})}{1 + W_t} \\ &\geq V(U^{(1)}). \end{aligned}$$

where the first inequality uses Claim D.10 to invoke Lemma C.2. The second inequality uses the fact that

$$w(A_{t,1}) \leq \sum_{\tau \in [t]} w_{\max} \cdot (2^{q_\tau - 1} - \max\{1, 2^{q_{\tau-1}-1}\}) \leq w_{\max} \cdot (2^{q_t-1} - 1) = W_t.$$

The last inequality follows because the total accumulated $\rho$ values within our dynamic program are rounded down, so we have that $\rho(A_{t,1}) \geq \rho^{(t)} + \rho(\mathrm{KNAP}^{(t)})$ for any $t \in [|\mathcal{Q}|]$.

**Proof of Claim D.10.** To prove the result, we consider arbitrary $\tau \in [t]$, and we show that product $i_{\max,t} = \max A_{t,1}$ (the lowest weight product within $A_{t,1}$) must be ordered by all customer types within $[k^{(t)}_{\min}, k^{(t)}_{\max}]$ For thi purpose, we have that

$$\begin{aligned} w_{i_{\max,t}} &\geq W_t \cdot (e^{f_{k^{(t)}_{\min}}} - 1) + (1 - e^{-f_{k^{(t)}_{\min}}}) \\ &\geq w(A_1 \cap [i_{\max,t} - 1]) \cdot (e^{f_{k^{(t)}_{\min}}} - 1) + (1 - e^{-f_{k^{(t)}_{\min}}}) \end{aligned}$$

where the first inequality follows because $i_{\max,t} \in N(U^{(t)})$, and the second inequality follows since $w(A_1 \cap [i_{\max,t} - 1]) \leq w(A_{t,1}) \leq \sum_{\tau \in [t]} w_{\max} \cdot (2^{q_t - 1} - \max\{1, 2^{q_{t-1}-1}\}) \leq w_{\max} \cdot (2^{q_t - 1} - 1) = W_t$, repeating the argument regarding the total weight of $w(A_{t,1})$ seen above. Since the last expression in the chain of inequalities above is decreasing in $k$, we get that

$$w_{i_{\max,t}} \geq w(A_1 \cap [i_{\max,t} - 1]) \cdot (e^{f_k} - 1) + (1 - e^{-f_k})$$

for any customer type $k \in [k^{(t)}_{\min}, k^{(t)}_{\max}]$, as desired.

### D.7 Proof of Lemma C.5 - property (i)

We prove the result by constructing a feasible sequence of states that leads to an accumulated reward of at least $(1-\epsilon)^2 \cdot R^{\triangleleft}_3(S^*)$. To do so, for each $t \in [|\mathcal{Q}|]$, we take actions $(\hat{k}^{(t+1)}_{\max}, \hat{j}^{(t+1)}_{\max})$, where

$$\hat{j}^{(t+1)}_{\max} = \begin{cases} \max S^*_{q_{t+1},3}, & \text{if } t < |\mathcal{Q}| \\ n+1, & \text{if } t = |\mathcal{Q}|, \end{cases}$$

and

$$\hat{k}^{(t+1)}_{\max} = \begin{cases} k^*_{q_{t+1},\max}, & \text{if } t < |\mathcal{Q}| \\ K+1, & \text{if } t = |\mathcal{Q}|. \end{cases}$$

Starting from state $\hat{U}^{(1)} = U^{(1)}$, this sequence of actions will lead to progressing over states $\hat{U}^{(1)}, \hat{U}^{(2)}, \ldots, \hat{U}^{(|\mathcal{Q}|+1)}$, where for $t \in [2, |\mathcal{Q}|]$, we have that

$$\hat{U}^{(t)} = (t, [k^*_{q_t,\min}, k^*_{q_t,\max}], [\max S^*_{q_{t-1},3} + 1, \max S^*_{q_t,3}], \hat{\rho}^{(t)}),$$

and the terminal state, we have

$$\hat{U}^{(|Q|+1)} = (|Q|+1, [K+1, K+1], [n+1, n+1], \hat{\rho}^{(|\mathcal{Q}|+1)}),$$

where $\hat{\rho}^{(t)} = G_\delta(\hat{\rho}^{(t-1)} + \rho_{j^*_{q_{t-1}}})$. We first show that this sequence of state/action pairs is feasible, and then proceed to argue that it accrues sufficient reward to establish property (i).

**Feasibility.** Follows using identical arguments to those seen for the proof of property (i) in Lemma C.4.

**Accumulated reward.** For ease of notation moving forward, we slightly abuse notation, using $\text{KNAP}^{(t)}$ to denote the blackbox knapsack FPTAS output in state $\hat{U}^{(t)}$. The total reward from following the the above-described sequence of actions is

$$\hat{V} = \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in [k^*_{q_t,\min}, k^*_{q_t,\max}]} \lambda_k \cdot \frac{\rho(\text{KNAP}^{(t)}))}{1 + W_t}.$$

We proceed to show that $\hat{V} \geq \frac{1}{3}(1-\epsilon)^2 \cdot R_3^{\lhd}(S^*)$, which implies property (i), since the feasibility argument above implies that $V(U^{(1)}) \geq \hat{V}$. Along this line, we will exploit the following intermediate claim, whose proof can be found at the end of this section.

Claim D.11. *For every $t \in [|\mathcal{Q}|]$, we have $\rho(\text{KNAP}^{(t)}) \geq (1-\epsilon) \cdot \rho(S^*_{q_t,3})$.*

Exploiting the above claim, along with the fact that $\mathcal{K}_{q_t} = [k^*_{q_t,\min}, k^*_{q_t,\max}]$, we get that

$$\begin{aligned}
\hat{V} &\geq \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in \mathcal{K}_{q_t}} \lambda_k \cdot \frac{(1-\epsilon)^2 \cdot \sum_{\tau \in [t-1]} \rho(S^*_{q_\tau,3})}{1 + W_t} \\
&\geq \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in \mathcal{K}_{q_t}} \lambda_k \cdot \frac{(1-\epsilon)^2 \cdot \sum_{\tau \in [t-1]} \rho(S^*_{q_\tau,3})}{1 + w(\mathcal{O}_k(S^*))} \\
&= (1-\epsilon)^2 \cdot R_3^{\lhd}(S^*),
\end{aligned}$$

as desired. The extra factor of $(1-\epsilon)$ in the first inequality comes out from the same inductive reasoning as in the last section.The second inequality uses the definition of the weight group $q_t$, for which the order weight of all customer types is at least $2^{q_t - 1} \cdot w_{\max} = W_t$.

**Proof of Claim D.11.** We can repeat the arguments in Claim D.9 to show that $S^*_{q_t,3}$ is feasible. Specifically, to show that the weight constraint in KNAP-($\lhd$) is satisfied, we appeal to property (iii) of Claim C.1. Next, to show that for each $i \in S^*_{q_t,3}$, we have that $i \in N(U^{(t)})$ just requires noting that $j^*_{q_{t+1}} > i$ is ordered by customer types $\mathcal{K}_{q_{t+1}}$ by construction. As a result,

$$\begin{aligned}
w_i &\geq w_{j^*_{q_{t+1}}} \geq w(S^* \cap [j^*_{q_{t+1}} - 1]) \cdot (e^{f_{k^*_{q_{t+1},\min}}} - 1) + (1 - e^{-f_{k^*_{q_{t+1},\min}}}) \\
&\geq (W_{t+1} - w_{\max}) \cdot (e^{f_{k^*_{q_{t+1},\min}}} - 1) + (1 - e^{-f_{k^*_{q_{t+1},\min}}}) \\
&= (W_{t+1} - w_{\max}) \cdot (e^{f_{k^*_{q_t,\max}+1}} - 1) + (1 - e^{-f_{k^*_{q_t,\max}+1}}),
\end{aligned}$$

where the third inequality follows by definition of the product $j^*_{q_{t+1}}$.

### D.8 Proof of Lemma C.5 - property (ii)

**Intermediate claims.** We again begin with a helpful intermediate claim, whose proofs appear at the end of this section. For each $t \in [|\mathcal{Q}|]$, define $A^{\lhd}_{t,3} = \biguplus_{\tau \in [t]} \text{KNAP}^{(\tau)}$ to be the cumulative assortment we have selected after the first $t$ stages of the dynamic program and $A^{\lhd}_3 = \biguplus_{\tau \in [|\mathcal{Q}|]} \text{KNAP}^{(\tau)}$

Claim D.12. *For any $t \in [|\mathcal{Q}|]$ and $k \in [k^{(t)}_{\min}, k^{(t)}_{\max}]$, we have that $A^{\lhd}_{t-1,3} \subseteq \mathcal{O}_k(A^{\lhd}_3)$.*

**Proof of property (ii).** With the above intermediate claim in-hand, we set out to show that $\mathcal{R}(S_3^{\triangleleft}) \geq V(U^{(1)})$. Observe that

$$\begin{aligned}
\mathcal{R}(S_3^{\triangleleft}) &= \sum_{t\in[|\mathcal{Q}|]} \sum_{k\in\mathcal{K}_{q_t}} \lambda_k \cdot \mathcal{R}_k(\mathcal{P}(A_3^{\triangleleft})) \\
&\geq \sum_{t\in[|\mathcal{Q}|]} \sum_{k\in\mathcal{K}_{q_t}} \lambda_k \cdot \mathcal{R}_k(A_3^{\triangleleft} \cap [\max A_{t-1,3}^{\triangleleft}]) \\
&= \sum_{t\in[|\mathcal{Q}|]} \sum_{k\in\mathcal{K}_{q_t}} \lambda_k \cdot \mathcal{R}_k(A_{t-1,3}^{\triangleleft}) \\
&= \sum_{t\in[|\mathcal{Q}|]} \sum_{k\in\mathcal{K}_{q_t}} \lambda_k \cdot \frac{\rho(A_{t-1,3}^{\triangleleft})}{1+w(A_{t-1,3}^{\triangleleft})} \\
&\geq \sum_{t\in[|\mathcal{Q}|]} \sum_{k\in\mathcal{K}_{q_t}} \lambda_k \cdot \frac{\rho(A_{t-1,3}^{\triangleleft})}{1+W_t} \\
&\geq V(U^{(1)}).
\end{aligned}$$

The last inequality follows because the total accumulated $\rho$-value is rounded down in our dynamic program. The first equality uses Claim D.12 combined with Lemma C.2. The second inequality uses the fact that

$$w(A_{t-1,3}^{\triangleleft}) \leq \sum_{\tau\in[t-1]} W_\tau = \sum_{\tau\in[t-1]} w_{\max} \cdot 2^{q_\tau - 1} = W_t - w_{\max},$$

where the first inequality directly above uses the weight constraint in KNAP-($\triangleleft$).

**Proof of Claim D.12.** To prove the result, we consider arbitrary $\tau \in [t-1]$, and we show that product $i_{\max,t-1} = \max A_{t-1,3}^{\triangleleft}$ is ordered by any customer type $k \in [k_{\min}^{(t)}, k_{\max}^{(t)}]$. We have that

$$\begin{aligned}
w_{i_{\max,t-1}} &\geq (W_t - w_{\max}) \cdot (e^{f_{k_{\min}^{(t)}}} - 1) + (1 - e^{-f_{k_{\min}^{(t)}}}) \\
&\geq w(A_{t-1,3}^{\triangleleft}) \cdot (e^{f_{k_{\min}^{(t)}}} - 1) + (1 - e^{-f_{k_{\min}^{(t)}}}) \\
&\geq w(A_3^{\triangleleft} \cap [i_{\max,t-1} - 1]) \cdot (e^{f_{k_{\min}^{(t)}}} - 1) + (1 - e^{-f_{k_{\min}^{(t)}}}),
\end{aligned}$$

where the first inequality follows because $i_{\max,t-1} \in N(U^{(t-1)})$ and because $k_{\max}^{(t-1)} + 1 = k_{\min}^{(t)}$, and the second inequality follows since $w(A_{t,3}^{\triangleleft}) \leq W_t$, repeating the argument regarding the total weight of $w(A_{t-1,3}^{\triangleleft})$ seen above.

### D.9 Proof of Lemma C.6 - property (i)

We prove the result by constructing a feasible sequence of states that leads to an accumulated reward of at least $\frac{1}{3}(1-\epsilon)^2 \cdot R_3^{\star}(S^*)$. To do so, for each $t \in [|\mathcal{Q}|]$, we take actions $(\hat{k}_{\max}^{(t+1)}, \hat{j}_{\max}^{(t+1)})$, where

$$\hat{j}_{\max}^{(t+1)} = \begin{cases} \max S_{q_{t+1},3}^*, & \text{if } t < |\mathcal{Q}| \\ n+1, & \text{if } t = |\mathcal{Q}|, \end{cases}$$

and

$$\hat{k}_{\max}^{(t+1)} = \begin{cases} k^*_{q_{t+1},\max}, & \text{if } t < |\mathcal{Q}| \\ K+1, & \text{if } t = |\mathcal{Q}|. \end{cases}$$

Starting from state $\hat{U}^{(1)} = U^{(1)}$, this sequence of actions will lead to progressing over states $\hat{U}^{(1)}, \hat{U}^{(2)}, \ldots, \hat{U}^{(|\mathcal{Q}|+1)}$, where for $t \in [2, |\mathcal{Q}|]$, we have that

$$\hat{U}^{(t)} = (t, [k^*_{q_t,\min}, k^*_{q_t,\max}], [\max S^*_{q_{t-1},3} + 1, \max S^*_{q_t,3}]),$$

and the terminal state, we have

$$\hat{U}^{(|Q|+1)} = (|Q| + 1, [K+1, K+1], [n+1, n+1]).$$

We first show that this sequence of state/action pairs is feasible, and then proceed to argue that it accrues sufficient reward to establish property (i).

**Feasibility.** Follows using identical arguments to those seen for the proof of property (i) in Lemma C.4.

**Accumulated reward.** For ease of notation moving forward, we slightly abuse notation, using $\text{KNAP}^{(t)}$ to denote the blackbox knapsack FPTAS output in state $\hat{U}^{(t)}$. The total reward from following the the above-described sequence of actions is

$$\hat{V} = \sum_{t \in [|\mathcal{Q}|]} \sum_{i \in \text{KNAP}^{(t)}} p_i(U^{(t)})$$

We proceed to show that $\hat{V} \geq \frac{1}{3}(1-\epsilon)^2 \cdot R_3^{\lhd}(S^*)$, which implies property (i), since the feasibility argument above implies that $V(U^{(1)}) \geq \hat{V}$. Along this line, we will exploit the following intermediate claim, whose proof can be found at the end of this section.

Claim D.13. *For every $t \in [|\mathcal{Q}|]$, we have* $\sum_{i \in \text{KNAP}^{(t)}} p_i(U^{(t)}) \geq \frac{1}{3}(1-\epsilon) \cdot R^{\star}_{t,3}(S^*)$.

Exploiting the above claim, along with the fact that $\mathcal{K}_{q_t} = [k^*_{q_t,\min}, k^*_{q_t,\max}]$, we get that

$$\hat{V} \quad \geq \quad \frac{1}{3}(1-\epsilon) \cdot R_3^{\star}(S^*),$$

as desired.

**Proof of Claim D.13.** We can repeat the arguments in Claim D.9 to show that $S^*_{q_t,3}$ would be feasible to KNAP-($\star$) if the total weight allowed was set to $W_t$ (in lieu of $\max\{w_{\max} \frac{W_t}{2}\}$), since property (iii) of Claim C.1 provides this exact upper bound on the weight of each assortment $S^*_{q_t,3}$. As such, if we can argue that by halving the size of the knapsack, its optimal objective degrades by no more than one third, we will get that

$$\sum_{i \in \text{KNAP}^{(t)}} p_i(U^{(t)}) \quad \geq \quad \frac{1}{3} \cdot \sum_{i \in S^*_{q_t,3}} p_i(U^{(t)})$$

$$
\begin{aligned}
&= \frac{1}{3} \cdot \sum_{i \in S^*_{q_t,3}} \sum_{k \in \mathcal{K}_i(U^{(t)})} \frac{\lambda_k}{1+W_t} \cdot \rho_i \\
&= \frac{1}{3} \cdot \sum_{k \in \mathcal{K}_{q_t}} \frac{\lambda_k}{1+W_t} \cdot \sum_{\substack{i \in S^*_{q_t,3}: \\ k \in \mathcal{K}_i(U^{(t)})}} \rho_i \\
&\geq \frac{1}{3} \cdot \sum_{k \in \mathcal{K}_{q_t}} \frac{\lambda_k}{1+W_t} \cdot \rho(\mathcal{O}_k(S^*) \cap S^*_{q_t,3}) \\
&\geq \frac{1}{3} \cdot \sum_{k \in \mathcal{K}_{q_t}} \frac{\lambda_k}{1+w(\mathcal{O}_k(S^*))} \cdot \rho(\mathcal{O}_k(S^*) \cap S^*_{q_t,3}) \\
&= \frac{1}{3}(1-\epsilon) \cdot R^\star_{t,3}(S^*).
\end{aligned}
$$

The second inequality follows from observing that if $i \in \mathcal{O}_k(S^*) \cap S^*_{q_t,3}$, then $w_i \geq W_t \cdot (e^{f_k} - 1) + (1 - e^{-f_k})$ by definition of weight group $q_t$. As result, we have that $k \in \mathcal{K}_i(U^{(t)})$. The final inequality follows again by definition of the weights groups, which implies that for any $k \in \mathcal{K}_{q_t}$, we have that $w(\mathcal{O}_k(S^*)) \geq W_t$.

To establish the first inequality above, we construct a feasible solution to KNAP-($\star$) from $S^*_{q_t,3}$ that garners an objective of at least $\frac{1}{3} \cdot \sum_{i \in S^*_{q_t,3}} p_i(U^{(t)})$ and whose weight is no more than $\max\{w_{\max}, \frac{W_t}{2}\}$. We start with the case where $q_t > 1$, and so $\max\{w_{\max}, \frac{W_t}{2}\} = \frac{W_t}{2}$. For ease of exposition, we re-index the products so that $S^*_{q_t,3} = \{1, 2, \ldots, |S^*_{q_t,3}|\}$. We then partition these products into the following three sets using the index $i^* = \max\{i \in S^*_{q_t,3} : w([i]) < \frac{W_t}{2}\}$, as follows:

$$
S^{(1)}_{q_t,3} = [i^*], \quad S^{(2)}_{q_t,3} = i^* + 1, \quad S^{(3)}_{q_t,3} = [i^* + 2, |S^*_{q_t,3}|].
$$

Clearly,

$$
\max\{\sum_{i \in S^{(1)}_{q_t,3}} p_i(U^{(t)}), \sum_{i \in S^{(2)}_{q_t,3}} p_i(U^{(t)}), \sum_{i \in S^{(3)}_{q_t,3}} p_i(U^{(t)})\} \geq \frac{1}{3} \cdot \sum_{i \in S^*_{q_t,3}} p_i(U^{(t)}),
$$

since these three assortments partition $S^*_{q_t,3}$. Consequently, all that is left to show is feasibility with respect to the weight constraint. By construction, the total weight of both $S^{(1)}_{q_t,3}$ and $S^{(3)}_{q_t,3}$ does not exceed $\frac{W_t}{2}$, since property (iii) of Claim C.1 establishes that $w(S^*_{q_t,3}) \leq W_t$. It remains to show that any singleton, i.e. $S^{(2)}_{q_t,3}$, must also be feasible with respect to the weight constraint. Since $q_t > 1$, we have that $\frac{W_t}{2} \geq w_{\max}$, and hence all singletons are feasible. For the case where $q_t = 1$, the total weight allowed in KNAP-($\star$) is $w_{\max}$, hence $S^*_{1,3}$ is feasible by property (iii) of Claim C.1, i.e., the above partition is not necessary.

### D.10 Proof of Lemma C.6 - property (ii)

**Intermediate claims.** We again begin with a helpful intermediate claim, whose proofs appear at the end of this section. For each $t \in [|\mathcal{Q}|]$, define $A^\star_{t,3} = \biguplus_{\tau \in [t]} \text{KNAP}^{(\tau)}$ to be the cumulative assortment we have selected after the first $t$ stages of the dynamic program and

$A_3^\star = \biguplus_{\tau \in [|\mathcal{Q}|]} \mathrm{KNAP}^{(\tau)}$. Finally, for each $t \in [|\mathcal{Q}|]$ and $k \in [k_{\min}^{(t)}, k_{\max}^{(t)}]$, define

$$\ell_{k,3}^{(t)} = \begin{cases} \max\{\ell \in [j_{\min}^{(t)}, j_{\max}^{(t)}] : k \in \mathcal{K}_\ell(U^{(t)})\}, & \text{if } k \in \mathcal{K}_{j_{\min}^{(t)}}(U^{(t)}) \\ j_{\min}^{(t)} - 1, & \text{otherwise,} \end{cases}$$

which is proxy for the last product ordered by customer type $k$ under $A_3^\star$.

Claim D.14. *For any $t \in [|\mathcal{Q}|]$ and $k \in [k_{\min}^{(t)}, k_{\max}^{(t)}]$, we have that $A_{t,3}^\star \cap [\ell_{k,3}^{(t)}] \subseteq \mathcal{O}_k(A_3^\star)$.*

**Proof of property (ii).** With the above intermediate claim in-hand, we set out to show that $\mathcal{R}(S_3^\star) \geq V(U^{(1)})$., and note that

$$\begin{aligned}
\mathcal{R}(S_3^\star) &= \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in \mathcal{K}_{q_t}} \lambda_k \cdot \mathcal{R}_k(\mathcal{P}(A_3^\star)) \\
&\geq \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in \mathcal{K}_{q_t}} \lambda_k \cdot \mathcal{R}_k(A_3^\star \cap [\ell_{k,3}^{(t)}]) \\
&\geq \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in \mathcal{K}_{q_t}} \lambda_k \cdot \frac{\rho(A_{t,3}^\star \cap [\ell_{k,3}^{(t)}])}{1 + w(A_{t,3})} \\
&\geq \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in \mathcal{K}_{q_t}} \lambda_k \cdot \frac{\rho(\mathrm{KNAP}^{(t)} \cap [\ell_{k,3}^{(t)}])}{1 + w(A_{t,3})} \\
&\geq \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in \mathcal{K}_{q_t}} \lambda_k \cdot \frac{\rho(\mathrm{KNAP}^{(t)} \cap [\ell_{k,3}^{(t)}])}{1 + W_t} \\
&\geq \sum_{t \in [|\mathcal{Q}|]} \sum_{k \in [k_{\min}^{(t)}, k_{\max}^{(t)}]} \lambda_k \cdot \frac{\rho(\mathrm{KNAP}^{(t)} \cap [\ell_{k,3}^{(t)}])}{1 + W_t} \\
&= V(U^{(1)}).
\end{aligned}$$

The first equality uses Claim D.12 combined with Lemma C.2. The third inequality drops the accumulated $\rho$-value for $\tau < t$, and the third inequality uses the fact that

$$w(A_{t,3}^\star) \leq \sum_{\tau \in [t]} \max\{w_{\max}, \frac{W_t}{2}\} \leq \sum_{\tau \in [q_t]} \max\{w_{\max}, w_{\max} \cdot 2^{\tau-2}) = w_{\max} \cdot 2^{q_t - 1} = W_t,$$

where the first inequality above uses the weight constraint in KNAP-($\star$). The final equality follows since for any $t \in [|\mathcal{Q}|]$, we have

$$\begin{aligned}
\sum_{k \in [k_{\min}^{(t)}, k_{\max}^{(t)}]} \lambda_k \cdot \frac{\rho(\mathrm{KNAP}^{(t)} \cap [\ell_{k,3}^{(t)}])}{1 + W_t} &= \sum_{k \in [k_{\min}^{(t)}, k_{\max}^{(t)}]} \frac{\lambda_k}{1 + W_t} \sum_{i \in \mathrm{KNAP}^{(t)} \cap [\ell_{k,3}^{(t)}]} \rho_i \\
&= \sum_{i \in \mathrm{KNAP}^{(t)}} \sum_{\substack{k \in [k_{\min}^{(t)}, k_{\max}^{(t)}]: \\ i \leq \ell_{k,3}^{(t)}}} \frac{\lambda_k}{1 + W_t} \cdot \rho_i
\end{aligned}$$

$$
\begin{aligned}
&= \sum_{i \in \mathrm{KNAP}^{(t)}} \sum_{k \in \mathcal{K}_i(U^{(t)})} \frac{\lambda_k}{1+W_t} \cdot \rho_i \\
&= \sum_{i \in \mathrm{KNAP}^{(t)}} p_i(U^{(t)}),
\end{aligned}
$$

where the third inequality follows by noting that $\mathcal{K}_i(U^{(t)}) = \{k \in [k_{\min}^{(t)}, k_{\max}^{(t)}] : i \le \ell_{k,3}^{(t)}\}$.

**Proof of Claim D.14.** To prove the result, we consider arbitrary $\tau \in [t]$, and we show that product $i \in A^\star_{\tau,3} \cap [\ell_{k,3}^{(t)}]$ must be ordered by all customer types within $[k_{\min}^{(t)}, k_{\max}^{(t)}]$. Let $i_{\max,t} = \max A^\star_{t,3} \cap [\ell_{k,3}^{(t)}]$ be the lowest weight product within $A^\star_{t,3} \cap [\ell_{k,3}^{(t)}]$, and we'll show that this product is ordered under $A_3^\star$ for any customer type $k \in [k_{\min}^{(t)}, k_{\max}^{(t)}]$. To do, we consider two cases based on the knapsack solution from which $i_{\max,t}$ was added.

- Case 1 - $i_{\max,t} \in \mathrm{KNAP}^{(t)}$: In this case we have

$$
\begin{aligned}
w_{i_{\max,t}} &\ge W_t \cdot (e^{f_k} - 1) + (1 - e^{-f_k}) \\
&\ge w(A^\star_{t,3}) \cdot (e^{f_k} - 1) + (1 - e^{-f_k}) \\
&\ge w((A_3^\star \cap [i_{\max,t} - 1]) \cdot (e^{f_k} - 1) + (1 - e^{-f_k})
\end{aligned}
$$

  where the first inequality follows by noting that since $i_{\max,t} \le \ell_{k,3}^{(t)}$, we have that $k \in \mathcal{K}_{i_{\max,t}}(U^{(t)})$, which implies the inequality. The third inequality follows since $w(A^\star_{t,3}) \le W_t$, as shown in the proof of property (ii) above.

- Case 2 - $i_{\max,t} \in \mathrm{KNAP}^{(t')}$ $(t' < t)$: In this case, for some $k' \in [k_{\min}^{(t')}, k_{\max}^{(t')}]$ we have

$$
\begin{aligned}
w_{i_{\max,t}} &\ge W_{t'} \cdot (e^{f_{k'}} - 1) + (1 - e^{-f_{k'}}) \\
&\ge W_{t'} \cdot (e^{f_k} - 1) + (1 - e^{-f_k}) \\
&\ge w(A^\star_{t',3}) \cdot (e^{f_k} - 1) + (1 - e^{-f_k}) \\
&\ge w((A_3^\star \cap [i_{\max,t} - 1]) \cdot (e^{f_k} - 1) + (1 - e^{-f_k}),
\end{aligned}
$$

  where the first inequality follows since if $i_{\max,t} \in \mathrm{KNAP}^{(t')}$, then there must be some $k' \in [k_{\min}^{(t')}, k_{\max}^{(t')}]$ such that $k' \in \mathcal{K}_{i_{\max,t}}(U^{(t')})$, since otherwise $p_{i_{\max,t}}(U^{(t')}) = 0$, and $i_{\max,t}$ would not be selected in the knapsack solution. The inequality follows since $k > k'$ by construction.

# E Additional Details of our Case Study

This section contains all missing details from the case study presented in Section 6.2.

## E.1 The Updated Bulk Returns model and assortment problem

In this section, we derive the new bulk returns assortment problem, where the customer experiences a disutility of $f_s$ for each return trip (note that there can be at most one under the Bulk Returns model), and a try-on disutility of $f_t$ for each returned item. Note that the dynamics of the Sequential Returns Model remained unchanged with this update, and if we set $f_s = 0$, the bulk model reverts back to our original framework. When $f_s > 0$, we start by formalizing the new choice process for bulk customers.

**Computing choice probabilities.** Similar to the layout of Section 2.2, we first derive the choice probabilities for a fixed order set $\mathcal{O} \subseteq [n]$, and then formalize the customers ordering process. To accomplish this first task, we consider two cases based on whether or not $|\mathcal{O}| = 1$. Note that if $|\mathcal{O}| > 1$, then customer is guaranteed to undergo a return trip, and thus experiences the associated disutility of $f_r$, whereas if $|\mathcal{O}| = 1$, then if the customer keeps the lone product, she experiences neither disutilities. The formal nature of this differences in formalized below.

- Case 1 - $|\mathcal{O}| = 1$: A customer who orders a single product $\mathcal{O} = \{i\}$ will keep the ordered product $i$, only if its total utility exceeds that of the no-purchase option:

$$\begin{aligned}\pi(i, \mathcal{O}) &= \Pr\left[U_i > U_0 - (f_t + f_s)\right] \\ &= \frac{w_i}{e^{-(f_t+f_s)} + W(\{i\})} \\ &= \frac{w_i}{e^{-(f_t+f_s)} + w_i},\end{aligned}$$

  where the first equality follows by noting that $f_t$ and $f_s$ are only experienced if the no purchase option is selected.

- Case 2 - $|\mathcal{O}| > 1$: In this case, the choice probabilities essentially revert back to the original model, with $f = f_t$. Specifically, the utility of keeping product $i \in \mathcal{O}$ is $\mathcal{U}_i(\mathcal{O}) = U_i - f_t \cdot (|\mathcal{O}| - 1 + \mathbb{1}_{i=0}) - f_s$, and thus we have that

$$\begin{aligned}\pi(i, \mathcal{O}) &= \Pr\left[\mathcal{U}_i(\mathcal{O}) \geq \max_{j \in \mathcal{O} \cup \{0\}} \mathcal{U}_j(\mathcal{O})\right] \\ &= \frac{w_i e^{-f_t \cdot (|\mathcal{O}|-1) - f_s}}{e^{-f_t \cdot |\mathcal{O}| - f_s} + w(\mathcal{O}) \cdot e^{-f_t \cdot (|\mathcal{O}|-1) - f_s}} \\ &= \frac{w_i}{e^{-f_t} + w(\mathcal{O})},\end{aligned}$$

  where we make explicit note of the fact that, in this second case, the "weight" of the no-purchase option is larger than in the first case, since the $f_s$ terms all cancel.

**Computing order sets.** Next, we step back and show how to derive the customer's optimal order set for a fixed assortment. It is important to note that the ordering decision is complicated by two distinct types of return costs, which renders the threshold cut-off point introduced in Lemma 2.2 moot. It is straightforward to see that the order set will still be some subset of the highest weight products, but one cannot merely trade-off ordering more to increase the chances of getting a desirable item with experiencing more return disutility. Now, with the introduction of the return disutility $f_s$, which can critically be avoided if only a single product is ordered then kept, the ordering decision is slightly more nuanced. Specifically, for fixed assortment $S$, define $i_{\min} = \min S$ to be the largest weight product offered within $S$. Moreover, for $\ell \in S$

define

$$\mathcal{U}_{\max}(\ell) = \begin{cases} \log\left(w_{i_{\min}} + +e^{-f_t - f_s}\right) + Q, & \text{if } \ell = i_{\min} \\ \log\left(w(S \cap [\ell]) \cdot e^{-f_t \cdot (|S \cap [\ell]|-1) - f_s} + e^{-f_t \cdot |S \cap [\ell]| - f_s}\right) + Q, & \text{o.w..} \end{cases}$$

to be the maximal expected utility if the order set is $S \cap [\ell]$, noting again that it is straightforward to see that the order set must take this form. Ultimately, we just need to explicitly check if ordering just $i_{\min}$ is better than any other multi-product order set. If not, then the ordering phase will proceed as in the original model. Putting things together, we get that

$$\mathcal{O}(S) = \begin{cases} i_{\min}, & \text{if } \mathcal{U}_{\max}(i_{\min}) > \mathcal{U}_{\max}(\ell(S)) \\ S \cap [\ell(S)], & \text{o.w.,} \end{cases}$$

where

$$\ell(S) = \max\{\ell \in S \cup \{0\} : w_\ell \geq w(S \cap [\ell - 1]) \cdot (e^{f_t} - 1) + (1 - e^{-f_t})\}.$$

Finally, we note that it is this hiccup in the order set that causes our previous algorithm for the Bulk Return assortment problem to no longer be valid (it is not clear how to update the approach to ensure that $S = \mathcal{O}(S)$), thus requiring us to develop a new IP-based approach. Before diving into this approach, we first formalize the profit function with the updated return cost structure.

**The updated assortment problem.** Having formalized the choice dynamics of the Bulk Returns Model with the new return disutility structure, we are now ready to formulate the assortment problem. Recall that the retailer incurs a deterministic refurbishing cost $c_r \geq 0$ for every returned product and a deterministic shipping cost $c_s \geq 0$ in the case of a return trip. As such the expected profit from offering assortment $S$ is

$$\mathcal{P}_B(S) = \sum_{i \in \mathcal{O}(S)} \pi(i, \mathcal{O}(S)) \cdot (r_i + c_r + c_s \cdot \mathbb{1}_{|\mathcal{O}(S)|=1}) - c_r \cdot |\mathcal{O}(S)| - c_s.$$

Similar to AO-Bulk, the goal in the assortment problem is to find the assortment $S \subseteq [n]$ that maximizes the expected profit. The problem remains NP-Hard, as we recover the original bulk returns assortment problem by setting $f_s = 0$. In what follows, we present an IP-based approach to uncover an optimal solution.

### E.2 The IP-based approach

In this section, we present our IP-based approach for the new Bulk Returns assortment problem. The approach begins by guessing two quantities related to the optimal assortment $S^*$ via efficient enumeration, and is then followed by nonlinear formulation of the problem, which is linearized via standard techniques to arrive at our final IP. We assume throughout that $S^* = \mathcal{O}(S^*)$.

**The initial guessing steps.** We first guess $i^*_{\min} = \min S^*$, which is the lowest index, highest weight product offered within $S^*$. Next we guess $|S^*|$, which is the number of products included within the optimal assortment, and we proceed to formulate an integer program, which finds the profit maximizing assortment of size at least two. The profit of this outputted assortment is then compared to that of offering $i^*_{\min}$ to ensure that we account for the possibility that the optimal assortment is a singleton.

**A nonlinear formulation.** Here, we present an initial nonlinear optimization, which yields the highest profit assortment with cardinality at least two. The decision variables $\{x_i\}_{i\in[n]}$ are simply binary indicators if product $i \in [n]$ is offered.

$$
\begin{aligned}
&\max_{x\in\{0,1\}^n} \sum_{i\in[n]} \frac{(r_i + c_r)\cdot w_i x_i}{e^{-f_t} + w(x)} - c_r \cdot |S^*| - c_s \\
\text{s.t.}\quad &(1)\quad w(x)\cdot e^{-f_t\cdot(|S^*|-1)-f_s} + e^{-f_t\cdot|S^*|-f_s} \geq w_{i^*_{\min}} + e^{-(f_t+f_s)} \\
&(2)\quad M\cdot(1-x_i) + w_i x_i \geq \sum_{j=1}^{i-1} w_j x_j \cdot (e^{f_t}-1) + (1-e^{-f_t}) \qquad \forall\, i\in[n] \\
&(3)\quad \sum_{i\in[n]} x_i = |S^*|
\end{aligned}
$$

where $M$ is a large constant and $w(x) = \sum_{i\in[n]} w_j x_j$. We also enforce that $x_{i^*_{\min}} = 1$. The constraint in (1) ensures that the customer will prefer to order the set of offered products over the singleton $\{i^*_{\min}\}$. The set of constraints in (2) further ensure that every offered product will be ordered.

**The IP formulation.** Using standard linearization tricks similar to those seen in Méndez-Díaz et al. (2014), we can linearize the objective as follows:

$$
\begin{aligned}
&\max \sum_{i\in[n]} (r_i + c_r)\cdot w_i z_i - \sum_{i\in[n]} c_r x_i - c_s \\
\text{s.t.}\quad &(1)\quad w(x)\cdot e^{-f_t\cdot(|S^*|-1)-f_s} + e^{-f_t\cdot|S^*|-f_s} \geq w_{i^*_{\min}} + e^{-(f_t+f_s)} \\
&(2)\quad M\cdot(1-x_i) + w_i x_i \geq \sum_{j=1}^{i-1} w_j x_j \cdot (e^{f_t}-1) + (1-e^{-f_t}) \qquad \forall\, i\in[n] \\
&(3)\quad \sum_{i\in[n]} x_i = |S^*| \\
&(4)\quad y\cdot e^{-f_t} + \textstyle\sum_{i\in[n]} w_i z_i = 1 \\
&(5)\quad y - z_i \leq M - M x_i \qquad \forall\, i\in[n] \\
&(6)\quad z_i \leq M x_i \qquad \forall\, i\in[n] \\
&(7)\quad z_i \leq y \qquad \forall\, i\in[n] \\
&(8)\quad x\in\{0,1\}^n, z\in[0,1]^n, y\geq 0.
\end{aligned}
$$

Together, constraints (4)-(7) ensure that if $x_i = 1$, then $z_i = \frac{e^{-f_t}}{e^{-f_t}+w(x)}$, which is precisely the no-purchase choice probability under the selected assortment. Ultimately, we compare the expected profit of the assortment produced by the above IP to the expected profit earned by $i^*_{\min}$, noting that the better of the two must be the optimal assortment.

#### E.2.1 Data Cleaning and Parameter Estimation

In this section, we detail the three-step procedure employed to estimate the key parameters of our model. This methodology is applied to a carefully selected subset of the ISMS Durable Goods Data Set 1 (Ni et al., 2012). It is important to note that our data does not include records of no-purchase events. Therefore, we generate no-purchase transactions based on the total number of transactions in our data. The specifics of generating no-purchase transactions are explained in Step 2 below.

**Step 1 - Data cleaning and organization.** Before estimating the model parameters, we perform an initial data cleaning process, following the key steps outlined below. Additionally, after completing the estimation process, we further clean the data to remove instances that do not align with our model assumptions.

1. *Transaction Volume:* To ensure the relevance and robustness of our analysis, we begin our data cleaning process by ranking each category in the original dataset based on the number of transactions. We then eliminate categories that primarily involve the products that are no longer in use. Finally, we retain only the top 2 categories, which are the Televisions and Speakers categories. The remainder of this discussion focuses on the former category, but can easily be extrapolated to the latter.

2. *Current Relevance:* To ensure our experiments accurately reflect current customer behavior, we meticulously review and remove outdated or obsolete products, such as TV antennas and telecorders, from the list of subcategories. To this end, we select the following 7 subcategories of TELEVISION category:

   - SPECIALTY TV,
   - 30 inch AND LARGER TVS,
   - 27 inch TELEVISIONS,
   - TELEVISION,
   - 19-20 inches COLOR TV,
   - 9-16 inches COLOR TV,
   - 25 inch TELEVISION.

   These subcategories were chosen based on their high transaction volume, indicating their relevance within the dataset, and their product diversity, which is crucial for understanding customer purchasing behavior. Additionally, these subcategories contain products that are still in use today, ensuring that our experiments reflect current customer behavior.

3. *Price Threshold:* To maintain the integrity of our analysis and exclude insignificant transactions, we systematically filter out products priced below $1 by using the 'unit price' column. This step ensures that the data reflects meaningful customer purchases and avoids potential skewing that could arise from minor, low-cost items.

4. *Unique Product IDs:* In addition to Product IDs, the original dataset includes subcategory, brand, and unit price information to identify purchased products. However, some transactions feature products with the same price, subcategory, and brand but different IDs across various transaction locations. These inconsistencies prevent us from using the original Product IDs to accurately identify products at specific transaction locations. To address this issue, we create Unique Product IDs (UPIDs) based on a product's brand and subcategory information.

5. *Assortment Assignments:* Each transaction is assigned to a store based on its location information. To refine each store's assortment, we systematically remove duplicate entries sharing the same UPIDs, retaining only the most recent transaction. This ensures an up-to-date and accurate representation of each store's assortment. Consequently, we end up with 635 distinct assortments, each assigned to a store according to its location data.

6. *Distinct Products:* Maintaining sufficient product variety is crucial for understanding preferences and decision-making processes, gaining insights into customer reactions to different assortments, and ensuring robust analysis of purchasing behavior. Therefore, we exclude any stores with fewer than five products in their assortment.

For the Television category, after cleaning the data, we are left with 4,318 transactions across 474 stores, which span 84 distinct UPIDs.

**Step 2 - Estimating the MNL-based weights.** We now move on to describe how we estimate MNL weights using maximum likelihood estimation (MLE). This process is outlined as follows:

- *Utility parameterization:* The deterministic component of the utility generated by an arbitrary product $j$ is:

$$v_{jt} = \beta_1 + \beta_2 \cdot \text{Price}_j, \tag{24}$$

  where $\beta_1$ represents the product fixed effect, $\beta_2$ is the price coefficient, and $\text{Price}_j$ is the price this product.

- *Organizing the data:* Consider a data set consisting of $\tau$ choice events, indexed by $t \in [\tau]$. Each event $t \in [\tau]$ represents the arrival of a single customer who is offered an assortment of products $S_t \subseteq [n]$. The variable $z_t$ denotes the option selected by the customer, which can be either a product $j \in S_t$ or the no-purchase option. Recall that the assortments are assumed to be the products previously purchased at the store where the given purchase was made. Since our data set does not include transactions with no-purchase events, we add no-purchase events at varying levels. Specifically, the total number of no-purchase transactions is set to $10\%, 20\%, 70\%, \text{and } 80\%$ of the total transactions at a particular store. All-in-all, the transactions in the dataset can be fully specified through the set:

$$DS = \{(z_t, S_t, \text{Price}_t) : t \in [\tau]\}$$

where $\text{Price}_t$ denotes the prices of the products offered in assortment $S_t$. We note that the data is aggregated across all stores.

- *The MLE problem.* Given the just described set-up, the log-likelihood function can be written as

$$\begin{aligned}\mathcal{LL}(\beta|DS) &= \log\left(\prod_{t=1}^{\tau}\pi(z_t, S_t, \text{Price}_t, \beta)\right)\\ &= \sum_{t=1}^{\tau}\log\left(\pi(z_t, S_t, \text{Price}_t, \beta)\right),\end{aligned}$$

where $\pi(z_t, S_t, \text{Price}_t, \beta) = \frac{e^{v_{z_t}}}{1+\sum_{j\in S_t} e^{v_{jt}}}$.

This expression gives the likelihood of observing data point $(z_t, S_t, \text{Price}_t)$ under the MNL model. Therefore, the general MLE problem of interest can be formulated as

$$\max_{\beta} \mathcal{LL}(\beta|DS)$$

The estimator obtained by solving the above problem is the vector of coefficients $\beta$ that maximizes the log-likelihood function for the available choice data. This optimization problem is convex and an optimal solution can be obtained using standard convex optimization tools.

**Step 3a - Deriving the return disutilities.** In this step, we first use the estimated deterministic utility components from the previous step to derive store-specific return disutility parameters. Initially, we compute a total return disutility $f \geq 0$ as a function of the log-sum of the product weights available in the given store. Assuming $n$ denotes the number of products offered in the store, $f$ is computed as:

$$f = \alpha \cdot \log\left(\sum_{i\in[\eta]} e^{v_i}\right)$$

where $\alpha \in \{0.05, 0.1, 0.15\}$ represents a scaling factor to introduce diversity across our instances. The log value represents the expected utility of ordering all available products, disregarding potential returns. Given that the log-sum of the weights is the expected maximal utility if all products are offered, we set $f$ to be some (small) fraction of this best-case utility. Later, when we generate the instances, we will divide the total disutility $f$ into the two components $f_s$ and $f_t$

**Step 3b - Deriving the return costs.** Finally, we derive an overall return cost $c$, which, similar to the total disutility $f$, will be partitioned in various proportions into $c_r$ and $c_s$ when we create the assortment instances. According to a 2022 Forbes article (Schneider, 2022b), the estimated cost of returning a \$50 product is around \$33, covering labor, liquidation losses, and transportation. To estimate these costs in our model, we calculate $c \geq 0$ as a percentage of the average price of products available in each store. Assuming $n$ denotes the number of products available in a particular store, we define the return cost $c$ for each returned product as:

$$c = \gamma \cdot \frac{1}{\eta} \sum_{i \in [n]} \mathrm{Price}_i$$

where $\gamma \in \{0.1, 0.2, 0.3\}$ is scale factor again used to create diversity in the instances we consider.